



Kentucky Children's Hospital

Department of Surgery
Division of Pediatric Surgery

PEDIATRIC TRAUMA CARE GUIDELINES

2011

Last Revision: July 2021

Disclaimer

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Pediatric Trauma Care Guidelines

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Preface

The University of Kentucky HealthCare Trauma Program faculty and staff developed and maintain the Pediatric Trauma Care Guideline Manual to provide resource material relevant to the variety of disciplines involved in the care of the multi-injured pediatric trauma patient. The UK Trauma Program strives to deliver timely and effective care to the injured patient utilizing evidence-based guidelines and pathways. This manual outlines the expectations and standards of care appropriate for Level I Pediatric Trauma Center designation.

To further enhance trauma care internally and in our service region, the Pediatric Trauma Care Guidelines will be published on the University of Kentucky Care Web. Link for the Care Web is listed here: www.hosp.uky.edu/careweb. When updates occur, the guidelines will be updated on the Care Web.

The UK Trauma Program Office recommends periodic review of the site. Please direct inquiries of this manual to the Trauma Program Office at 323-5022.

Other websites demonstrating the University of Kentucky and Kentucky Children's Hospital Pediatric Surgery Service information are as follows:

<http://surgery.med.uky.edu/surgery/sections/pediatric>

<https://ukhealthcare.uky.edu/kentucky-childrens-hospital/services>

<https://ukhealthcare.uky.edu/kentucky-childrens-hospital/services/surgery-pediatric>

<https://ukhealthcare.uky.edu/kentucky-childrens-hospital/services/surgery-pediatric#providers>

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Resolution

- The University of Kentucky HealthCare, Kentucky Children's Hospital is fully supported by the continued maintenance and improvement of the Trauma Program.
- The Trauma Program complies with the Level I Trauma Center criteria established by the American College of Surgeons Committee on Trauma.
- The Trauma Program addresses all components of the trauma system including injury prevention, prehospital care, acute hospital care, and rehabilitation.
- Support for the program is demonstrated through the commitment of resources to ensure the trauma program's compatibility with the mission of the University of Kentucky HealthCare Enterprise including patient care, research, and education.
- Support is further demonstrated through facilitation, when necessary, of the quality improvement process activities undertaken by the trauma program.
- The University of Kentucky Healthcare Medical Staff, through the clinical board, endorses the continued commitment of University of Kentucky HealthCare as the Level I Trauma Center for Central and Eastern Kentucky as determined by the American College of Surgeons Committee on Trauma criteria.

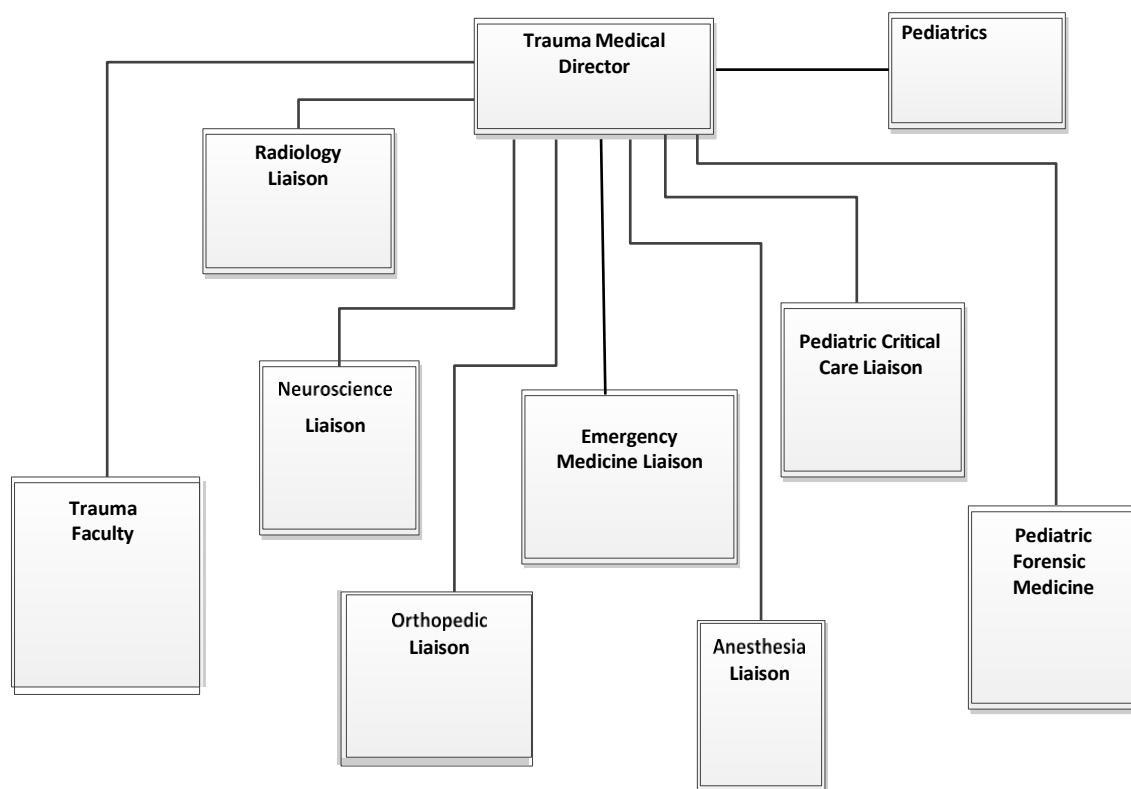
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Administrative Organization and Management of Pediatric Trauma Program

- Pediatric Surgery is a division of the Department of Surgery at the University Of Kentucky College Of Medicine.
- Kentucky Children's Hospital and the Division of Pediatric Surgery provide comprehensive pediatric trauma care for the most severely injured children locally, regionally, and across the Central and Eastern Kentucky referral area and beyond as needed.
- Pediatric Surgery provides oversight and management of trauma care for pediatric trauma patients at Kentucky Children's Hospital Level I Trauma Center.
- The Pediatric Trauma Medical Director (TMD) is an Attending Pediatric Surgeon who leads the multidisciplinary activities of the trauma program. The TMD maintains oversight and leadership of the pediatric trauma program in conjunction with the Pediatric Trauma Program Manager (Peds TPM), while working in conjunction with multidisciplinary services to assure the overall comprehensive care as required for a Level I Trauma Center by the American College of Surgeons.
- The Peds TPM assumes day to day responsibility for operational, quality improvement, and patient safety aspects as are related to nursing, ancillary departments, and physician functions as delineated by the TMD.
- Pediatric Trauma Service is the primary structure for providing care for injured patients and includes multidisciplinary services and resources to ensure efficient provision of care. The following chart depicts multidisciplinary services involved:



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Pediatric Surgical Service Team Overview

Attending Physicians

- Pediatric Surgical Service (PDS) attending physicians have overall responsibility for patient care.
- PDS Attending surgeons have oversight of Residents working on the pediatric surgery service.
- PDS Attending surgeons take calls from outside physicians requesting consults or referring patient to the service.
- Attending surgeons take all UKMDs calls 24/7.

Physician Assistant (PA)

- The role of the PA on the Pediatric Surgery Service is versatile.
- Rounds with team on weekday mornings.
- First assist in OR when needed.
- Assists with floor questions/problems, patient phone calls.
- Facilitates home health referrals and transfers to rehabilitation facilities.
- Functions as a primary provider in PDS clinic and is able evaluate follow-up patients & some new patients.
- Assist Attending and Resident physicians with other duties as requested.

Chief Resident

- The PGY-4 General Surgery Resident is in charge of the Pediatric Surgery Service during his or her rotation.
- Pre-rounds with the team Monday through Friday on all patients before first activity of the day – conferences, OR starts, etc.
- Has final responsibility for upkeep of daily patient census and morbidity/mortality list.
- Assigns case coverage for all residents.
- Responsible for fielding parent and patient phone calls after hours.
- Rounds with the Attending surgeon on call.
- Leads weekly educational sessions for students and residents on service.

Junior Residents

- Record medical history and physical exams on all new patients.
- Record daily exam and progress notes on all patients and consults.
- Maintain the daily patient list.
- Be prepared to present patient on rounds. Junior residents should present patients during teaching rounds.
- Follow all consults during hospitalization.
- Obtain operative permits and verify preoperative notes and orders. If unsure of operative plan, discuss the consent with the attending before talking to family.
- All patients MUST be seen in preoperative holding area for a final check of diagnosis, verify paperwork, and mark surgical site.
- Assist with operations and log your cases. Before entering OR, read about the procedure.

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4th Year Medical Students (Als)

- Perform the duties of the PGY-1 Resident, supervised by either a Resident or Attending.
- Record medical history and physical exams.
- Record daily exam and progress notes on patients and consults.
- Be prepared to present 2-3 patients during Teaching Rounds.
- Present 15-minute talk on topic in pediatric surgery to the team.
- Participate in rounds and procedures with team.
- Read about your patients.

Third Year Medical Students

- Record medical history and physical exam on at least 1 patient per week.
- Record daily exam and progress notes on at least 1 patient per day.
- Be prepared to present at least 1 patient during Teaching Rounds.
- Participate in rounds and procedures with team.
- Read about your patients.

Pediatric Surgery Conferences

- All team members are expected to attend unless precluded by patient care responsibilities.
- The Pediatric Surgery Conference is a mandatory weekly conference. Typically held Mondays at 0700.
- Teaching Rounds are held Monday and Friday mornings and begin in the NICU.

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Patient Care Specifics

- The Pediatric Trauma Service accepts pediatric trauma patients from age **0 to 15 years**.
- Trauma patients age ≥ 16 years are to be evaluated by the Adult Trauma Service.
- Pediatric trauma activations are based on established criteria. Patients may arrive via EMS ground or air medical, POV/walk in, or have an acute deterioration event while in the Emergency Department.
- **Pediatric trauma activation criteria and care are NOT the same as the adult service.** Residents are required to be familiar with Pediatric Trauma System Processes, Activation Criteria, and Roles and Responsibilities outlined in this document.
- Patients referred to the Pediatric Trauma Service from another hospital are considered trauma expects.
- Pediatric trauma expects with minor injuries can be evaluated by the Emergency Medicine Service.
- Patients in the Emergency Department become the responsibility of the PDS Residents either by consult request from the Emergency Medicine Service or by pediatric trauma activation (TA or TAR).
- Initial assessment and evaluation of pediatric trauma patients shall proceed according to ATLS protocol.
- The PDS Chief Resident assumes responsibility for timely evaluation, management, and disposition of all pediatric trauma patients; including prompt notification of the PDS Attending surgeon.
- The PDS Chief Resident is allowed to take call from home. However, the PDS Chief Resident is expected to return to the hospital for pediatric activations after hours and on weekends.
- The in-house Trauma Surgery Chief Resident (PGY-4 or PGY-5) will begin evaluation and management of the patient until the PDS Chief Resident arrives and assumes care.
- Patient disposition should be determined within **30 minutes** of ER arrival.
- Entire diagnostic evaluation/disposition should not exceed **120 minutes**.
- If the patient requires surgical intervention, the OR Desk is to be called immediately.
- A surgical resident will accompany all hemodynamically unstable patients outside the ER for all diagnostic procedures. Physicians are not required to accompany stable patients.
- ED nursing staff will accompany trauma patients for any testing or procedures performed outside the ED.
- During the highest level trauma activation, an operating room is placed on hold. **The Chief Resident is responsible for releasing the OR suite as soon as it is determined that the OR is not needed.**

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- **Documentation, Consults/Referrals, Orders, and Legal Responsibility**

Documentation

- Remember the medical record is a legal document. Think before you type.
- Attending physicians assume liability for Resident actions and words.
- Trauma flow sheets are completed by the Scribe for all trauma activation patients.
- The Scribe will document the names and arrival times of the Attending and Resident physicians who respond to trauma activations. **State your name and service for the Scribe upon arrival to the trauma bay.**
- The pediatric trauma admission form is to be completed IN FULL on ALL pediatric trauma patients admitted to the Pediatric Trauma Service or those receiving consultation from the Pediatric Trauma Service. This includes ED and OR mortalities.
- Consultation templates prompt for documentation of the date and time the physician was called to see patient and the date and time that you actually saw the patient.
- Documentation of consultation request and time to patient care are American College of Surgeons Performance Improvement criteria that require ongoing monitoring.
- The following data are frequently missing from the documentation:
 - Injury time
 - Loss of consciousness and length of time (if known)
 - Laboratory results including ETOH and UDS
 - Procedures
 - Primary diagnosis in detail (MVC is **NOT** considered an adequate medical diagnosis and **WILL NOT** be accepted)
 - Admitting PDS Attending surgeon
 - Referring physician and referring hospital
 - For patients transported directly from the scene, the county where the injury occurred
- All procedures should have a detailed procedure note completed in the medical record.
- Unless an emergent situation necessitates otherwise, an Attending physician will be notified prior to performing any procedure.
- For procedures performed in the ED after hours and on weekends, document the Emergency Medicine Attending as the supervising physician for the procedure note, unless the Pediatric Surgery or Adult Trauma Surgery Attending was present and oversaw the procedure.
- Daily progress notes and any acute changes in patient condition should be accurately documented in the medical record. **DO NOT** copy and paste.

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Consults/Referrals

- Pediatric Surgery Service is responsible for the timely evaluation of pediatric patients when consulted by the Emergency Medicine Service.
- PDS Chief Resident is in-house during working hours should be contacted for consult request.
- The ESS Resident is to be contacted for consult request after hours and on weekends.
- Chief Resident or ESS Resident assumes ultimate responsibility for the timely evaluation, management, and disposition of all pediatric surgery emergency patients. This includes timely notification of the Attending surgeon.
- On occasion, there will be direct inter-facility transfers from referring hospitals to the PICU that will bypass ED evaluation.
- Consult request date and time and time and date at patient bedside should be documented by the resident and or attending in the consult note. ACS requires monitoring of consult request in accordance with facility policy and procedures.
- Consultants should be notified promptly following secondary survey (< 20 minutes from patient arrival).
- Consultants **should respond to a page within 10 minutes** and should be **present for patient evaluation within 20 minutes of notification**. Hospital Policies related to consultation timeframes:
 - Inpatient Consults: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A08-85%20Inpatient%20Consultations.docx?Web=1>
 - Admissions: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A02-000%20Admissions%20Hospital%20Inpatient.docx?Web=1>
 - Emergency Department: <https://ukhealthcare.mc.uky.edu/policies/departmental/layouts/15/WopiFrame.aspx?sourcedoc=/policies/departmental/Emergency%20Department/ED02-01%20ED%20Admissions.docx&action=default>
- Consultation should be performed by an upper level resident (PGY-2 or higher) or faculty. Interns should not be notified for ED patient evaluation unless all other members of the consultant team are involved in priority patient care that precludes their presence.
- The decision regarding admission, level of care, and admitting service should be made within 30 minutes on most patients, with exception of patients going directly to OR.
- Pediatric Trauma Service is dependent on referrals from outside physicians and accepts referrals from a wide geographic area.
- ALL pediatric trauma patients referred by an outside physician are to be accepted for transfer unless otherwise instructed by an Attending physician. NO EXCEPTIONS.
- Maintain a professional, cooperative, and friendly relationship with referring physicians. Most referring hospitals do not have the resources to care for these patients.
- Document the name and phone number of referring physicians for prompt follow up.
- Initial discussion with referring physicians should include appropriate recommendations and suggestions regarding patient care prior to transport in order to ensure optimal transfer (e.g. additional I.V. access, fluid resuscitation, immobilization, splinting, and mode of transport).
- After accepting patients for transfer, notify the ED triage & relay pertinent patient information: *name/age/sex, referring physician and ETA; Residents to be page on arrival and pager number; patient history/chief complaint; specific diagnostics/interventions to be prepared.*

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Orders

- Pediatric trauma order sets are available in the electronic medical record.
- Review your orders. Assure that you are using the correct orders for each patient.
- Orders must be completed in a timely manner. Patient admissions and transfers cannot proceed until orders are completed.
- **Blood alcohol (BA)** and **urine drug screen (UDS)** are mandatory on all trauma activation patients **age ≥ 12 years**.
- BA and/or UDS may be obtained on other pediatric trauma patients at the discretion of the Chief Resident or Attending physician.
- **Urine pregnancy test** should be obtained for all females of childbearing age.

Legal Responsibility

- Some pediatric trauma patients become involved in custody, civil, or criminal cases. Residents may receive a subpoena to testify in these cases. In most circumstances, it is the Attending physician that supervised the case who is responsible to testify.
- Bring all subpoenas to the immediate attention of the Chief of Pediatric Surgery.
- All other medical practice/legal issues should be brought to the immediate attention of the Attending Physician and/or the Chief of Pediatric Surgery.

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SECTION I: ADMINISTRATIVE

Pediatric Surgical Service and Resident Rotation

- **OR Scheduling, Service Transfers, Discharges, and Pediatric Surgery Clinic**

OR Scheduling

- When booking operative case assure the following is done:
 - H&P
 - Consent from parent or guardian
 - Call the OR desk to schedule case
- OR desk will require the patient's name, MRN, location, procedure, Attending surgeon, classification, and how long NPO.
- Booking of emergent surgical cases is the primary responsibility of the Chief Surgical Resident.
- **Operation Classifications are:**
 - **Class A Emergency** (require surgery **within 1 hour**)
 - **Class B Emergency** (require surgery **within 4 hours**)
 - **Class C Emergency** (require surgery **within 8 hours**)

Service Transfers

- Patients with single system injury may be transferred to the appropriate subspecialty service after the patient is stable. This is typically after the first 24 hours from admission.
- Coordination of service transfer process is the primary responsibility of the PDS Chief Resident.
- After patients are transferred to another service, the PDS Service should function as a consultative service as appropriate.

Hospital Discharges

- All discharge summaries are to be completed at time of discharge.
- Specify the name of the child's primary physician.
- Discharge summaries are to be succinct and correspond to timeline of patient care during hospitalization.
- Document the patient's weight at the time of discharge.
- Discharge process is to include:
 - Follow up appointments scheduled (date/time/specialty). All appointments are to be made prior to patient discharge, including subspecialty appointments, and documented in the summary.
 - Assure patients are given adequate supplies and medication to last until their first clinic appointment.
 - Assure patients/guardians are given applicable written discharge instructions and instructions are documented in the discharge summary.
 - PDS clinic appointment should be scheduled 2-4 weeks after discharge unless otherwise specified.
 - Not all patients will require a PDS clinic appointment. Verify the need for appointments on all patients with the Chief Resident, PA, or Attending physician before scheduling.
 - The Attending physician of record named on the discharge summary should be the Attending physician who is on service at the time the patient is discharged.

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Pediatric Surgery Clinic

Pediatric Surgery clinic is located on second floor of the Kentucky Clinic in the Pediatric Specialties clinic. Clinic hours are Monday 1pm – 4pm and Friday 9am – 1pm

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SECTION I: ADMINISTRATIVE

Pediatric Surgical Service and Resident Rotation

- **Tiered Call Escalation Guideline for Pediatric Surgery Patients**

Junior Surgical Resident to notify Chief Resident:

- Persistent hypoxemia
- Persistent or worsening lactic acidosis
- New or unresponsive oliguria or anuria
- Increasing ventilator needs
- Significant worsening of vital signs, physical exam, or laboratory / radiographic tests
- Consideration of consultation from another service
- Inadequate pain control
- Persistent nursing calls on a single patient
- Difficult family interactions
- Any other questions or concerns

Surgical Resident to notify PDS Attending:

- New admissions / consults
- Transfer to PICU
- Major family conflict
- Conflicts with consulting services or KCH nursing
- Procedure complications
- New, refractory organ failure
- Code
- Death
- Any other questions or concerns

KCH Nursing to notify PDS Attending:

- If no answer with notification of Junior Resident and Chief Resident
- Any concerns not adequately addressed by PDS Surgical Residents

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SECTION I: ADMINISTRATIVE

Pediatric Level I Trauma Program Performance Improvement and Patient Safety Plan

Introduction

The Trauma Performance Improvement Master Plan for the Trauma Program Office outlines the overall general processes for performance improvement and patient safety for all trauma services. This plan outlines the processes specific to the Pediatric Trauma Program.

Pediatric Trauma Program Overview

The Kentucky Children's Hospital Level I Pediatric Trauma Performance Improvement and Patient Safety program (PIPS) is directed by the Pediatric Trauma Medical Director (TMD). The Pediatric Trauma Coordinator (PTC) and the Pediatric Trauma Registrar (PTR) work in collaboration with the TMD to provide daily oversight of the PIPS. The Orthopedic Trauma Coordinator (OTC) collaborates with the TMD, PTC, and PTR regarding the orthopedic care of the pediatric trauma patients. This collaborative oversight provides means to assure safe, efficient, and effective care to injured pediatric patients. The Kentucky Children's Hospital Pediatric Trauma Program defines the pediatric trauma population as patients 15 years and under. The American College of Surgeons (ACS) defines the pediatric trauma population as patients age 14 years and under.

Performance Improvement and Patient Safety Process (PIPS)

The PIPS process is primarily responsible for monitoring and improving structures, processes, and outcomes within the institution. The PIPS program and activities also support participation and integration in the overall facility quality program, state, and regional programs. The PIPS program and processes integrate all departments, resources, and professions internally and externally that are involved in the patient's care to assure the multidisciplinary system. PIPS process delineates identifiable opportunities for improvement through, provider, systems, and/or disease. PIPS activities and documents are managed in accordance with all regulatory and institutional requirements for privacy and confidentiality. The multidisciplinary PIPS program provides collaborative efforts for the quality improvement and patient safety that assures an effective continuum of care throughout the institution via evidenced based practice guidelines and American College of Surgeons Committee on Trauma (ACS) criteria. The Pediatric Trauma Program's participation in the ACS Trauma Quality and Improvement Program (TQIP) provides a means to have risk adjusted benchmarking data that compares this trauma center to other like facilities for patient outcomes.

PIPS Operations

The daily concurrent and retrospective PIPS activities are primarily performed by the TMD, PTC, PTR, and OTC. The following information describes methodologies and processes each perform to provide PIPS oversight and integration with the enterprise quality programs. This is an overview of daily operations and is not intended to be all inclusive as each case potentially presents unique circumstances.

Trauma Medical Director (TMD)

- Attends weekly General Surgery Grand Rounds.
- Attends weekly General Surgery Morbidity and Mortality Conference.
- Conducts weekly educational sessions to medical students and residents on the pediatric surgery service.
- Participates in the UK Enterprise and Kentucky Children's Hospital Organ Procurement Committee provided by the Kentucky Organ Donor Affiliates (KODA).
- Meets monthly with Chief Medical Officer to provide updates on the trauma program.
- Member of the UK Enterprise and KCH Transfusion Committee.
- Chairs the Pediatric Inter-Departmental Trauma Quality Assurance Committee (IDTQA).
- Chairs the Child Protection Team Committee (CPT).
- Collaborates with PTC to conduct Pediatric Trauma Patient Care Committee (PTPCC).
- Meets weekly with PTC to review cases and operational needs for the trauma program.

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- Actively works with PTC to develop and implement action plans for identified needs, including re-evaluation and loop closure.
- Participates in trauma program Pre-hospital Quality Assurance Committee.
- Conducts ongoing communications with internal and external professionals as required for identified process improvements.
- Member of the KCH Quality and Patient Safety Committee.
- Conducts annual evaluations of trauma surgeons, critical care, anesthesia, orthopedic surgery, radiology, neurosurgery, and emergency medicine liaisons.
- Works collaboratively with PTC and program liaisons to assure guidelines are maintained and based on evidenced-based practice.
- Conducts quarterly educational simulations and debriefings for Trauma Team members.
- Works with PTR on data for identifying information for research, publications, abstracts, and poster presentations.
- Works with PTR and Injury Prevention to support state legislation related to children.
- Serves as member of the Kentucky Trauma Advisory Committee (KyTAC) providing oversight to the Kentucky Trauma System.
- Serves as member of the Pediatric Trauma Society.
- Presents annual Trauma Quality Report to KCH Quality and Patient Safety Committee.
- Actively participates in review of TQIP reports with PTC and PTR.
- Attends Adult IDTQA meetings for informational purposes as needed.

Pediatric Trauma Coordinator (PTC)

- Performs daily review of the Pediatric Surgical Service census.
- Performs daily reviews of the Emergency Department departures, readmissions, and admission reports for identification of new patients.
- Performs daily individual PI primary reviews of each new pediatric trauma patient.
- Maintains a monthly log of all identified trauma patients for the trauma registry.
- Maintains reviews of pediatric trauma patients throughout their hospital stay.
- Meets with TMD weekly to conduct secondary reviews, establish plans for further review, and discusses oversight activities and events of the program.
- Daily communications with multiple types of professions, services, in relation to PIPS
- Coordinates, compiles, and maintains all information for committee presentation including case reviews and associated PI.
- Maintains PI records by means of monthly logs, daily review forms, PI log, and registry under the enterprise Risk Management Patient Safety Organization process.
- Maintains minutes for IDTQA, CPT, and Patient Care Committees.
- Attends adult IDTQA for informational purposes.
- Participates in Pre-hospital Quality Assurance Committee (PHQAC).
- Works collaboratively with PTR as needed.
- Developed, maintains, and distributes Pediatric Trauma Dashboard.
- Prepares annual Quality Report and presentation of the Pediatric Trauma program.
- Reviews TQIP reports and drills down results as needed.
- Participates in monthly COTVRC webinars.
- Member of Pediatric Trauma Society and participates in monthly meetings and offerings.
- Maintains Pediatric Trauma Guideline manual and assures guidelines are reviewed, updated, and available throughout enterprise via intranet (CareWeb).
- Serves as a resource internally and externally daily.
- Follows up with referring centers for PI, outreach, and educational purposes.

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- Assures referral centers have portal access to receive follow ups on their transferred patients.
- Attends annual TQIP Conference.
- Assist with planning and implementation of annual statewide Trauma Symposium.
- Participates ad hoc on EMSC advisory committee.
- Provides trauma education to new KCH nursing orientation (intro to TPO, guidelines, etc.)
- Maintains routine communication with POR liaison, through regular meetings or emails, calls, etc.
- Assures PI recommendations, action plans, re-evaluation, and loop closures are maintained and communicated to the multidisciplinary committee members.
- Develops and implements needed PI action plans, performs re-evaluations and loop closures as accomplished.
- Routes Trauma PI up through KCH Quality and Patient Safety Committee as needed and requested.
- Performs reviews and follow ups at the request of KCH Administrative Quality Review committees.
- Collaborates with UK MD Physician Liaison to assure a continuum of outreach needs to referral area.
- Instituted, plans, and participates in statewide Trauma Manager Meetings.
- Attends Kentucky Trauma Advisory Committee meetings.
- Collaborates with POR, NS, EM, or other liaisons to conduct educational in-services for staff as needed.
- Collaborates with KCH Staff Development in coordination of KCH Nursing Grand Rounds.

Pediatric Trauma Registrar (PTR)

- Compiles daily list of new pediatric trauma patients relevant to entry into trauma registry.
- Collects all pediatric trauma activation records for data entry.
- Collects, abstracts, and enters all pediatric trauma cases into registry via manual collection.
- Maintains pediatric trauma database in accordance with ACS NTDB and TQIP standards.
- Assists with the Pediatric Statistical Dashboard monthly and annually.
- Conducts data reports upon requests.
- Maintains competencies related to participation in IRB research activities.
- Meets with PTC and TMD as needed.
- Maintains open communications to TMD regarding provider documentation needs.
- Collaborates with PTC to compile data for annual quality report.
- Monitors trends in data and notifies TMD and PTC of significant trends.
- Annually reviews data elements for registry inclusion with TMD and PTC.
- Participates in statewide Trauma Registrar meetings.
- Presents educational offerings to statewide trauma registrars.
- Serves as informational resource upon request.
- Attends annual TQIP conference.
- Participates in online webinars to maintain knowledge related to pediatric registry.
- Member of pediatric trauma IDTQA, CPT, and Patient Care committees.
- Maintains registrar certifications and continuing education in accordance with ACS requirements.
- Participates in annual statewide Trauma Symposium.

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Committees and Reporting Structure

The Pediatric Trauma Program structure for the PIPS program includes the Pediatric Trauma IDTQA, CPT, and PTPCC. All committees meet on a bi-monthly schedule to the equivalent of six (6) meetings annually. Additionally, scheduled meetings may be conducted as needed.

- **The Child Protection Team (CPT) Committee** is a subcommittee of the IDTQA committee. The Pediatric Trauma Program has a significant number of identified non-accidental trauma (NAT) cases that require the dedication of a separate committee to review cases and operations for our NAT population. This is a multidisciplinary committee chaired by the TMD with presentations by the Pediatric Forensic Medicine Team. Risk Management is an active member of this committee. All recommendations from committee regarding care and operations are routed up through the IDTQA committee which meets immediately following the CPT meeting.
- **The Pediatric Interdepartmental Quality Assurance (IDTQA) Committee** is the multidisciplinary peer review committee that provides review and oversight to pediatric trauma care and program operations. This committee meets bi-monthly and convenes after CPT meetings. Surgeons and Physician Liaisons are required to have a minimum of 50% attendance at this committee. All case reviews and individual PI opportunities are routed through this committee. Risk Management is an active member of this committee. Recommendations from this committee impacting the operations of the trauma program are routed to the Pediatric Trauma Patient Care Committee.
- **The Pediatric Trauma Patient Care Committee (PTPCC)** meets bi-monthly on the fourth Monday of the month and the same months as IDTQA and CPT. This committee is the multidisciplinary operations meeting for the pediatric trauma program. This committee receives recommendations from the CPT and IDTQA meetings and proceeds with necessary action plans for systems improvements. This committee receives reports from all multidisciplinary departmental representatives to assure communications and continuum of care throughout KCH. Opportunities for improvement brought forth through this committee are then communicated back through the committee structure beginning with IDTQA and or CPT if applicable. Members of this committee have the responsibility to assure dissemination of information to their respective departments and staff.

Additional means of quality review are performed through the **KCH Quality and Patient Safety Committee**, enterprise **SWARMS**, and via **specific requests** from provider services and KCH administration. This process assists in the integration of the trauma program into the overall institution quality programs. This collaboration allows for a structured flow for the quality program, timely reviews, and timely recommendations for improvement. Any activities conducted outside the committee structure is routed through the next scheduled committee meetings as follow up or continued action, re-evaluation, and ultimately goal of loop closure.

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Daily Trauma Guideline Compliance Initiative for Peak Performance in the Care of the Pediatric Trauma Patient

Description:

This initiative is a daily process for which the Pediatric Trauma Coordinator, PDS PA, and Social Worker Director will collaborate to assure compliance and follow up on pediatric trauma guidelines that are not routine and are easily overlooked and become incomplete. This is a means for which these three individuals will work with the PDS Attending's, Residents, and Pediatric Trauma Social Workers to help assist them in being informed of patient needs/information they may not be aware of.

Goal:

To assure our pediatric trauma patients are receiving the full benefit of clinical care during their stay at Kentucky Children's Hospital, assuring interventions and results are ordered, performed, completed, reviewed, followed up on, and documented in accordance with the pediatric trauma clinical guidelines.

Process:

- The Pediatric Trauma Coordinator (PTC) reviews pediatric trauma records daily.
- The PTC maintains the daily trauma quality and patient safety reviews based on ACS criteria for a Level I Pediatric Trauma Center and the KCH Trauma Guidelines.
- Based on these guidelines, should there be an element incomplete; the PTC will notify the PDS PA and SW Manager.
- This notification will then alert the appropriate parties of new patient care information and or patient needs.
- Examples include but not all inclusive are: *positive UDS that hasn't been addressed according to pediatric trauma guidelines; a positive BHCG on patient that may be preparing for OR and isn't noted in record that results have been discussed or that the provider is aware.*
- These notifications will not be all inclusive, but shall only be implemented when assurance of provider acknowledgement of pertinent information regarding patient's care appears to be necessary based on performance improvement and patient safety daily quality reviews.
- Primary means of communication will be via email to the appropriate attending.
- PDS PA and or SW will be copied on all correspondence of this nature to assure completion of communication and processing of patient needs.

*Note: This process is an additional means of assistance to the care providers, as available, and **Does Not** relinquish overall responsibility of the patient's care from the care providers. This process is an element of performance improvement and patient safety.

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SECTION II: OPERATIONS

Pediatric Trauma Process Guidelines

Overview

- This section is designed to clarify the pediatric trauma process for admissions to the University of Kentucky HealthCare, Kentucky Children's Hospital Level I Trauma Center.
- The Pediatric Trauma Operations section is composed of requirements from the American College of Surgeons Committee on Trauma, our Division of Pediatric Surgery, and University of Kentucky, Kentucky Children's Hospital that guides our Level I Pediatric Trauma Center.
- The Pediatric Surgical Service has oversight of the Pediatric Trauma Program and the patients therein.
- The PDS surgeons collaborate with multiple subspecialty services which are involved in the care of the pediatric trauma patients.
- The Trauma Medical Director and the Pediatric Trauma Program Manager from the Trauma Program Office, maintain day to day oversight of the programs operations and patient care systems to assure quality and safety in accordance with all governing standards/criteria.
- Pediatric trauma patient is defined as any patient **age ≤ 15 years** who has a significant injury resulting from a traumatic mechanism.
- A major pediatric trauma patient is any patient age ≤ 15 years who has a significant injury to 2 or more systems, or the high potential for injury to 2 or more systems due to mechanism of injury.
- Major pediatric trauma patients may include those with evidence of physiological and compromise cannot be attributed to only one organ system.
- Major pediatric trauma patients require pediatric trauma activations – either Trauma Alert or Trauma Alert Red.
- Not all trauma patients will require trauma team activation, but they are still considered pediatric trauma patients. The PDS Service should be consulted and will be responsible for the care of these patients.
- All pediatric trauma patients with injury to more than one system, or who are otherwise included under the definition of major trauma, will be admitted to the PDS Service.
- Admission to services other than Pediatric Surgery does not preclude (prevent) close consultation with the PDS service for trauma patients.
- Significant alterations in the condition of trauma patients admitted to other services will require evaluation by the PDS Service and any other appropriate consultative service.
- After 24 hours, the PDS Service may transfer a patient with a single system injury to another service.
- The PDS Service will be available as a consultative service after transfer of a trauma patient to another service.

Trauma Activations

- Pediatric Trauma Activation criteria are developed by the Pediatric Trauma Service in conjunction with the American College of Surgeons Committee on Trauma requirements for a Level I Pediatric Trauma Center.
- The activation system is designed to provide rapid and efficient mobilization of personnel and resources essential for the resuscitation, evaluation, diagnosis and treatment of the multiply injured pediatric trauma patient.
- There are 2 levels of activation:
 - **Lowest level** of activation is a **Pediatric Trauma Alert (TA)**.
 - **Highest level** of activation is a **Pediatric Trauma Alert Red (TAR)**.
- Pediatric Trauma Activation criteria and personnel requirements **ARE NOT** the same as Adult Trauma criteria.

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SECTION II: OPERATIONS

Pediatric Trauma Process Guidelines (continued)

- TARs (highest activation) may have a pre-hospital report indicating the need for rapid OR access for patients sustaining life threatening injuries.
- An operating room will be held for 30 minutes once the pediatric Trauma Alert Red has been activated.
- Key OR personnel have trauma pagers and will be notified of TAR activations.
- The PDS Chief Resident is responsible for releasing the OR suite as soon as possible after patient arrival.
- Decision of activation level for pediatric trauma patients is based on patient condition and established criteria for each level.
- Trauma activations arrive by ground or air; transported by Emergency Medical Transport or by private vehicle.
- Trauma activations may already be present in the ED and thus activated as condition deteriorates.
- Trauma lab panels are established for the pediatric trauma patients based on age and level of activation; they are automatically ordered by use of the appropriate computerized order set.
- **ALL** pediatric activations (**TA & TAR**) **age ≥ 12 years** are to have **Blood Alcohol** and **Urine Drug Screens** ordered and performed.
- All **female** activations (**TA & TAR**) of **child bearing age** are to have a **urine pregnancy test** ordered and performed.

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SECTION II: OPERATIONS

Pediatric Trauma Activation Criteria

Trauma Alert Criteria (TA)

A Trauma Alert will be issued if a patient exhibits any of the following criteria:

- Pediatric trauma score < 13
- Respiratory rate < 10 or > 30
- Glasgow Coma Score (EMV) < 10
- Any intubated trauma patient
- Penetrating trauma to head
- > 15% BSA with 2nd or 3rd degree burn
- Age ≤ 15 years with significant mechanism of injury
- Falls > 15 feet
- Rollover MVC
- MVC with ejection of the patient
- Extrication time > 20 minutes
- Motor Vehicle Crash at speeds > 40 MPH
- Motorcycle or ATV crash speed > 20 MPH with separation of rider
- Same vehicle occupant fatality
- Pedestrian struck by a motor vehicle at > 20 mph
- Blast injury
- Two or more proximal extremity fractures and/or open fractures
- Amputation above the wrist/ankle
- Multiple system trauma requiring immediate evaluation/intervention by more than one surgical specialty
- Emergency Medicine Attending Discretion
- Pediatric Surgery Attending Discretion

Trauma Alert Red Criteria (TAR)

A Pediatric Trauma Alert Red will be issued if a patient exhibits any of the following criteria:

- Gunshot or penetrating wound to the neck, chest, back, or abdomen or extremities proximal to the elbow/knee
- Confirmed age specific hypotension at any time
- Any patient who remains hemodynamically unstable after 2 boluses of 20 ml/kg of isotonic crystalloid
- Any pediatric trauma patient receiving/having received blood transfusion to maintain vital signs
- Any pediatric patient with pulseless extremity or tourniquet in place
- GCS < 9 with mechanism attributed to trauma
- Intubated patients transported directly from the scene
- Patient with respiratory compromise or obstruction
 - Includes intubated patients transferred from referring hospital with ongoing respiratory compromise
 - Does not include intubated patients from referring hospital who are stable from a respiratory standpoint
- Smoke inhalation, electrical burns, and all intubated burn patients
- Trauma Codes
- Emergency Medicine Attending Discretion

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University of Kentucky HealthCare Pediatric Level I Trauma Center Pediatric Trauma Activation Criteria

Age ≤ 15 years

Trauma Red Criteria

One or more of the following:

- Gunshot wounds or penetrating trauma to the neck, chest or abdomen or extremities proximal to the elbow/knee.
- Confirmed age specific hypotension at any time.
- Any pediatric trauma who remains HDUS after 2 Boluses of 20 ml/kg Isotonic Crystalloid*
- Any pediatric patient receiving/received blood transfusion.
- Any pediatric patient with pulseless extremity or tourniquet in place
- GCS <9 with mechanism attributed to trauma.
- Intubated patients transferred directly from scene.
- Patients with respiratory compromise or obstruction (Includes intubated patients who are transferred from another facility with ongoing respiratory compromise; Not intubated patients from referring facility who are stable from a respiratory standpoint).
- Smoke Inhalation, electrical burns, and all intubated burn patients
- Trauma codes
- Emergency Medicine Attending discretion.

*HDUS (Hemodynamically Unstable)

Age	SBP	Heart Rate
0-1 month	<70	<80 or >180
1 mo – 1 yr	<80	<80 or >150
1-5 yrs	< 80 + 2Xage in yrs	<60 or >130
5-12	< 80 + 2Xage in yrs	<60 or >110
12-15	<90	<60 or >100

Response/Resources activated:

These Resources are only team members required in patient room during initial assessment upon patient arrival. Check in with Recorder on arrival. EMS (1) may be in room waiting to give report.

- Pediatric Surgery Attending **IN PT Room**
- Pediatric Surgery &/or Trauma Surgery Chief Resident **IN PT RM**
- Emergency Medicine Resident **IN PT RM**
- ED Nurses (Primary, Secondary, Recorder) **IN PT ROOM**
- ED Technician/ED Paramedic: (1), may need to assist in room if resources limited, **IN PT ROOM**
- Respiratory Therapist (1) at **HOB w EM Resident**
- Anesthesiologist Attending (1) **HOB w EM Resident & RT**

These individuals respond and wait outside trauma bay to be available as needed. Check in with Recorder on arrival. Should not be in room (with limited space) during initial assessment.

- Supply Tech/ED Tech: **Outside PT. Rm w Trauma Carts**
- Pharmacy (**Available as needed outside patient room**)
- Radiology Technologist: **Outside PT. Room w Portable Machine**
- Ultrasound Technologist: **Outside PT. Room w US Machine**
- CT Scan Technologist: **Await notification in CT**
- Blood Bank confirms availability of uncross-matched blood in ED refrigerator
- Chaplain: **Outside PT ROOM**
- Child Life: **Outside PT ROOM**
- Social Worker: **Outside PT ROOM**
- PICU RN (or MD): **Outside Pt Room**; will respond to assist in providing rapid access admission to PICU on Trauma Alert Red Patients only). PICU Attending assumes responsibility for vent pt. after secondary survey
- Operating Room Nurse calls regarding room availability. Does not respond to ED.

Trauma Alert Criteria

One or more of the following:

- Trauma Score <13.
- Respiratory Rate <10 or >30.
- Glasgow Coma Score <10.
- Any intubated trauma patient.
- Penetrating trauma to the head.
- High Suspicion of Spinal Cord Injury
- >15% BSA with 2ND or 3RD degree burns
- Emergency Medicine Attending discretion.
- Pediatric Surgery Attending discretion.
- Age ≤15 with significant mechanism of injury.
- Falls > 15 feet.
- Rollover MVC.
- Ejection of patient.
- Extrication > 20 minutes.
- Motor vehicle crash speed changes >40 mph.
- Motorcycle crash speed > 20 mph & separation of rider.
- Same vehicle occupant fatality.
- Pedestrian struck by a motor vehicle.
- Blast Injury
- Two or more proximal extremity fractures and/or open fractures.
- Amputation above ankle or wrist.
- Multiple system traumas involving more than one surgical specialty.

Response/Resources Activated:

These Resources are only team members required in patient room during initial assessment upon patient arrival. Check in with Recorder on arrival. EMS (1) may be in room waiting to give report.

- Emergency Medicine Attending: **IN PT Room**
- Pediatric Surgery &/or Trauma Surgery Chief Resident: **IN PT RM**
- Pediatric Surgery (TS) Attending –**notified/IN PT ROOM**
- Emergency Medicine Resident: **IN PT ROOM**
- ED Nurses: (Primary, Secondary, & Recorder) **IN PT ROOM**
- ED Technician/ED Paramedic: (1), may need to assist in room if resources limited, **IN PT ROOM**
- Respiratory Therapist: (1) at **HOB w EM Resident/Attending**

These individuals respond and wait outside trauma bay to be available as needed. Check in with Recorder on arrival. Should not be in room (with limited space) during initial assessment.

- Supply Tech/ED Tech: **Outside PT. Room w Trauma Carts**
- Pharmacy (**Available as needed outside patient room**)
- Radiology Technologist: **Outside PT. Room w Portable Machine**
- Ultrasound Technologist: **Outside PT. Room w US Machine**
- CT Scan Technologist: **Await notification in CT**
- Blood Bank confirms availability of uncross-matched blood in ED refrigerator
- Chaplain: **Outside PT ROOM**
- Child Life: **Outside PT ROOM**
- Social Worker: **Outside PT ROOM**

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SECTION II: OPERATIONS

Pediatric Trauma Activation Criteria

Authorized personnel will initiate a Pediatric Trauma Alert for any of the following scenarios:

- A patient who has exhibited one or more activation criteria and is expected to arrive at hospital within 15 minutes.
- A patient who exhibits one or more activation criteria and arrives at hospital without previous notification.
- A patient whose condition deteriorates acutely while in the Emergency Department.

Who Can Activate a Pediatric Trauma Alert?

- When activation criteria are met during transport or upon arrival, the Pediatric Trauma Service authorizes the following people to initiate a Pediatric Trauma Activation:
 - Emergency Medicine Attending physician
 - Pediatric Surgery Attending physician
 - Adult Trauma Attending physician
 - Pre-hospital ambulance and air medical personnel
 - Emergency Department Charge Nurse or ED RN
 - Surgical or Emergency Medicine Resident
- Activations based on physician discretion regardless of mandatory criteria met may be initiated by:
 - Emergency Medicine Attending physician
 - Pediatric Surgery Attending physician
 - Adult Trauma Attending physician

Who Can Cancel or Downgrade a Trauma Activation?

- Only the Attending PDS surgeon, Adult Trauma surgeon, or Emergency Medicine Physician can deactivate a Pediatric Trauma alert.
- Deactivations are defined as TA's cancelled or TAR's downgraded to TA or cancelled.
- The recording nurse will document the deactivation on the nursing trauma flow sheet including the time of decision.
- The ED Patient Clerical Assistant (PCA) shall be notified of the deactivation, who will cancel the orders in the medical record and place the appropriate information in the ED Trauma Log book to prevent unwarranted patient billing.
- When a Trauma Activation is cancelled, the Pediatric Surgery Resident who responded to the activation should write a brief note in the medical record.

**** Think about amount of time spent and utilization of resources in conjunction with the patient status before making decision to cancel a Trauma Alert.**

Who Can Dismiss Trauma Staff/Personnel?

- Only the Attending PDS surgeon, Adult Trauma surgeon, or Emergency Medicine physician can dismiss trauma personnel from the Emergency Department after a deactivation, as dictated by needed resources.

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SECTION II: OPERATIONS

Pediatric Trauma Activation Criteria (continued)

Activation Process

- Once a trauma has been activated, the ED Charge Nurse or ED Pediatric Team Leader will notify the PCA of the appropriate activation level, estimated time of arrival (ETA), patient age (if known) and brief patient descriptors for mechanism of injury (MOI).
- The PCA will notify the Blood Bank by phone (TAR and TA).
- The PCA will document all activations in the Trauma Log and denote TA or TAR, as directed by the Charge Nurse or Pediatric Team Leader.
- The PCA will activate the trauma pager system with the appropriate information to notify the trauma team members.
- The in-house PDS Chief Resident and Junior Residents are expected to respond within 15 minutes from time of page.
- If PDS Chief Resident is at home (after hours), the response time is 15 minutes from time of page.
- The Adult Trauma Service Chief Resident (PGY-4 or PGY-5) will respond to pediatric trauma pages and provide direction of the resuscitation until the PDS Chief Resident arrives.
- Care of the pediatric patient and direction of the resuscitation will be assumed by the PDS Chief Resident upon his/her arrival.

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SECTION II: OPERATIONS

Pediatric Trauma Center Bypass

Purpose

- To provide a process for trauma patient flow when, in the extremes of circumstances, should it occur, this facility and or Pediatric Surgical Resources are unable to care for Pediatric Trauma patients. Kentucky Children's Hospital is one of only 2 Level I Pediatric Trauma Centers in the state of Kentucky.
- The American College of Surgeons on Trauma criteria (CD3-5 Resources for Optimal Care) for Level I Trauma Centers states "when diversion becomes necessary, there must be a specific process and procedure in place to alert prehospital personnel and surrounding trauma centers". The Trauma Medical Director must be involved in the decision process for instituting trauma bypass.

UK HealthCare Diversion

- The enterprise does have formalized diversion policies in place and can be accessed here:
https://ukhealthcare.mc.uky.edu/policies/departmental/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/departmental/Emergency%20Department/ED01-08%20ED%20Diversion%20Guidelines.docx&action=default
- https://ukhealthcare.mc.uky.edu/policies/enterprise/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/enterprise/Enterprise/A02-115%20Emergency%20Department%20Diversion.docx&action=default
- Pediatric Trauma or Pediatric patients are NOT diverted in accordance with this institutions policy.
- In the event of extenuating catastrophic circumstances, for which pediatric trauma would be deferred, the Trauma Medical Director (TMD) will be involved in the decision making process.
- The TMD will communicate with enterprise administration and the emergency department prior to making said decision.
- As this would only happen in the extreme of circumstances, the state's Kentucky Hospital Association has mutual aid agreements/MOUs with all its members that allow hospital facilitates to cover for one another in extreme, hazardous/disaster like situations. This allows for the flow of patients and on occasion staff, to other trauma centers within the state of Kentucky providing emergency approvals, acceptances, and credentialing.
- Refer to Care Web for Enterprise Policies and Procedures, and Emergency Preparedness plans:
<https://spwww.ukhc.org/proceduresandsafetyplans/UK%20HealthCare%20Plans/Forms/AllItems.aspx>

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SECTION II: OPERATIONS

University of Kentucky Healthcare Trauma Bypass Protocol

As the University of Kentucky Healthcare provides the only regional Level 1 Adult and Pediatric Trauma Center, we accept for evaluation trauma patients from the scene as well as transfers from other hospitals needing a higher level of care. Our trauma center does not provide limits on the inpatient trauma census thereby making any diversion for high volume unnecessary. Possible reasons for bypass:

- Destruction of the physical hospital plant including the emergency department and operating room rendering it inoperable
- Barrier to the physical hospital plant due to external weather damage, law enforcement activities, or other.
- Internal disaster.

Hospital administration together with the Emergency Department and/or Trauma Surgery teams will recognize any need for diversion and institute the protocol.

- Surrounding EMS and dispatch companies will be notified of bypass conditions.
- Surrounding local hospitals with Memorandum of Understanding on record will be notified of bypass conditions and discern ability to accept diverted patients.
- Regional Level 1 Trauma Centers such as University of Louisville and University of Cincinnati will be notified of bypass conditions and evaluate for secondary distribution if needed.

When bypass conditions resolve as determined by the Hospital administration together with the Emergency Department director and/or Trauma Surgeon, we will notify all the above entities and direct secondary distribution back to our facility if needed.

An After Action Report will be completed and evaluated by the Emergency Preparedness Team and Hospital Administration for any improvement in process for future events.

Draft 8/17/17, ct

Edited 8/29/17, emergency preparedness team

Edited 9/22/17, ct

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SECTION II: OPERATIONS

Neurotrauma Bypass/Diversion

Neurotrauma is not diverted. The on-call Faculty Neurosurgeon is dedicated to UK HealthCare for the emergency evaluation and management of neurosurgical trauma. If the on-call Neurosurgical Faculty becomes encumbered and additional neurosurgical resources are required the back-up Neurosurgeons will be notified. In the event that all neurosurgery faculty are encumbered, the Trauma Surgery Faculty are able to provide initial evaluation and stabilization of the neurotrauma patient. In the event back-up resources have been exhausted, the hospital trauma bypass/diversion protocol will be implemented.

All cases requiring transfer, bypass or diversion as a result of this policy will be reviewed by the Trauma PI process.

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SECTION III: TRAUMA TEAM ROLES AND RESPONSIBILITIES

Overview of Responding Services

Pediatric Trauma Team Physicians

- Physicians assigned to the Trauma Team are notified of an activation and will report to the Emergency Department within 15 minutes of notification.
- Assume roles outlined in Trauma Individual Roles and Responsibilities.
- The PDS Chief Resident/Trauma Chief Resident/Senior Emergency Medicine Physician will coordinate response activity.

Emergency Department Nursing

- The Charge Nurse, Nurse Team Leader, or Primary Nurse will designate Trauma Nurse #1 and Trauma Nurse #2.
- Notify any other nursing personnel of patient's pending arrival.
- Designate a Trauma Nursing Care Technician and or Emergency Department Paramedic.
- Assign other roles as indicated.
- Prepare pediatric trauma resuscitation room for patient: Broselow on stretcher
- Document names, service, and arrival times of responding trauma team personnel

Radiology Personnel

Radiology Technologist and Ultrasound Technologist will respond within 5 minutes to assure:

- Delivery of portable radiology machine to the trauma bay.
- Delivery of portable ultrasound machine to the trauma bay.
- Perform and process radiographs of trauma patients as quickly as possible.
- Perform and process abdominal ultrasound (FAST exam).
- Ensure CT scan availability, provide oral contrast if requested.
- Notify radiologist that trauma patient's radiographs are ready for evaluation.

Blood Bank Personnel

- Ensure that the ED blood refrigerator is stocked with O-negative and O-positive PRBCs for use with unstable trauma patients.
- Prepare requested number of units of type-specific blood with 15 minutes after receipt of patient blood sample and notify ED when blood is ready.
- Prepare requested number of units of cross-matched blood within 45 minutes after receipt of patient blood sample, and notify ED when blood is ready.
- Implement the Massive Transfusion Guideline upon request of the Pediatric Trauma Team Leader.

Respiratory Personnel

- Report to the Emergency Department within 15 minutes of notification.
- Obtain, set up, and manage ventilator.
- Assist physicians with airway management and ventilator support.

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Laboratory Personnel

- Expedite analysis of all trauma labs.
- **Call** Emergency Department to report **ALL** panic level lab results. All other results are reported electronically.

Operating Room Personnel

- Determine Operating Room availability.
- Evaluate potential need for Operating Room Scrub and Circulating Teams.
- Evaluate potential need for Anesthesiology Team.

Social Workers/Chaplain Personnel

- Respond to area within 15 minutes of notification.
- Maintain availability outside of patient room.
- Assist with family waiting outside patient room and or lobby.
- Ensure completion of assessment on all pediatric activations either during ED visit or via communication to inpatient SW//CM3

Pharmacy Personnel

- Respond to Emergency Department upon notification.
- Maintain availability outside of patient room.

PICU Personnel

- Respond to Trauma Alert Reds (TAR) only.
- Nurse Team Leader/Charge and/or Attending will respond and wait outside room to assure expedited rapid access of the patient to the PICU.

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SECTION III: TRAUMA TEAM ROLES AND RESPONSIBILITIES

Trauma Resuscitation Staff Roles & Responsibilities

Pediatric Surgery Attending, Trauma Attending, or Trauma Fellow

- Checks in with Scribe upon arrival.
- Oversees trauma resuscitation and provides guidance as needed.
- Performs and/or assist with procedures.
- Stands at foot of bed.
- The PDS Attending surgeon is required to be present for all TARs and receive notification for TAs.
- Pediatric Trauma Alert Reds will be managed by the Pediatric Surgery Service and/or the Adult Trauma Service with the assistance of the Emergency Medicine Service.

Trauma Team Leader

- **Surgery Chief Resident (Peds/Trauma), EM Attending, or EM Resident**
- Stands at patient's left-hand side.
- Checks in with Scribe upon arrival.
- In charge of the pediatric trauma resuscitation under the guidance of the Pediatric Surgery, Trauma, or EM Attending physician.
- Speaks clearly and loud enough for all in the room to hear.
- Responsible for overall patient evaluation and management.
- Ensures completion of primary and secondary surveys.
- Communicates physical findings and assessment to Scribe.
- Determines priority of procedures and necessary tests.
- Determines need for operative intervention and contacts the OR desk.
- Determines need for appropriate consultations.
- Makes physician assignments.
- Dismisses ancillary personnel when services are no longer needed.
- Cancels trauma activations when appropriate.
- Ensures control of room and traffic.

Emergency Medicine MD or MD #2

- **EM Resident, EM Attending, Anesthesiology Attending, or Surgical Resident**
- Stands at head of bed.
- Checks in with Scribe upon arrival.
- Responsible for airway management and assuring cervical spine precautions.
- Coordinates initial assessment of airway and breathing.
- Communicates findings clearly to Team Leader and Scribe.
- Conducts secondary assessment of HEENT and clearly communicates findings.

Primary MD or MD #3

- **Second most senior Resident – preferably PGY-2 or above**
- Stands at patient's right-hand side, shoulder level.
- Checks in with Scribe upon arrival.
- Assists Trauma Team Leader in patient evaluation and management.

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- Performs primary and secondary examination.
- Speaks clearly and loud enough for all in the room to hear.
- Performs necessary procedures under direction of Trauma Team Leader.
- Coordinates orders.

MD #4

- **Third most senior Resident, preferably PGY-1 or above**
- Stands inside the trauma bay at the computer station.
- Checks in with Scribe upon arrival.
- Assist in patient evaluation and management under direction of MD #3.
- At direction of Team Leader, contacts Radiology for CT scan and special procedures.
- Enters orders and complete Trauma H&P under direction of MD #3.

Emergency Department Charge Nurse

- **ED Charge Nurse or ED Peds Team Leader**
- Notifies ED staff of impending arrival.
- Delegate's responsibility of equipment and supply maintenance in trauma bay.
- Disseminates pre-hospital patient report to appropriate MD's/RN's.
- Assist with traffic control.
- Acts as family liaison if needed, in absence of Social Work or Chaplain.
- Maintain position outside trauma bay unless serving in direct care capacity as part of team.

Primary Patient Care Nurse: Nurse #1

- **Pediatric ED RN or ED RN**
- Prepares Trauma Resuscitation area:
 - Supplies/equipment available
 - Broselow tape placed on stretcher
 - Trauma bay doors closed to warm environment
 - Trauma bay thermostat turned up
 - Monitoring equipment ready
- Stands at patient's right-hand side, mid stretcher.
- Assists with initial assessment.
- Monitors vital signs – announces updates every 5 minutes.
- Places patient on cardiac monitor.
- Establish PIV's or ensures function of pre-existing IV lines.
- Obtains blood specimens and sends to lab.
- Places NG tube and/or Foley catheter as indicated.
- Accompanies patient outside ED.
- Ensures initial notification of family and provides updates.
- Notifies Capacity Command Center with bed requirements ASAP.
- Ensures documentation from admission to disposition complete.
- Ensures report handoff of patient.
- May serve as the Circulating Nurse/Paramedic and assume the duties thereof.

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Circulating Nurse/EM Paramedic: Nurse #2

- **RN or Paramedic**
 - Stands at patient's left-hand side, shoulder level.
 - Checks in with Scribe upon arrival.
 - Assists with initial assessment.
 - Obtains first **manual** blood pressure.
 - Assist with medication administration.
 - Assists physicians with procedures.
 - Assists in removal of patient's clothing.
 - Inserts PIV's and draws labs.
 - Assists with nursing procedures.
 - Assist with administration of IV fluids and/or blood products (fluid warmer, as needed).
 - Paramedics will not administer IV fluids.

Emergency Department Nursing Care Technician

- **ED Nursing Care Tech or Paramedic**
 - Assists with trauma bay set up prior to patient arrival.
 - Checks in with Scribe upon arrival.
 - Stands at patient right-hand side, near feet.
 - Assists in removal of patient cloths as needed.
 - Obtains blood from Blood Bank if required.
 - Obtains IV fluids and blankets from warmer.
 - Processes lab specimens.
 - Obtains additional supplies as direct by nurse and/or in absence of Supply Technicians.
 - Ensures patient has an identification band.
 - Processes patient clothing/valuables.
 - Assists with CPR.
 - Assists with patient transport.

Scribe or Documentation Nurse

- **Registered Nurse**
 - Maintains documentation on nursing trauma flow sheet.
 - Stands in corner of trauma bay to have overview of room.
 - Documents all initial assessments and procedures of resuscitation.
 - May assist with procedures as needed.
 - Assists with crowd control.
 - Accompanies patient to CT or other procedures during initial resuscitation.
 - Assures report of patient given in handoff to admitting/receiving area.
 - May serve as Nurse #2.

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Supply Technician

- **ED Supply Technician or ED Nursing Care Technician**
- Positions supply cart and Broselow cart just outside trauma bay doors.
- Checks in with Scribe upon arrival.
- Establishes position outside trauma bay with supply carts.
- Retrieves requested supplies and/or equipment during resuscitation.
- Assures supply carts are adequately stocked before and after patient care.

Respiratory Therapist

- **Pediatric Respiratory Therapist**
- Checks in with Scribe upon arrival.
- Stands at patient's left-hand side, lateral to head of bed – aside or behind MD #2.
- Assists with airway management.
- Ventilates patient.
- Prepares, monitors, and documents all required aspects of mechanical ventilation.
- Connects ventilator circuit to end tidal CO₂ detector. Assures documentation of initial in-line CO₂ reading per Scribe.
- Assist during patient transport.

Radiology Technologist

- Responds to area with portable x-ray machine.
- Checks in with Scribe upon arrival.
- Maintains position outside trauma bay until requested by Team Leader.
- Responsible for obtaining radiography studies in timely manner as ordered.

Ultrasound Technologist

- Responds to area with portable ultrasound machine.
- Checks in with Scribe upon arrival.
- Maintains position outside trauma bay until requested by Team Leader.
- Performs Focus Abdominal Sonogram for Trauma (FAST) in trauma bay as ordered.
- Removes ultrasound machine from trauma bay after FAST completed.

Pharmacist

- **ED Pharm D**
- Checks in with Scribe upon arrival.
- Assumes unobtrusive position outside the trauma bay.
- Retrieves, calculates dosage, and administers medication as ordered by Team Leader.

Social Worker

- **ED Social Worker** responds to trauma activations during business hours.
- Checks in with Scribe upon arrival.
- Assumes unobtrusive position outside the trauma bay.
- Assists with patient/family with social needs.

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- Evaluates all cases for history and need to report to DCBS and Pediatric Forensic Medicine Service.
- Assists in providing families with safety instructions as needed.
- Assists in providing patient/family with available resources.
- Follows up on all after hour consults on admitted trauma patients.
- Communicates identified needs to KCH SW/CM appropriately on admitted patients

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Chaplain

- **ED Chaplain** responds to all trauma activations.
- Checks in with Scribe upon arrival.
- Assumes unobtrusive position outside the trauma bay.
- Assist with patient and or family needs, emotional and or social.

Pediatric Trauma Medical Director (TMD)

- TMD has oversight of the Pediatric Trauma Program and the care of pediatric trauma patients.
- May observe trauma activations to assure patient quality and safety.
- Will conduct crowd control and direction in accordance with team roles & responsibilities.
- Assumes unobtrusive position outside of the trauma bay.
- Assist in patient care if requested by Team Leader.
- Will provide direction and communication as observation of resuscitation indicates.

Pediatric Trauma Program Manager (TPM)

- TPM has oversight of day to day operations and quality improvement of the Pediatric Trauma Program in collaboration with the TMD.
- May observe trauma activations to assure patient quality and safety.
- Will conduct crowd control and direction in accordance with team roles & responsibilities.
- Assumes unobtrusive position outside the trauma bay.
- Checks in with Nursing Team to assure assistance not needed.
- May assume role of Scribe if needed.

Child Life Specialist

- ED Child Life responds to trauma activations.
- Assist with patient emotional and or social.
- Checks in with Scribe upon arrival.
- Assumes unobtrusive position outside the trauma bay.

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SECTION III: TRAUMA TEAM ROLES AND RESPONSIBILITIES

Additional Responders for Pediatric Trauma Alert Red

Anesthesiology Attending Physician

- Responds to **Pediatric TAR** activations only.
- Checks in with Scribe upon arrival.
- Assists or assumes role of MD #2 at head of bed to maintain airway and breathing. (See role of MD #2)
- Alerts OR of needed operating room and facilitates availability.

PICU Charge or Team Leader RN

- Responds to **Pediatric TAR** activations only.
- The PICU Staff maintain a pediatric trauma pager to alert them of any pediatric TARs.
- Checks in with Scribe upon arrival.
- Assumes unobtrusive position outside the trauma bay.
- Maintains communication with PICU staff/MD on patient status and possibility of rapid access.
- Informs Scribe, Primary Nurse, and Team Leader of assigned PICU bed number.
- Accompanies team and patient to CT.
- Alerts PICU staff of patients pending arrival.
- Assumes patient report after CT if going straight to PICU.

PICU MD

- Critical Care Attending may respond to **Pediatric TAR** activations with PICU RN.
- Assumes unobtrusive position outside the trauma bay, unless otherwise instructed by Team Leader.
- May assume position alongside Pediatric Surgery Trauma Attending as needed.
- Checks in with Scribe upon arrival.
- Provides medical assistance and expertise as needed or requested by Team Leader.
- Assumes management of ventilator for intubated patients after secondary survey

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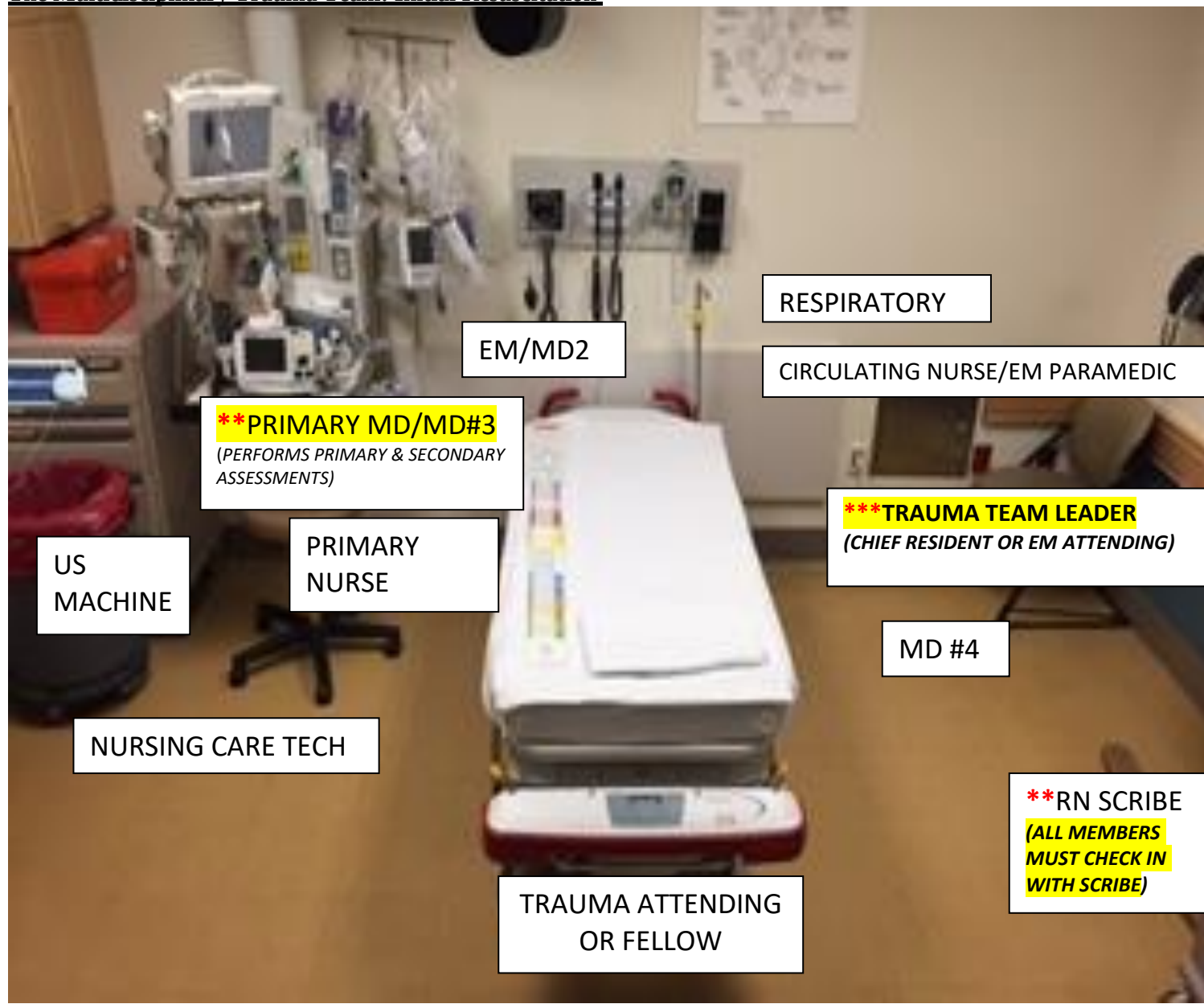
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SECTION III: TRAUMA TEAM ROLES AND RESPONSIBILITIES

Trauma Bay Diagram

The Multidisciplinary Trauma Team: Initial Resuscitation



Trauma Bay Doors Closed

Supply Tech

Supply & Broselow Carts

US Technician	Radiology Technician w portable
Chaplain	Pharmacy
Social Worker	Students/Residents
Anesthesiologist (TARs)	
PICU RN & PICU MD (TARs)	

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Pediatric Trauma Lab Requirements

- Pediatric trauma patients shall have trauma labs drawn upon arrival.
- Utilize PIV insertion as opportunity to obtain specimen for labs to prevent multiple sticks.
- The following charts are a reference for required lab specimens.
- Additional labs may be ordered at the physicians discretion based on patient condition.
- Patients age 12 or greater who is TAs or TARs are to have an ETOH and UDS (Quantitative) ordered.
- Urine Pregnancy test are required on female patients if are of child bearing age.
- The referenced charts below are posted in the trauma rooms.

Pediatric Trauma Venipuncture Reference

Trauma Alert Patients:

- Must draw at least 5 ml
- **Type and Screen** use pink topped vacutainer (adult sized) with minimum 3 ml for children > 4 months old. May send two (2) lavender topped microtainers (shown below) for infants less ≤ 4 months of age.



- **EPOC**
- **CBC** use lavender microtainer, minimum of 0.25ml required.



- **BMP** use green microtainer, minimum of 0.5ml required.



- If you are unable to get enough blood for all tests, send what you have in the order of priority listed above.
- The above are minimum amounts of blood required for the ordered labs. Only if additional blood is available (e.g., if patient has deep line or a-line in place), then use larger vacutainers (adult sized) for other ordered tests.

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Pediatric Trauma Venipuncture Reference

Trauma Alert Reds

- Must draw at least **7.5 ml** for ordered tests.
- **Type and Cross** use pink topped vacutainer (adult sized) with minimum **3 ml** for children > 4 months old. May send two (2) lavender topped microtainers (shown below) for infants less $\leq$ 4 months of age.



- **EPOC**



- **CBC** use lavender microtainer, minimum of **0.25ml** required.
- **PT/PTT/Fibrinogen** – use adult sized blue topped vacutainer, minimum **1.8 ml**.



- **Lactate** – use grey microtainer, minimum of **0.25 ml** required.



- **CMP** use green microtainer, minimum of **0.75ml** required.

- **Lipase**

- **Urinalysis**

- If you are unable to get enough blood for all tests, send what you have in the following priority:

1. Type and Cross
2. POC EPOC
3. CBC
4. PT/PTT/Fibrinogen
5. Lactate
6. CMP
7. Lipase
8. UA

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Guide for Pediatric Body CT in the Emergency Department

- **Image acquisition:** 0.625 mm slices
Kv: 100 (pt < 36 kg); 120 (pt > 36 kg)
mA: modulates by patient weight
Gantry rotation: 0.4 seconds
Noise: 12
- **Reconstruction:** First series: 5mm slices at 2.5 mm intervals (std algorithm)
Second series: 1.25 mm slices at 0.625 mm intervals (soft algorithm)
Third series: 5mm slices at 2.5 mm intervals (lung algorithm)
- **Contrast:** IV: volume per pt weight (2-3 cc/kg) power-injected over 1 minute.
- **Contrast timing:** Nontrauma:
Chest: 70 seconds
Abdomen/Pelvis: 85 seconds
C/A/P: 80 seconds
Trauma:
Chest: 55 seconds
Abdomen/Pelvis: 80 seconds
C/A/P: chest at 55 seconds and A/P 20-25 seconds later Delayed scanning (at discretion of radiologist): 70-120 seconds.
- **Oral Contrast:** None
- **Parameters:**
 1. All pediatric CTs from the ED will be performed with IV contrast unless:
 - a. Known contrast allergy
 - b. Creatinine >1.5
 - c. Question is urinary tract stone (CT urogram)
 - d. Question of retroperitoneal hemorrhage (particularly in hemophilia pts)
 2. Oral contrast is not routinely given to pediatric patients. Requests for oral contrast must have approval from pediatric radiologist or call resident, and may be considered in the question of pelvic cystic mass or abscess.
 3. Except in the case of a CT urogram, a pediatric CT ordered without contrast should be discussed with the radiology resident or attending on call as another imaging study may be more appropriate.
 4. All chest CTAs for PE should be discussed with the tier 2 resident or the peds radiology attending on call. Given the extremely high radiation dose required for this exam, the patient should have suggestive labs and risk factors for PE.

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Guide for Pediatric Body CT in the Emergency Department

- **Parameters (continued)**
 5. CTs on children less than 2 year old (except in the case of significant trauma) should be discussed with tier 2 resident or peds radiology attending. These exams tend to be poor without sedation, and another imaging study may be more appropriate.
- **Special considerations for trauma:**
 1. Leads should be moved to the extremities and support apparatus/lines should be moved above or below the chest, abdomen, and pelvis. These cause significant streak artifact, which can lead to equivocal interpretations as to vascular injury.
 2. Trauma scans should be checked by the tier 2 resident or peds radiology attending immediately after the scan is performed to determine if delayed images are needed. The pediatric surgery attending or SAR may choose to forego the delayed imaging if the patient is unstable, but the assessment for active bleeding may be incomplete. Technologist should document in memo pad section of ISUITE if patient is removed from CT table before requested delays are performed.
 3. CT technologists in the EDP have been asked to verify whether or not the chest portion of a body CT is wanted prior to performing the scan. Once the dose of IV contrast is administered, the patient cannot receive more contrast for 24 hour...so late request for the chest portion will be limited (and cannot assess vascular structures).

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Care of Patients Needing Immediate OR

- To provide criteria and procedures to manage the needs of patients who require access to the OR at times when the Operating Room is at a maximum capacity follow the guidelines in the following Emergency Department Policy ED08-75 link:

Care of Patients in Need of Immediate Surgery When the Operating Room is at Maximum Capacity

<https://ukhealthcare.mc.uky.edu/policies/departmental/Emergency%20Department/ED08-75%20Care%20of%20Patients%20in%20Need%20of%20Immediate%20Surgery%20When%20the%20Operating%20Room%20is%20at%20Maximum%20Capacity.docx?Web=1>

- The American College of Surgeons requires verified Level I Trauma Centers must an operating room that is adequately staffed and available within 15 minutes (Resources for Optimal Care, CD 11-14).
- Resources for Optimal Care further outlines Level I Trauma Center operating room availability :
 - In Level I if the first operating room is occupied and adequately staffed additional room must be available (CD 11-15)
 - Availability of the operating room personnel and timeliness of starting operations must be continuously evaluated by the trauma PIPS process and measures must be implemented to ensure optimal care (CD 11-16).
 - The Pediatric Trauma Coordinator monitors all timeliness measures of the operating room through the Pediatric Trauma Program Performance Improvement Process.
- The Chief Resident is to contact the OR within 30 minutes of patient arrival to inform of need or release of the held OR room. See *Roles and Responsibilities* section of this manual for further information. The following link provides guidelines for all perioperative surgical scheduling including life threatening events.

https://ukhealthcare.mc.uky.edu/policies/departmental/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/departmental/Perioperative%20Services/OR%2001-07%20Procedure%20for%20Surgical%20Scheduling%20and%20Unused%20Block%20Time2.docx&action=default

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Documentation of Pediatric Trauma Alert/Trauma Alert Red Patients

- Provision of guidelines for use of the paper Trauma Care Nursing Record will allow a standardized method for use of the assessment tool resulting in more effective utilization and accuracy of the tool and the data collected.
- Pediatric patients that do not meet Trauma Alert/Trauma Alert Red Criteria will have documentation completed on the SCM Computerized Chart
- Effective and timely management of the multiple trauma pediatric patient demands a rapid, organized, systematic approach to assessment, planning, implementation and evaluation.
- The paper trauma nursing record provides the framework for organization of the approach and provides a documentation tool to facilitate establishment of a permanent record of comprehensive baseline and monitoring assessment data as well as facilitating the documentation of the implementation and evaluation phases of the nursing process.
- The nursing record organizes specific data points to assist in determination of patient status and detection of potential or actual human response to injury.
- The trauma nursing record is included in the patient's permanent medical record.
- Use of the assessment tool is initiated by the nurse when it is determined that the patient meet Pediatric Trauma Alert or Trauma Alert Red criteria.
- The tool must be used with all critically injured pediatric patients who require ongoing monitoring and are classified as acuity Level I or II.
- After completion of the initial resuscitation, when the patient condition has stabilized, the documentation can be transitioned to the SCM computer chart.
- As assessments, interventions and/or evaluations are completed the corresponding time and information will be recorded in the appropriate area.
- Evaluative notes will be recorded under the Nursing Evaluation section of the nursing care record.
- The record must be signed and initialed by all nurses delivering care to the patient including the recording nurse. Assurance of adequate signatures to document care is the responsibility of the patient care nurse.
- A mechanism for quality assurance will be implemented to audit charts for completeness. Staff will receive feedback surrounding quality assurance audit filters regarding form completion and/or documentation issues.
- Use of Black or blue ball point pen is essential.

Completing the Nursing Trauma Flow Sheet

- Legibly print patient name in addressograph area.
- Record demographic information:
 - Classification
 - Time of arrival - in military time
 - Date
 - Age
 - Sex
 - Weight - in kilograms or pounds
 - Mode of arrival:
 - If Air Med, specify: service.
 - If Lexington Fire Department, specify unit number: EC-1, EC-5, etc.
 - If ambulance, specify: Scott County, Jessamine County, etc.
 - Recording nurse: Specify by name
 - Patient care nurse: Specify by name
 - Circulating nurse names in appropriate areas.
 - Physician: Specify by name

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Completing the Nursing Trauma Flow Sheet (continued)

- Chief complaint/pre-hospital report - include as many details as are available concerning:
 - Mechanism of injury: driver, passenger, restrained, unrestrained, ejected, roll-over, speed traveling, air bag, carseat or booster seat use, etc.
 - Description of injury, object (if available) for penetrating trauma - caliber of gun, length or type of knife, removal of impaled objects, etc. Height of fall and surface patient landed on are important.
 - Pre-hospital treatment: IV, intubation, immobilization, meds given and fluids infused (type and amount).
 - Chief complaints: particularly important to help identify injuries and assess initial level of consciousness.
 - Injury time: Actual time of MVC, fall, GSW, etc. This information is available from patient, pre-hospital personnel, referring hospital, paperwork, etc.
 - Scene flight: Check box if patient arriving via helicopter directly from injury scene.
 - Past medical history - check the appropriate box and include any other conditions you identify.
- Record information in boxes of trauma team response section.
 - If trauma activation is called, check the appropriate box for *Pediatric* and box for *Trauma Alert* or *Trauma Alert Red*.
 - For each team responder (e.g., Pediatric Surgery attending, Pediatric Surgery or trauma chief, Peds surgery or trauma resident, emergency medicine resident, anesthesia, neurosurgeon, U/S tech, RT, X-ray, etc.) complete time called and time arrived section under each responder.
 - Complete name section under each responder by writing in responder's first initial and last name.
- Initial Assessment
 - Complete initial assessment section by documenting the primary and secondary survey findings.
 - Document time assessment of system occurs in space provided.
 - All sections should be completed in full.
 - If you are unable to assess a particular area or it is not applicable document accordingly.
 - All pediatric trauma patients must have a completed Glasgow coma scores, pupil evaluation & trauma score on arrival and upon discharge from the ED.
- After completion of the initial assessment, shade injured areas on diagram using legend at bottom of page.
- Document lab draws with time, site, prep and person obtaining labs.
- Vital Signs
 - Vital signs should have a PTA set documented.
 - Then the frequency of the vital signs should correspond to the acuity of the patient.
 - Vital signs will include Temp, BP, HR, RR, SpO2, GCS, and pupils.
 - Critical Care monitoring should be documented every hour at minimum e.g. Codman, ICP, CPP, CVP, PCWP, etc.
- Intake and Output should be documented at regular intervals or as ordered by MD. This should include chest tube, NG tube, and visible external hemorrhage
- Drips will be documented at regular interval for dosage and rate or when titrated.
- Ventilator Settings will be completed initially and when changes occur.
- etCO2, should be continuously monitored in-line and documented every hour at minimum for intubated pediatric trauma patients.

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SECTION IV: PEDIATRIC TRAUMA PATIENT PROCESS MANAGEMENT GUIDELINES

Completing the Nursing Trauma Flow Sheet (continued)

- Intervention/Procedures - record time of procedures, size of tubes & lines placed along with sites and removal of backboard and/or changing of C-Collar in appropriate spaces. Other procedure data can be entered in the nursing evaluation section.
- Resuscitation Fluids
 - Record fluids patient received prior to arrival in shaded area.
 - Record all fluids patient receives by using
 - Time
 - Fluid Type
 - Crystalloids - liter number (#)
 - IV site
 - "Y" for fluid warmer used
 - Colloids – document Type & unit number (#) & donor number (#) for blood products
 - Totals must be documented
- Rhythm strip must be posted and interpreted for all pediatric patients with chest pain or a cardiac event.
- Document all medication administration by entering time administered, name of med, dose, route, your initials and ordering MD.
- Documentation of evaluation of patient response to interventions should be documented under Nursing Evaluation.
- Nurses, EMT- Ps, and NCTs should include their initials after documenting and signature should be legible. If additional space is needed use nursing continuation record.
- Patient transports to diagnostics should be documented in this space.

Discharge Information

- Document D/C vital signs, admitting service, total I&Os and disposition of valuables.
- Transfer form should be completed if patient is transferred to another facility.

Notifications

- All notifications should be documented by checking the box and inserting time notified in space provided.

Signatures

- All nursing staff involved in care of the patient should sign and initial the nursing care record. All signatures should be legible.

Disposition

- Document time report is called and last name of RN receiving report.
- *Time of actual patient movement from the department should be noted as well as destination of patient. If "floor" is checked, enter room number.*

Place patient stickers in upper right corner of patient chart on white and yellow sheets. The white copy is to accompany the patient to the in-patient area and yellow stays in ED.

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SECTION IV: Pediatric Trauma Patient Process Management Guidelines

Transport of Pediatric Trauma Patients by UK Peds Transport Crew

Information

- The pediatric surgeon on call or the Pediatric EM Attending physician will be responsible for activating the pediatric ground transport team to ensure timely inter-facility transport of trauma patients 0-5 years of age when other methods of transport are unavailable. The attending pediatric surgeon or Peds EM Attending physician will assume medical control for the duration of the transport.

Procedure

- Once a pediatric trauma patient has been accepted from another facility and it has been determined that other methods of transport are unavailable, the UK attending pediatric surgeon on call may contact the UK Emergency Communications office (323-6215) to request the pediatric transport team to transport the patient.
- The UK Emergency Communications office will advise the physician if the transport team is available and will take patient information to relay to the transport nurses.
- If the team is available the emergency communications tech will tone a ground transport and relay report information to the transport nurses.
- The attending pediatric surgeon who activated the pediatric team will maintain medical control of the transport and will be available to the transport nurses through UKMDs.
- Upon arrival at UK the patient will be admitted to the pediatric emergency department.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Pediatric Endotracheal Intubation

- Endotracheal intubation is a means of providing an airway in a patient with an inadequate or compromised airway.
- Endotracheal intubation provides an airway only and patients may require assisted ventilation.
- Insure that adequate ventilation and oxygenation are in progress.
- Open airway by using the head tilt, chin lift method. If spinal injury is suspected use the jaw thrust method to open the airway.
- A towel may be placed beneath an infant's shoulders or older child's head to facilitate positioning.
- Maintain open airway by using an oropharyngeal or nasopharyngeal airway.
- Consider RSI (see guideline)
- Endotracheal tube size may be chosen according to the formula: $16 + \text{age in years} \div 4$ or by the size of the child's little finger or nares; or reference to the table below.
- Have suction equipment available.
- Hold the laryngoscope in the left hand.
- A Miller (straight) blade is preferred for patients <2 years of age.
- A Miller or Macintosh (Curved) blade may be used for patients > 2 years
- Insert the laryngoscope in the right side of the mouth moving the tongue to the left.
- Observe the epiglottis, then the vocal cords.
- Cricoid pressure may make visualizing of the vocal cords easier
- Insert the endotracheal tube carefully watching the ETT pass through the vocal cords. Inflate cuff if applicable. (Use of a stylet may increase the chance of success)
- Check the placement of the endotracheal tube by observing chest expansion and auscultating the chest and gastric area with a stethoscope.
- Use the E-Z cap CO2 detector as an additional verification of placement.
- Secure the endotracheal tube with tape or commercial device.
- Insert oral or nasal gastric tube to decompress stomach.
- Obtain a chest film to ascertain exact placement of the ET tube.
- Document number of intubation attempts.
- Place patient on monitor for continuous in-line monitoring of etCO2.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Formula for ETT Size and Depth

Age	Weight	ET tube	Blade
< 28 weeks	<1000 g	2 to 2.5	0 Miller
28-34 weeks	1000-2000 g	2.5 - 3	0 Miller
34-38 weeks	2000 - 3000 g	3 - 3.5	0 Miller
term	> 3000 g	3.5 - 4	1 Miller
3 mo	5.5 kg	4	1 Miller
6 mo	7	4	1 Miller
1 yr	10	4.5	1 Miller
2 yr	12	4.5	1.5 Miller
3	14	5	2 Miller/Mac
4	16	5.5	2 Miller/Mac
5	18	5.5	2 Miller/Mac
6-7	20-22	6	2 Miller/Mac
8-10	25-30	6.5	2 Miller/Mac
Adult Female	60	7	
Adult Male	70	8	

Formulas for size and depth of ETT

- Estimation ETT size:
$$\frac{16 + \text{age in years}}{4}$$
- Use uncuffed ETT for child under 6-8 years of age.
- Subtract 0.5 from size of tube when using cuffed ETT
- ETT Depth: *(at gum line) should be approximately = 3 X ETT size*

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

RSI Intubation: (Julia E Martin M.D., Assistant Professor, Department of Emergency Medicine)

General information:

- Airway control is always the most important objective in the initial resuscitation and stabilization. It takes the highest priority in primary assessment. The trauma team must be prepared for any airway emergency.
- RSI involves the use of neuromuscular blocking agents and sedatives to facilitate endotracheal intubation.
- Rapid Sequence induction technique is used to prevent regurgitation and aspiration of gastric contents.
- Requires preoxygenation and denitrogenation by using 100% oxygen via non-rebreather face mask to prevent apnea related hypoxia during the procedure.
- Once paralytic is on board, mask ventilation is not attempted at this point.
- During induction, a skilled assistant provides manual in-line axial stabilization of the head while a second assistant presses the cricoid cartilage to prevent gastric aspiration.
- Cricoid pressure is maintained until the cuff on the ET tube is inflated and tube placement is confirmed.
- Main disadvantage is once anesthesia has been induced there is no turning back.
- **The only contraindication to RSI intubation is a practitioner who is not skilled in airway management.**
- Indication for surgical airway is the inability to intubate the trachea.
- In neck trauma, intubation may be difficult or impossible and surgical airway may be required.
- Short acting agents are used to allow patient to resume spontaneous respirations and to allow close monitoring of neurological status.
- Oral endotracheal intubation is usually the preferred method.
- If the head and neck are stabilized by an assistant there is almost no risk of spinal cord injury by oral tracheal intubation.
- Always anticipate vomiting. Even patients, who otherwise seem relatively unresponsive, may vomit during attempted intubation without RSI. This may result in loss of airway control and aspiration of gastric contents
- Struggling patients increase muscle activity making hypoxemia worse and increase ICP.
- As a general rule, presume all trauma patients have just eaten.
- Risk for aspiration is greatest during anesthesia induction and instrumentation of the upper airway. This risk is minimized by applying cricoid pressure.
- Patients with severe closed head injury are of major concern because intracranial pressure can rise precipitously during intubation.
- Rapid sequence induction of anesthesia and oral intubation are now recommended for patients with head injuries to minimize the rise in ICP.
- Remember, rendering patient apneic, when endotracheal intubation is beyond the skill of the operator, may be rapidly fatal.

Indication for RSI Endotracheal Intubation of the Acute Trauma Patient:

- Trauma patients with GCS ≤ 8
- Significant facial trauma with poor airway control
- Airway obstruction
- Closed head injury or hemorrhagic CVA
- Burn patients with airway involvement and inevitable airway loss
- Class 3-4 hemorrhagic shock
- Failure to maintain adequate oxygenation (PaO₂ < 60 despite 100% FiO₂)
- Paralysis due to high spinal cord injury
- Need for positive pressure ventilation
- Blunt chest trauma with compromised ventilatory effort
- Mandibular fractures with loss of airway muscular support

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RSI Intubation: (*Julia E Martin M.D., Assistant Professor, Department of Emergency Medicine*)

Evaluation

- “Talking patient” usually indicates airway is patent for the moment.
- Respiratory distress associated with trauma to the upper airway is frequently made worse by blood or gastric contents in the airway and requires prompt action. These patients are often combative because of hypoxia.
- When evaluating an awake patient with severe facial trauma ask them if they are getting enough air.
- If they cannot answer, stick out their tongues fairly easily, or are hyperventilating, they should probably be intubated.
- In unconscious patients, it is probably best to intubate.
- Tachypnea may be subtle but an early sign of airway or ventilatory compromise.
- Tachypnea is often also associated with pain and/or anxiety.
- Signs of Airway Obstruction:
 - Agitation = hypoxia
 - Obtundation = suggests hypercarbia Cyanosis = hypoxia
 - Retractions and use of accessory muscles
 - Snoring, gurgling, stridor = partial obstruction at pharynx
 - Hoarseness = laryngeal obstruction

Drugs

Sedatives:

Versed:

Benzodiazepine
Rapid onset (1-2 min) and short duration (20 min)
Amnesic
Anticonvulsant
Muscle relaxant
Slight decrease in blood pressure and increase in pulse rate. No increase in ICP.
Dose: 0.1 mg/kg

Etomidate:

Nonbarbiturate, nonnarcotic sedative-hypnotic induction agent.
Good agent in multisystem trauma patient because it evokes minimal change in HR and CO compared to Thiopental. (ideal agent in any patient in shock including cardiogenic and septic shock)
Decreases ICP and IOP during procedure
Rapid onset (<1 min) and short acting (5 min)
Vomiting, esp. when combined with a narcotic
Dose: 0.3 mg/kg

Thiopental:

Ultrashort acting barbiturate sedative
Dose: 3-5 mg/kg
Onset 30-40 sec
Last 10 min.
Does not cause increase in ICP but can cause severe hypotension, therefore avoid in multi-traumatized patients.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

RSI Intubation: (*Julia E Martin M.D., Assistant Professor, Department of Emergency Medicine*)

Drugs (continued)

Can also induce bronchospasm.

Fentanyl:

Narcotic

Little or no histamine release

Rarely causes hypotension

Consider in head-injured patients as a premedication to prevent increase in ICP (blunts pressor response)

Rapid injection may cause chest wall rigidity.

Dose: 3-5 mcg/kg

Onset in 2 min with 30-40 min duration.

Paralytic Agents:

Vecuronium:

Nondepolarizing agent

1/3 more potent and pancuronium and duration of action is 1/3 to 1/2 as long (25-40 min vs Pancuronium which last 2-3 hours)

Onset 2-3 minutes

Dose not cause the degree of tachycardia seen with pancuronium.

No histamine release.

Defasciculating dose: 0.01 mg/kg

Paralytic dose: 0.1 mg/kg

Succinylcholine:

Depolarizing agent, which causes muscle fasciculations which can be prevented by pretreatment with a non-depolarizing neuromuscular agent.

Rapid onset (30-60sec) with short duration of action (5-7 min).

Dose:

Adult: 1.5 mg/kg

Pediatric (<10 y.o): 2.0 mg/kg

Contraindications:

Burns > 7 days old

Extensive crush injuries > 7 days old.

Paraplegia > 7 days old.

Narrow-angle glaucoma

Neuromuscular Diseases:

Guillain-Barre, myasthenia gravis, Multiple sclerosis, muscular dystrophy, Parkinson's disease.

Others susceptible to increased potassium:

Renal failure (no real evidence that RSI increases K+)

Rhabdomyolysis

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RSI Intubation: (*Julia E Martin M.D., Assistant Professor, Department of Emergency Medicine*)

Drugs (continued)

Rocuronium:

Non-depolarizing agent
Onset < 1 min.
Duration 20-30 min.
Dose: 0.9-1.2 mg/kg

Adjunctive:

Atropine:

Succinylcholine will cause bradycardia in infants and children therefore they should be premedicated with atropine.
Also pretreat any adult who is already bradycardic.
Children < 8 y.o.
Dose: 0.01 mg/kg up to 0.5 mg (minimum dose of 0.1 mg)

Lidocaine:

Dose: 1.5 mg/kg
Some studies recommend intravenous Lidocaine to blunt the pressor response of increased pulse, increased blood pressure, increased intracranial pressure, and increased intraocular pressure associated with intubation, its usefulness is controversial. However, because a single dose of lidocaine is unlikely to cause harm, it seems reasonable to use in the patient who has a known or suspected head injury.
Should be administered 2-3 min prior to intubation.

Procedure:

- The 5 P's of rapid sequence intubation:
 - Preparation
 - Preoxygenation
 - Pretreatment
 - Paralysis (with anesthesia)
 - Placement (of the endotracheal tube)
- Preoxygenation with 100% oxygen for 3-5 minutes via NRB mask (or 3 vital capacity breaths, avoid BVM if possible).
- Secure IV's, ECG, pulse oximeter.
- Prepare intubation equipment: ETT with stylet, suction, BVM, laryngoscope.
- Premedication:
 - Lidocaine (head injury) 1.5 mg/Kg
 - Vecuronium (defasciculating dose) 0.01 mg/Kg
 - Versed 0.1 mg/Kg
 - Atropine (peds) 0.01 mg/Kg
 - Etomidate 0.3 mg/Kg

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

RSI Intubation: (*Julia E Martin M.D., Assistant Professor, Department of Emergency Medicine*)

Drugs (continued)

- Perform Sellick's maneuver, maintain maneuver until after confirmation of tube placement.
- Succinylcholine 1.5 mg/Kg (Peds: 2.0 mg/Kg)
- Wait 30-60 sec, place ETT.
- Confirm ETT placement by: listening for bilateral breath sounds, chest rise and fall, tube fogging, & positive ETCO₂. Final confirmation by CXR.
- Release Sellick maneuver.
- Secure ETT

Basic RSI

Time (min)

-5.00	Preparation and Preoxygenation
-2.00	Administer Vecuronium 0.01 mg/kg IV Lidocaine 1.5 mg/kg IV (Fentanyl 3 mcg/kg IV) (Atropine 0.01 mg/kg in kids < 10)
-1.00	Apply cricoid pressure and in-line cervical stabilization
-1.00	Administer Versed 0.1 mg/kg IV or Etomidate 0.3 mg/kg IV
-55 sec	Administer Succinylcholine 1.5 mg/kg IV (2mg/kg if <10 years old)
	Intubate

RSI without Succinylcholine

Time (min)

-5.00	Preparation and Preoxygenation
-4.45	Lidocaine 1.5 mg/kg IV (Fentanyl 3 mcg/kg IV) (Atropine 0.01 mg/kg in kids <10)
-4.45	Apply cricoid pressure and in-line cervical stabilization
-3.00	Administer Versed 0.1 mg/kg IV or Etomidate 0.3 mg/kg IV
-2.45	Administer Rocuronium 0.9 – 1.2 mg/kg
	Intubate

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Pediatric Mechanical Ventilation

Guideline

- Mechanical ventilation will be utilized to assist breathing once the airway has been established but respiration is still inadequate.
- Mechanical ventilation can be supplied either by mechanical ventilator or Ambu bag.
- Mechanical ventilatory assistance can be provided to either the intubated or non-intubated patient.
- In all intubated patients, the Ambu bag or the mechanical ventilator will be the source of ventilatory assistance.
- In the non-intubated patient, mechanical ventilation can be provided by the bag-valve-mask.

Procedure

- The **bag valve-mask** technique may be employed to provide ventilatory assistance to patients prior to establishment of an airway, when temporary ventilatory assistance is needed or when an airway is not possible.
- Procedure is conducted according to the Pediatric Advanced Life Support (PALS) recommendations.
- The **Ambu bag** can be used to provide ventilation of the intubated patient and, when used with increased FIO₂ can supplement oxygenation.
 - The Ambu bag is connected securely to the endotracheal tube and a baseline tidal breath is provided at 8 ml/kg of patient body weight.
 - Inspired FIO₂ should be at 1.0 until satisfactory oxygenation is proven, after which FIO₂ may be decreased.
 - Adequacy of ventilation should be determined by monitoring the rise and fall of the patient's chest wall, the presence of bilateral breath sounds and pulse oximetry. These parameters should be frequently reassessed to insure proper tube placement and ventilation of both lungs.
 - In-line monitoring of etCO₂ is required on all pediatric patients in accordance with PALS recommendations.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Mechanical Ventilation of the Intubated Pediatric Patient

- The mechanical ventilator provides several modes of ventilatory assistance to the patient.
- The mechanical ventilator can be of great assistance when fatigue prevents adequate ventilator assistance with the Ambu bag or when other patients' needs must be tended to.

Procedure

- Provide adequate ventilatory assistance by use of the Ambu bag while ventilator settings are being made.
- Inspired FIO₂ should be at 1.0 until satisfactory oxygenation is proven, after which FIO₂ should be decreased to maintain saturations of 93 to 95% unless specifically contra-indicated (i.e. pulmonary hypertension, congenital cyanotic heart disease, etc.)
- An end tidal CO₂ detection device should be used
- Set desired tidal volume. This should be 8 ml/kg of body weight.
- Set desired tidal volume. This should be 8 ml/kg of body weight.
- Select mode of ventilatory assistance. This should be synchronized intermittent mandatory ventilations (SIMV).
- Select ventilator rate:
 - 0 – 6 months 30-40 bpm I-time 0.5 secs
 - 6 – 12 months 25-35 bpm I-time 0.55 secs
 - 12 months – 5 years 20-30 bpm I-time 0.6 secs
 - 5 years – up 15-25 bpm I-time 0.7 secs
- Select positive end expiratory pressure (PEEP). PEEP should be set at 5 cm of water.
- Peak airway pressure (PAP) alarm should be set at 40 cm of water.
- Peak airway pressures should be noted and recorded at the initiation of mechanical ventilatory support.
- Increasing levels of peak inspiratory pressure may indicate:
 - Airway obstruction
 - Worsening pulmonary congestion
 - Development of tension pneumothorax
 - Development of hemopneumothorax with respiratory compromise
 - Increased patient anxiety/agitation
 - Pulmonary over-inflation (air trapping)
- Signs of decompensation such as desaturation, mottling, and lack of chest rise, shall be evaluated using PALS mnemonic:
 - **D**isplacement of the endotracheal tube
 - **O**bstruction of the endotracheal tube
 - **P**neumothorax or pneumopericardium
 - **E**quipment malfunction or failure

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Ventilator Reference Information

Goals:

- Adequate oxygenation and ventilation to maintain normal blood gases without causing significant barotrauma.
- Oxygenation: Ventilator settings to maintain appropriate oxygen saturations
- Ventilation: Ventilator settings to appropriately clear carbon dioxide

Types of ventilation

Conventional

- SIMV: Ventilator set to give a certain number of breaths per minute, child may over breathe and extra breaths supported. Can set Rate, PEEP, pressure over PEEP. Does not set a tidal volume.
- PRVC: Ventilator set to give a certain tidal volume but limits the pressure delivered, child may over breathe and extra breaths supported. Can set Tidal volume, PEEP, and peak pressure, Rate.
- Oscillator (HFOV)
- Ventilator to give high frequency and low tidal volumes, the tidal volumes are set just above the dead space so different mechanisms for gas exchange used.

Settings

- MAP: Mean Airway Pressure (cmH₂O)
- Amplitude: Variation around MAP
- Frequency: High frequency rate (0.5-5 ml/kg)

Indications for HFOV

- Failure of conventional ventilation
- Air leak syndromes (Pneumothorax, interstitial emphysema)
- Attempt to limit barotrauma from high conventional ventilation settings

Blood Gas Ranges

	<u>ABG</u>	<u>CBG</u>	<u>VBG</u>
pH	7.3-7.45	7.25-7.35	7.25-7.35
pCO ₂	35-45	40-50	40-50
pO ₂	50-70	35-50	35-45

Ventilator Management

- Improve oxygenation: May increase FiO₂, PIP, PEEP, iTime, MAP
- Decrease CO₂: Increase rate, PIP, tidal volume, PEEP, Amplitude, lower Hz

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Sedation for the Intubated Pediatric Patient

- The goals of sedation of the intubated patient are to relieve discomfort and to increase the likelihood that the patient will not remove airway access. This concept must be balanced with the increasing responsibility of maintaining the patient en route.
- The major side effects of all of the drugs listed below are **HYPOTENSION** and **DECREASED RESPIRATORY EFFORT**.

Sedative agents to consider:

- **Versed** – short acting benzodiazepine. May need large doses to be effective in the toddler population. Usual dose is 0.1mg/kg IV every 30 minutes prn.
- **Fentanyl** – narcotic. Fentanyl is not a sedative per se; but may be useful in the control of pain associated with orotracheal intubation. Usual dose is 1-2 mcg/kg IV every 30 minutes prn.
- **Chloral hydrate** – excellent hypnotic. Usual dose is 25-40 mg/kg PO/NG/PR every 6 hours prn. Produces apnea and hypotension when combined with other sedatives in high dose.
- **Propofol** – anesthetic. Propofol is an excellent anesthetic that carries VERY frequent hypotensive results with its administration. There is a known correlation in the pediatric population between propofol sedation and sudden death from profound metabolic acidosis. Propofol will not be initiated by the transport team but if previously started by the referring physician it may be continued with a direct order from the transporting physician. Usual dose is 10 mg/kg/hr IV drip.
- **Neuromuscular blocking agents** are not sedatives or analgesics. These agents should **NOT** be used until the patient's sedative needs are accounted for. The primary use for neuromuscular agents in the intubated patient is to decrease oxygen demand and to synchronize work of breathing around mechanically ventilated patients.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Cricothyroidotomy: Needle

- Surgical airway may be utilized in providing airway access in patients in whom orotracheal intubation is impossible or contraindicated.
- Surgical airway should be used only after consideration of other routes for airway access.

Needle Cricothyroidotomy

- Needle cricothyroidotomy is the most rapid and preferred method for obtaining a surgical airway.
- This method allows adequate oxygenation in most instances.
- Patients can be supported for 30-45 minutes with needle cricothyroidotomy and jet insufflation.
- This method should be attempted prior to performing an incisional cricothyroidotomy.
- Place the patient in a supine position.
- Palpate the cricothyroid membrane, anteriorly, between the thyroid cartilage and cricoid cartilage.
- Surgically prep the area using betadine swabs.
- Assemble a #12- or #14-gauge catheter over the needle apparatus to a 6 to 12 ml syringe.
- Puncture the skin midline and directly over the cricothyroid membrane (i.e., midsagittal).
- Direct the needle at a 45 degree angle caudally.
- Carefully insert the needle through the lower half of the cricothyroid membrane, aspirating as the needle is advanced.
- Aspiration of air signifies entry into the tracheal lumen.
- Withdraw the stylet while gently advancing the catheter downward into position, being careful not to perforate the posterior wall of the trachea.
- Attach the catheter hub to oxygen tubing, with an inline Y-connector at 15 liters/minute (50 PSI) oxygen flow.
- **Note:** Adequate PaCO₂ removal can be maintained for only 30-45 minutes.
- Insufflate O₂ for one second and allow four seconds for deflation.
- Observe lung inflations and auscultate the chest for adequate ventilation.
- Secure the apparatus to the neck

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: AIRWAY

Cricothyroidotomy: Surgical

Surgical Cricothyroidotomy should rarely be done in children. For children ≤ 7 years of age this should be airway of last resort!

- Surgical cricothyroidotomy is indicated when airway access cannot be obtained or is contraindicated via the orotracheal route.
- Surgical cricothyroidotomy may also be indicated when needle cricothyroidotomy proves inadequate.
- Place the patient in a supine position with the neck in a neutral position.
- Palpate the thyroid notch, cricothyroid interval, and the sternal notch for orientation.
- Surgically prep and anesthetize the area locally, if patient is conscious.
- Stabilize the thyroid cartilage with the left hand.
- Make a transverse skin incision over the lower one-half of the cricothyroid membrane.
- Carefully incise through the membrane, being careful to remain in the midline.
- Insert the scalpel handle into the incision and rotate 90 degrees to open the airway. (A tracheal spreader may be used instead of the scalpel handle.)
- Insert an appropriately sized (typically 6 mm), cuffed, ET tube into the cricothyroid membrane incision, directing the tube distally into the trachea.
- Inflate the cuff and ventilate the patient.
- Observe chest expansion and auscultate the chest for adequate ventilation.
- Secure the ET tube to the patient to prevent dislodging.
- Caution: Do not cut or remove cricothyroid cartilage.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Head CT Scanning of Pediatric Patients

- **Absolute Indications for Head CT Scan in Pediatric Trauma Patients:**
 - GCS $\leq$ 14 on two separate exams by Medical Personnel
 - Focal neurological deficit
 - Seizures after trauma
 - Persistent Vomiting
 - Extensive Facial Injury
 - Signs or suspicion of skull fracture

- **Relative Indications for Head CT Scan in Pediatric Trauma Patients**
 - History of change in level of consciousness
 - Drug or alcohol intoxication
 - Suspected child abuse
 - Unreliable or inadequate details of mechanism of injury
 - Age < 2 years, unless reliable adult witnessed the event and can confirm that the mechanism of injury was minor
 - Post Concussive Amnesia

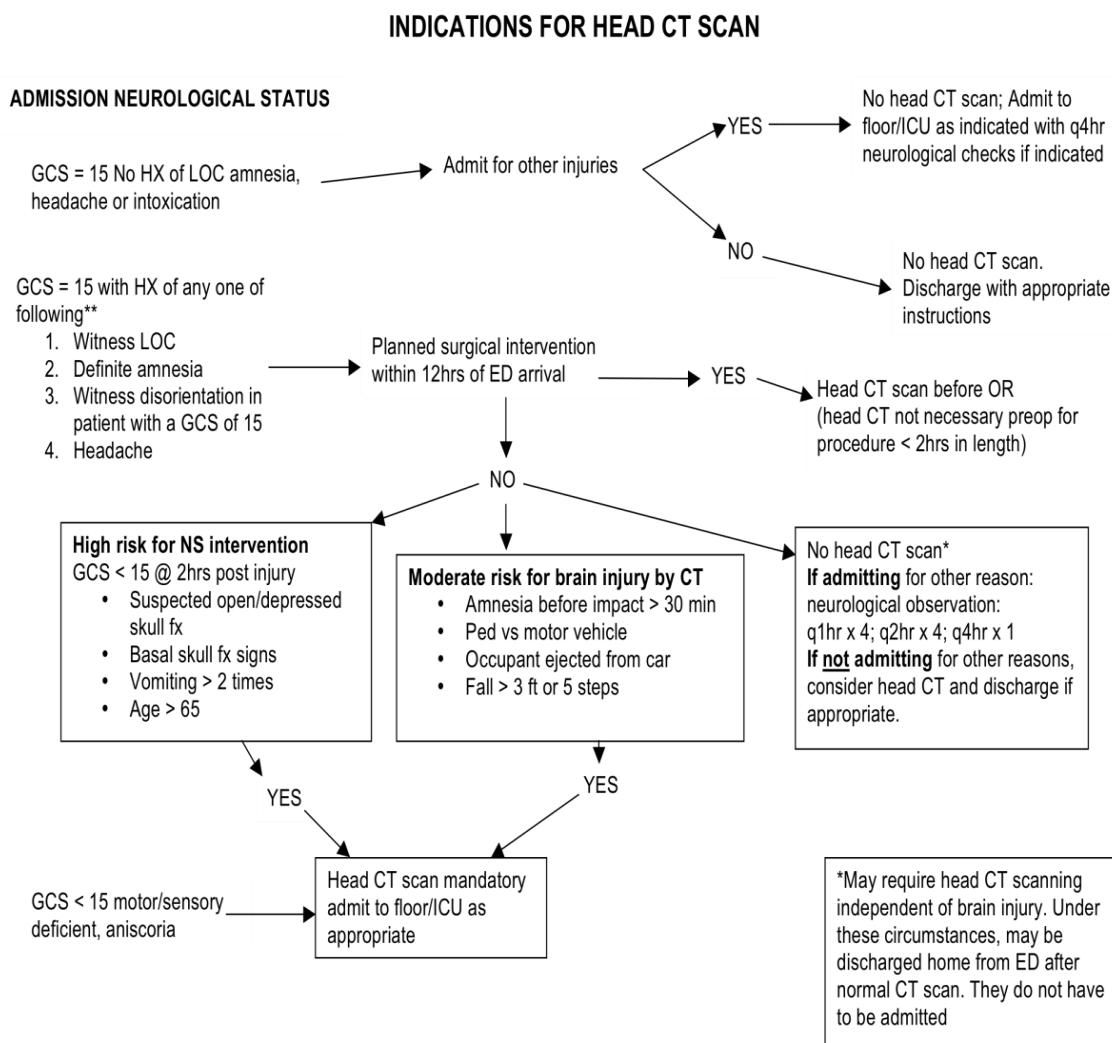
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Indications for Head CT Algorithm



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Pediatric Severe Traumatic Brain Injury Clinical Practice Guideline

Revision: May 2020

Overview

Traumatic brain injury (TBI) is a major cause of morbidity and mortality in pediatric patients. The goal of severe TBI care is to have a standardized clinical guideline approach that will enhance overall care and outcomes of the pediatric severely injured TBI patient. Evidenced based practice has successfully demonstrated adherence to guidelines and best practices can improve outcomes. The severe treatment algorithm is intended to be used in the care of **ALL** pediatric severe TBI patients by **ALL** services involved in the care of the traumatically injured pediatric patient. Usage of the severe TBI guideline is a tracked quality parameter of the pediatric trauma program. In addition to the algorithm, a Severe TBI Daily Checklist is to be utilized by **ALL** care providers. This checklist is to be posted discretely, yet conveniently, as a prompt and reminder of daily care needs and diagnostics required for the patient.

Initial Care & Resuscitation

The majority of the severe TBI patient's entry to facility will be via the Emergency Department and in accordance to pediatric trauma activation criteria, should be a Trauma Alert Red activation for which the Pediatric Surgery Attending is present and in charge of overall care. The overall goal of initial resuscitation is:

- Primary/secondary assessment upon arrival to trauma bay
- Neurosurgery consulted with presence at bedside prior to or immediately after patient arrival to the trauma bay.
- PDS or Neurosurgeon guidance implementation of initial treatment components of the severe TBI algorithm.
- Decision of CT based on PECARN and consultation of Neurosurgeon.
- Trauma bay to CT scanner within 15 minutes of patient arrival.
- Timely decision by neurosurgeon for CT scanner to OR or rapid access to PICU for definitive care.

Critical Care Treatment

- Upon patient arrival to the PICU the standardized severe TBI pediatric guideline shall be followed in conjunction with utilization of the daily severe TBI checklist.

Discharge Follow up Care

- Discharge will be based on patient outcome needs and standard UK policies and procedures.
- Patient follow up care shall be outlined by the neurosurgical service.

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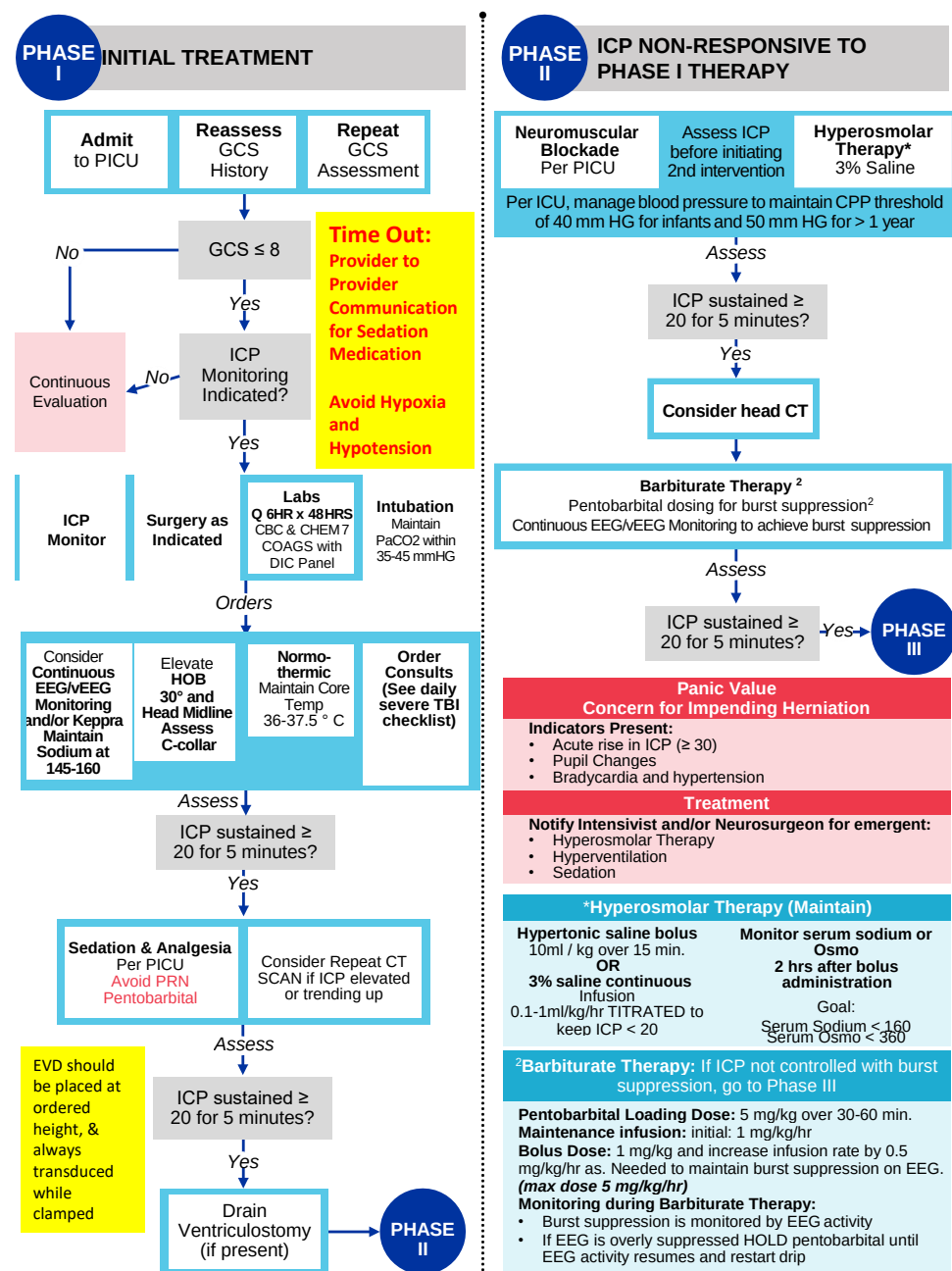
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Pediatric Severe Traumatic Brain Injury Clinical Practice Guideline (continued)

Revision: Nov 2020

Pediatric Severe TBI Treatment Algorithm

[Pediatric Severe TBI Treatment Algorithm \(printable form page 1\)](#)



Reisner, Andrew; Chern, Joshua J, et al. Introduction of severe traumatic brain injury care protocol is associated with reduction in mortality for pediatric patients; a case study of Children's Healthcare of Atlanta's neurotrauma program. J Neurosurg Pediatr. August 2018; 22:165-172. DOI: 10.3171/2018.2.PEDS17562

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

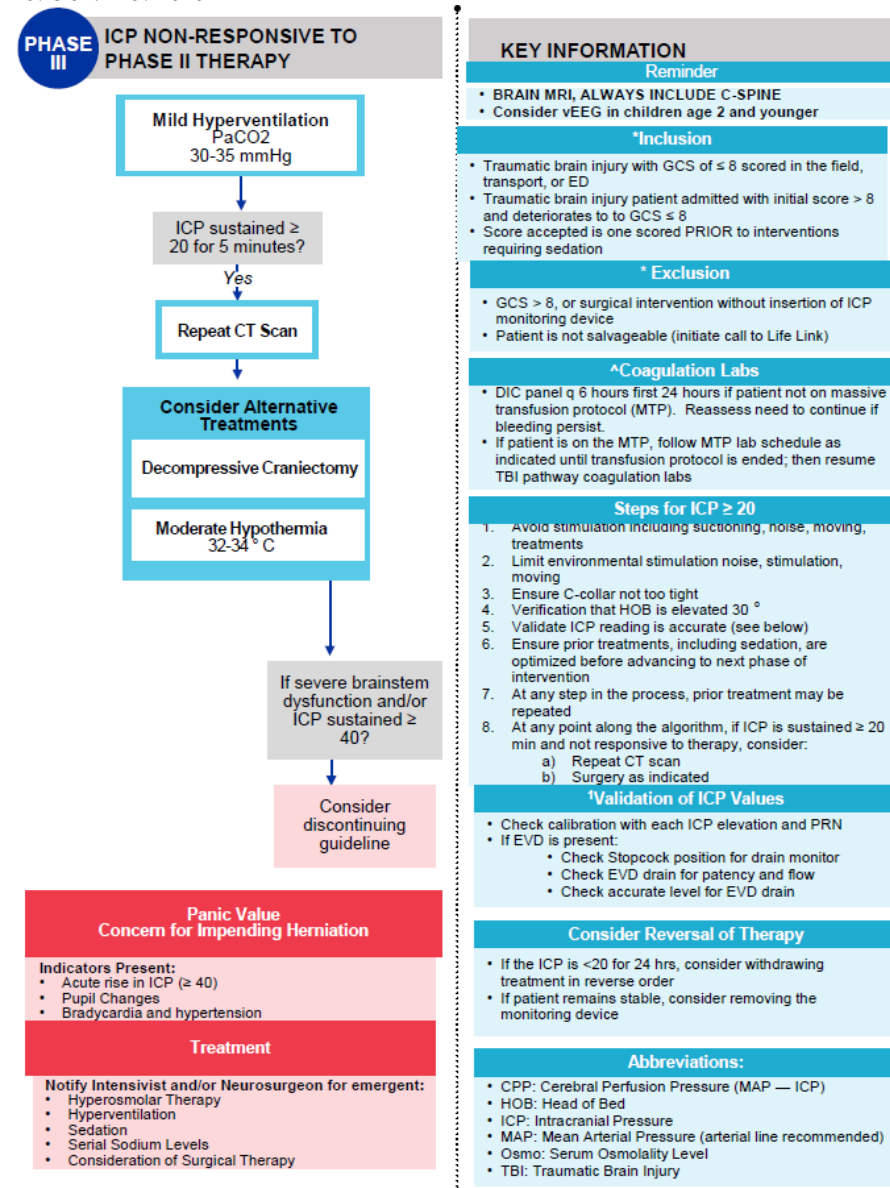
Pediatric Severe Traumatic Brain Injury Clinical Practice Guideline (*continued*)

Revision: May 2020

Pediatric Severe TBI Treatment Algorithm (*continued*)

[Pediatric Severe TBI Treatment Algorithm \(printable form page 2\)](#)

Revision: Nov 2020



Reisner, Andrew; Chern, Joshua J, et al. Introduction of severe traumatic brain injury care protocol is associated with reduction in mortality for pediatric patients; a case study of Children's Healthcare of Atlanta's neurotrauma program. J Neurosurg Pediatr. August 2018; 22:165-172. DOI: 10.3171/2018.2.PEDS17562

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Pediatric Severe Traumatic Brain Injury Clinical Practice Guideline (*continued*)

Revision: May 2020

KCH SEVERE TBI DAILY CHECKLIST

Daily Severe TBI Care Requirements

Completed	Requirement	Plan/Comments
	Thermoregulation needs	
	End-tidal CO2 monitoring in place	
	Sedation Plan	
	AED Plan	
	C-Collar Plan & Status	
	Opening Pressures Documented	
	Cath Site Assessment	
	EVD Catheter: - o zeroed every shift - o Check EVD for patency and flow - o Check accurate level for EVD drain	
	Manual ICP check to ensure transducer accuracy every shift	
	Na+ Level due when?	
	ICP Transduced or to Drain	
	Sedation Time Out:	
	Provider to Provider Communication	
	EEG or vEEG	
	CT Scans	
	MRIs (Reminder get C-spine with MRI)	
	Known C-spine Fractures?	
	Consults Completed - o PT/OT: 24hrs (Preferred by PT) - o Nutrition (within 72 hrs are evaluated in PICU during rounds) - o Speech: Prefer early placement of orders for preparation - o Rehabilitation: PMR prefer early placement of orders	
	KODA Notified per enterprise policy	
	Coroner Notified per state law & enterprise policy	

KCH SEVERE TBI CHECKLIST: February 28, 2020
PEDIATRIC TRAUMA GUIDELINES

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Pediatric Severe Traumatic Brain Injury Clinical Practice Guideline (*continued*)

Revision: Nov 2020

References

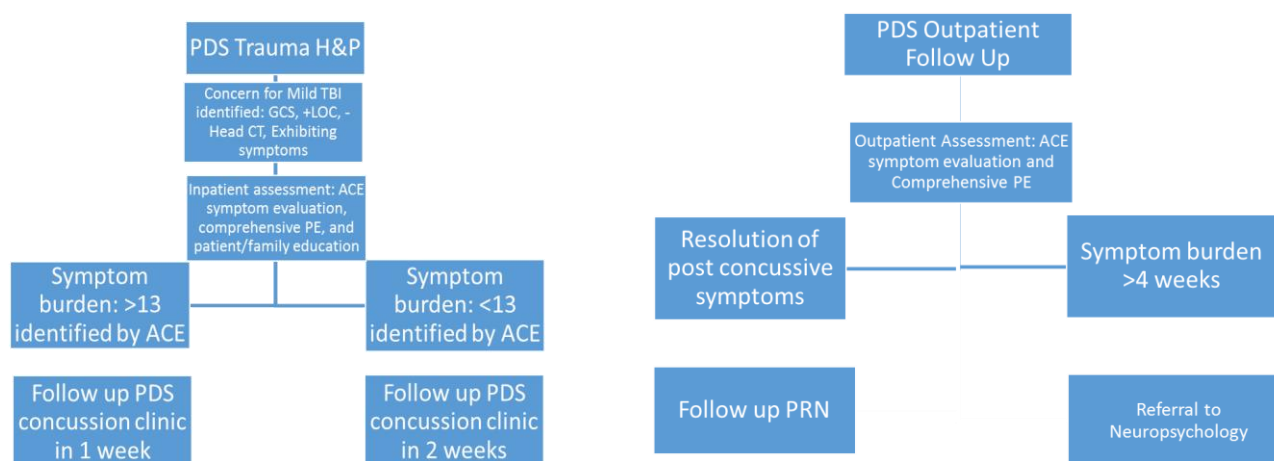
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Pediatric Mild-Moderate Traumatic Brain Injury Clinical Practice Guideline



- All families are provided with KRAMS guideline prior to hospital discharge
- PDS PA-C or on-call resident completes ACE form during inpatient stay

A concussion (or mild traumatic brain injury (MTBI)) is a complex pathophysiologic process affecting the brain, induced by traumatic biomechanical forces secondary to direct or indirect forces to the head. Disturbance of brain function is related to neurometabolic dysfunction, rather than structural injury, and is typically associated with normal structural neuroimaging findings (i.e., CT scan, MRI). Concussion may or may not involve a loss of consciousness (LOC). Concussion results in a constellation of physical, cognitive, emotional and sleep-related symptoms. Symptoms may last from several minutes to days, weeks, months or even longer in some cases.

ACE Instructions

The Acute Concussion Evaluation (ACE) is intended to provide an evidence-based clinical protocol to conduct an initial evaluation and diagnosis of patients (both children and adults) with known or suspected MTBI. The research evidence documenting the importance of these components in the evaluation of an MTBI is provided in the reference list.

A. Injury Characteristics:

- Obtain **description of the injury** - how injury occurred, type of force, location on the head or body if force transmitted to head. Different biomechanics of injury may result in differential symptom patterns (e.g., occipital blow may result in visual changes, balance difficulties).
- Indicate the **cause of injury**. Greater forces associated with the trauma are likely to result in more severe presentation of symptoms.

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3/ 4. **Amnesia**: Amnesia is defined as the failure to form new memories. Determine whether amnesia has occurred and attempt to determine length of time of memory dysfunction – before (retrograde) and after (anterograde) injury. Even seconds to minutes of memory loss can be predictive of outcome. Recent research has indicated that amnesia may be up to 4-10 times more predictive of symptoms and cognitive deficits following concussion than is LOC (less than 1 minute).¹

- **Loss of consciousness (LOC)** - If occurs, determine length of LOC.
- **Early signs**. If present, ask the individuals who know the patient (parent, spouse, friend, etc) about specific signs of the concussion/ MTBI that may have been observed. These signs are typically observed early after the injury.
- Inquire whether **seizures** were observed or not.

B: Symptom Checklist: ²

- Ask patient (and/ or parent, if child) to report presence of the four categories of symptoms since injury. It is important to assess all listed symptoms as different parts of the brain control different functions. One or all symptoms may be present depending upon mechanisms of injury.³ Record 1 for Yes or 0 for No for their presence or absence, respectively.
- For all symptoms, indicate presence of symptoms as experienced within the past 24 hours. Since symptoms can be present premorbidly/at baseline (e.g., inattention, headaches, sleep, sadness), it is important to assess change from their typical presentation.
- **Scoring**: Sum total number of symptoms present per area, and sum all four areas into Total Symptom Score (score range 0-22). (Note: most sleep symptoms are only applicable after a night has passed since the injury. Drowsiness may be present on the day of injury.) If symptoms are new and present, there is no lower limit symptom score. Any score > 0 indicates positive symptom history.
- **Exertion**: Inquire whether any symptoms worsen with physical (e.g., running, climbing stairs, bike riding) and/or cognitive (e.g., academic studies, multi-tasking at work, reading or other tasks requiring focused concentration) exertion. Clinicians should be aware that symptoms will typically worsen or re-emerge with exertion, indicating incomplete recovery. Over-exertion may protract recovery.
- **Overall Rating**: Determine how different the person is acting from their usual self. Circle 0 (Normal) to 6 (Very Different).

C: Risk Factors for Protracted Recovery: Assess the following risk factors as possible complicating factors in the recovery process.

- **Concussion history**: Assess the number and date(s) of prior concussions, the duration of symptoms for each injury, and whether less biomechanical force resulted in re-injury. Recent research indicates that cognitive and symptom effects of concussion may be cumulative, especially if there is minimal duration of time between injuries and less biomechanical force results in subsequent concussion (which may indicate incomplete recovery from initial trauma).⁴⁻⁸

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- **Headache history:** Assess personal and/or family history of diagnosis/treatment for headaches. Recent research indicates headache (migraine in particular) can result in protracted recovery from concussion.⁸⁻¹¹
- **Developmental history:** Assess history of learning disabilities, Attention-Deficit/Hyperactivity Disorder or other developmental disorders. Recent studies indicate the possibility of a longer period of recovery with these conditions.¹²
- **Psychiatric history:** Assess for history of depression/mood disorder, anxiety, and/or sleep disorder.¹³⁻¹⁶

D: Red Flags: The patient should be carefully observed over the first 24-48 hours for these serious signs. Red flags are to be assessed as possible signs of deteriorating neurological functioning. Any positive report should prompt strong consideration of referral for emergency medical evaluation (e.g. CT Scan to rule out intracranial bleed or other structural pathology).¹⁷

E: Diagnosis: The following ICD-10 diagnostic codes may be applicable.

S06.0X0A (Concussion, with no loss of consciousness) – Positive injury description with evidence of forcible direct/ indirect blow to the head (A1a); plus evidence of active symptoms (B) of any type and number related to the trauma (Total Symptom Score >0); no evidence of LOC (A5), skull fracture or intracranial injury (A1b).

S06.0X1A (Concussion, with brief loss of consciousness < 30 minutes) - Positive injury description with evidence of forcible direct/ indirect blow to the head (A1a); plus evidence of active symptoms (B) of any type and number related to the trauma (Total Symptom Score >0); positive evidence of LOC (A5), skull fracture or intracranial injury (A1b).

S06.0X9A (Concussion, unspecified) - Positive injury description with evidence of forcible direct/ indirect blow to the head (A1a); plus evidence of active symptoms (B) of any type and number related to the trauma (Total Symptom Score >0); unclear/unknown injury details; unclear evidence of LOC (A5), no skull fracture or intracranial injury.

Other Diagnoses – If the patient presents with a positive injury description and associated symptoms, but additional evidence of intracranial injury (A 1b) such as from neuroimaging, a moderate TBI and the diagnostic category of **S06.890A (Intracranial injury)** should be considered.

F: Follow-Up Action Plan: Develop a follow-up plan of action for symptomatic patients. The physician/clinician may decide to (1) monitor the patient in the office or (2) refer them to a specialist. Serial evaluation of the concussion is critical as symptoms may resolve, worsen, or ebb and flow depending upon many factors (e.g., cognitive/ physical exertion, comorbidities). Referral to a specialist can be particularly valuable to help manage certain aspects of the patient's condition. (Physician/clinician should also complete the ACE Care Plan included in this tool kit.)

- **Physician/clinician serial monitoring-** Particularly appropriate if number and severity of symptoms are steadily decreasing over time and/or fully resolve within 3-5 days. If steady reduction is not evident, referral to a specialist is warranted.

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- **Referral to a specialist** – Appropriate if symptom reduction is not evident in 3-5 days, or sooner if symptom profile is concerning in type/severity.
- Neuropsychological Testing can provide valuable information to help assess a patient's brain function and impairment and assist with treatment planning, such as return to play decisions.
- Physician Evaluation is particularly relevant for medical evaluation and management of concussion. It is also critical for evaluating and managing focal neurologic, sensory, vestibular, and motor concerns. It may be useful for medication management (e.g., headaches, sleep disturbance, depression) if post-concussive problems persist.

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ACUTE CONCUSSION EVALUATION (ACE)

Physician/Clinician Office Version

Gerard Gioia, PhD¹ & Micky Collins, PhD²

¹Children's National Medical Center
²University of Pittsburgh Medical Center

Patient Name _____
DOB: _____ Age: _____
Date: _____ ID/MR# _____

A. Injury Characteristics Date/Time of Injury _____ Reporter: ☐ Patient ☐ Parent ☐ Spouse ☐ Other _____

1. Injury Description _____

- 1a. Is there evidence of a forcible blow to the head (direct or indirect)? ☐ Yes ☐ No ☐ Unknown
 1b. Is there evidence of intracranial injury or skull fracture? ☐ Yes ☐ No ☐ Unknown
 1c. Location of Impact: ☐ Frontal ☐ Lft Temporal ☐ Rt Temporal ☐ Lft Parietal ☐ Rt Parietal ☐ Occipital ☐ Neck ☐ Indirect Force
 2. Cause: ☐ MVC ☐ Pedestrian-MVC ☐ Fall ☐ Assault ☐ Sports (specify) _____ Other _____
 3. **Amnesia Before (Retrograde)** Are there any events just BEFORE the injury that you/ person has no memory of (even brief)? ☐ Yes ☐ No Duration _____
 4. **Amnesia After (Anterograde)** Are there any events just AFTER the injury that you/ person has no memory of (even brief)? ☐ Yes ☐ No Duration _____
 5. **Loss of Consciousness:** Did you/ person lose consciousness? ☐ Yes ☐ No Duration _____
 6. **EARLY SIGNS:** ☐ Appears dazed or stunned ☐ Is confused about events ☐ Answers questions slowly ☐ Repeats Questions ☐ Forgetful (recent info)
 7. **Seizures:** Were seizures observed? No ☐ Yes ☐ Detail _____

B. Symptom Check List* Since the injury, has the person experienced any of these symptoms any more than usual today or in the past day?
 Indicate presence of each symptom (0=No, 1=Yes). *Lovell & Collins, 1998 JHTR

PHYSICAL (10)		COGNITIVE (4)		SLEEP (4)	
Headache	0 1	Feeling mentally foggy	0 1	Drowsiness	0 1
Nausea	0 1	Feeling slowed down	0 1	Sleeping less than usual	0 1 N/A
Vomiting	0 1	Difficulty concentrating	0 1	Sleeping more than usual	0 1 N/A
Balance problems	0 1	Difficulty remembering	0 1	Trouble falling asleep	0 1 N/A
Dizziness	0 1	COGNITIVE Total (0-4) _____		SLEEP Total (0-4) _____	
Visual problems	0 1	EMOTIONAL (4)		Exertion: Do these symptoms <u>worsen</u> with: Physical Activity ☐ Yes ☐ No ☐ N/A Cognitive Activity ☐ Yes ☐ No ☐ N/A Overall Rating: How <u>different</u> is the person acting compared to his/her usual self? (circle) Normal 0 1 2 3 4 5 6 Very Different	
Fatigue	0 1	Irritability	0 1		
Sensitivity to light	0 1	Sadness	0 1		
Sensitivity to noise	0 1	More emotional	0 1		
Numbness/Tingling	0 1	Nervousness	0 1		
PHYSICAL Total (0-10) _____		EMOTIONAL Total (0-4) _____			
(Add Physical, Cognitive, Emotion, Sleep totals)		Total Symptom Score (0-22) _____			

C. Risk Factors for Prolonged Recovery (check all that apply)

Concussion History? Y ☐ N ☐	Headache History? Y ☐ N ☐	Developmental History	Psychiatric History
Previous # 1 2 3 4 5	Prior treatment for headache	Learning disabilities	Anxiety
Longest symptom duration Days _____ Weeks _____ Months _____ Years _____	History of migraine headache ____ Personal ____ Family	Attention-Deficit/ Hyperactivity Disorder	Depression
If multiple concussions, less force caused reinjury? Yes ☐ No ☐		Other developmental disorder _____	Sleep disorder
			Other psychiatric disorder _____

List other comorbid medical disorders or medication usage (e.g., hypothyroid, seizures) _____

D. RED FLAGS for acute emergency management: Refer to the emergency department with sudden onset of any of the following:

- | | | | |
|--------------------------|--|--|------------------------------------|
| * Headaches that worsen | * Looks very drowsy/ can't be awakened | * Can't recognize people or places | * Neck pain |
| * Seizures | * Repeated vomiting | * Increasing confusion or irritability | * Unusual behavioral change |
| * Focal neurologic signs | * Slurred speech | * Weakness or numbness in arms/legs | * Change in state of consciousness |

E. Diagnosis (ICD-10): ☐ Concussion w/o LOC S06.0X0A ☐ Concussion w/ LOC S06.0X1A ☐ Concussion (Unspecified) S06.0X9A ☐ Other (854) _____
☐ No diagnosis

F. Follow-Up Action Plan Complete ACE Care Plan and provide copy to patient/family.

- ☐ No Follow-Up Needed
☐ Physician/ Clinician Office Monitoring: Date of next follow-up _____
☐ Referral:
☐ Neuropsychological Testing
☐ Physician: _____ Neurology _____ Sports Medicine _____ Psychiatrist _____ Other _____

ACE Completed by: _____ MD RN NP PhD ATC

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References:

- 1) McCrory P, et al. Consensus Statement on Concussion in Sport- the 5th international conference on concussion in sport held in Berlin, October 2016. Br J Sports Med 2017;0:1–10. doi:10.1136/bjsports-2017-097699
- 2) Gioia, G. A., Collins, M., & Isquith, P. K. (2008). Improving Identification and Diagnosis of Mild Traumatic Brain Injury With Evidence. Journal of Head Trauma Rehabilitation, 23(4), 230–242. doi: 10.1097/01.htr.0000327255.38881.ca
- 3) Broglio, S. P.; Collins, M. W.; Williams, R. M.; Mucha, A.; Kontos, A. P. Current and emerging rehabilitation for concussion: A review of the evidence. Clinics in Sports Medicine. 2015;34(2), 213-231. doi:10.1016/j.csm.2014.12.005
- 4) Davis, G. A.; Anderson, V.; Babl, F. E.; Gioia, G. A.; Giza, C. C.; Meehan, W.; Moser, R. S.; Purcell, L.; Schatz, P.; Schneider, K. J.; Takagi, M.; Yeates, K. O.; Zemek, R.;. What is the difference in concussion management in children as compared to adults? A systematic review. British Journal of Sports Medicine. 2017;51(949), 957.
- 5) Hung, R.; Carroll, L. J.; Cancelliere, C.; Côté, P.; Rumney, P.; Keightley, M.; Donovan, J.; Stålnacke, B. M.; Cassidy, J. D.. Systematic review of the clinical course, natural history, and prognosis for pediatric mild traumatic brain injury: Results of the International Collaboration on Mild Traumatic Brain Injury Prognosis. Archives of Physical Medicine and Rehabilitation. 2014;95(3), S174-S191. doi:https://doi.org/10.1016/j.apmr.2013.08.301
- 6) Kapadia, M.; Scheid, A.; Fine, E.; Zoffness, R. Review of the management of pediatric post-concussion syndrome-a multi-disciplinary, individualized approach. Current Reviews in Musculoskeletal Medicine. 2019;12(1), 57-66. doi:10.1007/s12178-019-09533-x

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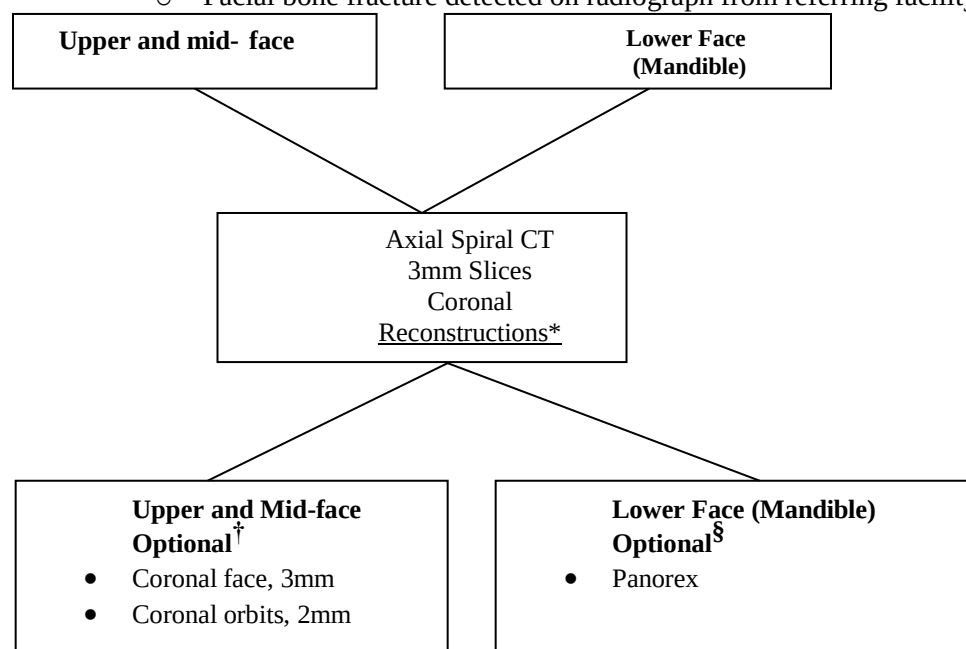
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: HEAD & FACE

Facial Bone Fractures

- **Indications for ED Radiological studies for facial bone fractures are:**
 - Obvious facial bone fracture or fractures suspected on physical exam
 - Facial bone fracture detected on head CT
 - Facial bone fracture detected on radiograph from referring facility



- *Axial spiral CT with 3mm slices provides sufficient detail for diagnosis and treatment planning for facial fractures (forehead to mandible).
- †These films are not often needed urgently and should be obtained at the request or discretion of the maxillofacial trauma consultant.
- Facial plain films are sometimes helpful for treatment planning.
- Formal coronal CT is usually only needed when coronal reconstructions do not provide adequate detail for surgical treatment planning.
- An urgent coronal CT may sometimes be needed to resolve an equivocal CT with regards to optic nerve integrity or compression.
- Mandibular Panorex is a useful study in patients who can sit upright and cooperate with the exam.
- Panorex is not often logistically obtainable in multi-trauma patients.
- Axial CT with coronal reconstructions, which provides satisfactory detail for treatment planning, is the usual diagnostic of choice.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: SPINE EVALUATION

Spine Evaluation and Clearance: Basic Principles

- **General**
 - Entire spine is immobilized during primary survey.
 - Radiographic clearance of the spine is not required before emergent surgical procedures. Presence of a spinal column injury is simply assumed until excluded.
 - Secondary and tertiary exams include examination of the spine for tenderness as well as testing all motor roots, sensation and reflexes.
 - Tertiary exams are performed only on alert and unimpaired patient without distracting injuries.
 - If any spine fractures are found, entire spine must be radiographed.
 - Patients with radiographic injury will have spine consultation for focused pre-operative evaluation regarding relative instability and severity of injury prior to intubation when possible.
 - Patients remain on spine precautions until spine is cleared.
- **Cervical**
 - C-spines are not cleared until after the tertiary exam is completed.
 - Cervical CT scan is the preferred radiographic modality when physical exam is not adequate in patients > 8 years of age.
 - Children under age 8 should have spines and undergo clinical exam. If unable to clear spines by radiographs and clinical exam, maintain spinal precautions and perform an MRI of the spines.
 - If spine clearance cannot be achieved within 2 hours, rigid collars should be removed and replaced with semi-rigid pressure reducing collar.
 - Enter patients in cervical algorithm for C-Spine clearance
- **Thoraco-Lumbar**
 - CT scan of thoracic and lumbar spines if there are clinical findings on secondary or tertiary exams or an unreliable exam.
 - Multi-detector CT-scan with reformatted axial collimation is superior to plain films.
 - Radiographic Thoraco-Lumbar clearance is not needed prior to OR for non-spine surgery.
 - Thoracic & Lumbar clearance may however be required for some non-supine positioning in the OR, depending upon acuity and case type.
 - Tertiary exam is necessary to clear thoracic and lumbar spines.

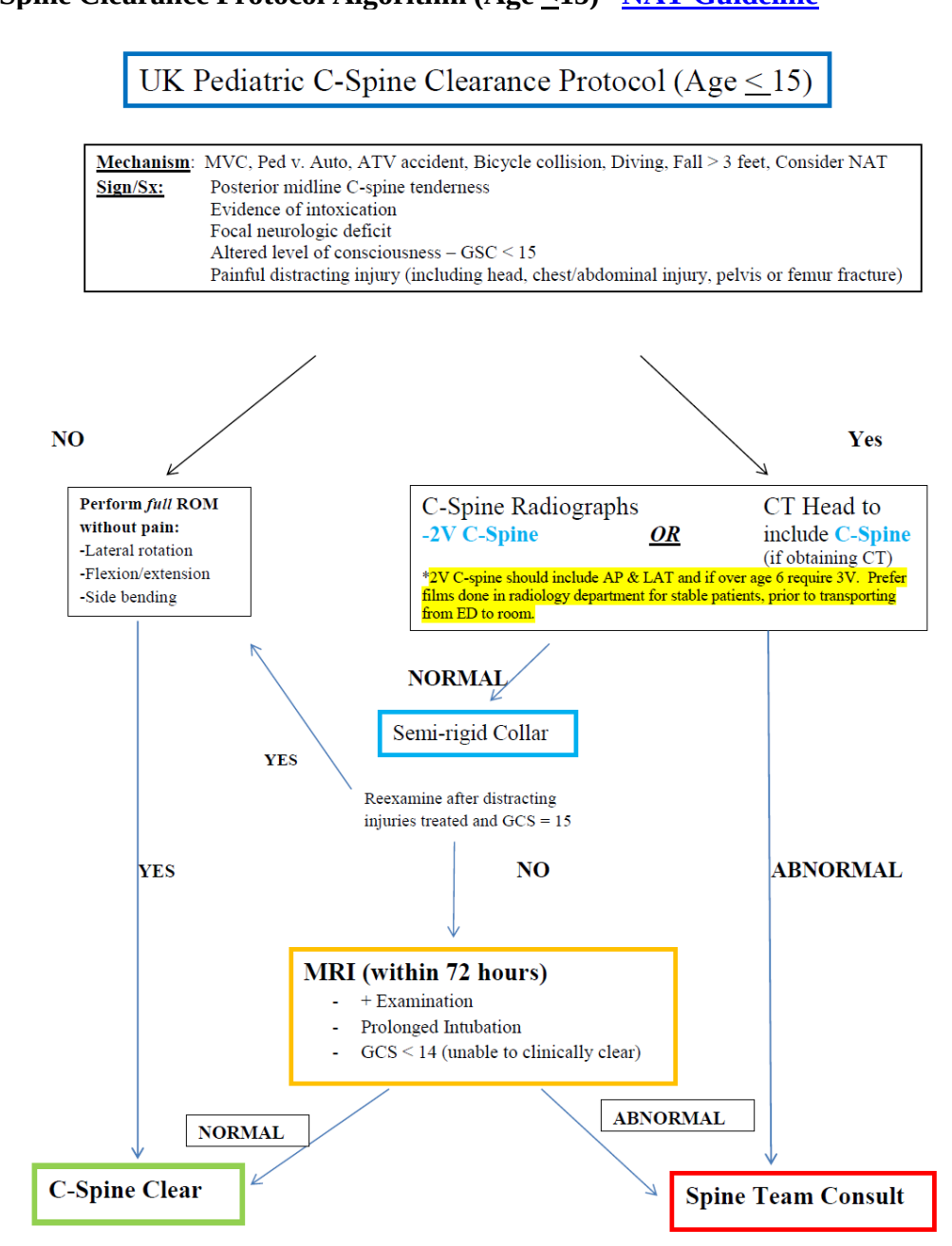
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: C-SPINE CLEARANCE

UK Pediatric C-Spine Clearance Protocol Algorithm (Age ≤15) [NAT Guideline](#)



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: C-SPINE CLEARANCE

Continued:

References:

1. Stiell, IG., et al., The Canadian C-spine rule for radiography in alert and stable trauma patients. JAMA, 2001 October. 286(15): p. 1841-8. (Link: [JAMA.](#))
2. Viccellio, P., et al., A Prospective Multicenter Study of Cervical Spine Injury in Children. <http://www.pediatrics.org/cgi/content/full/108/2/e20> PEDIATRICS Vol. 108 No. 2 August 2001.
3. Patel, JC., et al., Pediatric Cervical Spine Injuries: defining the disease. Journal of Pediatric Surg. 2001 Feb; 36(2): p.373-6.

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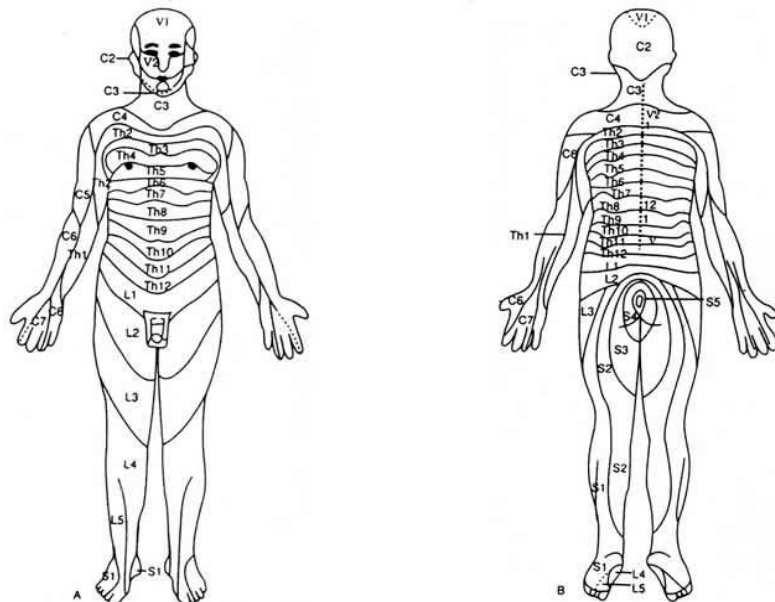
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: SPINE EVALUATION

Initial Management of Spinal Cord Injury

- Priorities: Airway, Breathing & Circulation
- Maintain complete spine immobilization using:
 - Semi-rigid cervical collar
 - Modified logroll – maintaining spine in neutral position at all times
 - Remove patient from long board within 2 hours.
- *If patient is hypotensive – determine cause and treat hypovolemia with fluids and definitive surgical intervention as directed.
- If hypotension due to Neurogenic Shock confirmed, consider inotropic agents to maintain blood pressure.
 *Effort must be made to reduce secondary injury.
 *Methylprednisolone use – insufficient evidence to support routine use
- Patient should be removed from long spine board & placed on pressure reducing surface within 2 hours of trauma room arrival.
- Radiographic studies to determine location of injury include:
 - Plain films
 - Spiral CT scan
 - MRI
- Determine if injury is complete or incomplete and fracture is stable or unstable.
- Fully document complete neurological exam during secondary survey & prior to OR if possible.
- Document level of sensory and motor function using dermatomes.



- Obtain Spine Surgery (Orthopedic or Neurosurgery) consult.
- Place urinary catheter to monitor urinary output & prevent bladder distension.
- Place gastric tube to prevent gastric distension & aspiration.
- Physical Medicine and Rehab consult on admission to begin a timely transfer to Acute Spinal Cord Injury

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Rehabilitation.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NECK

Blunt Neck Trauma/Blunt Cerebrovascular Injury Algorithm

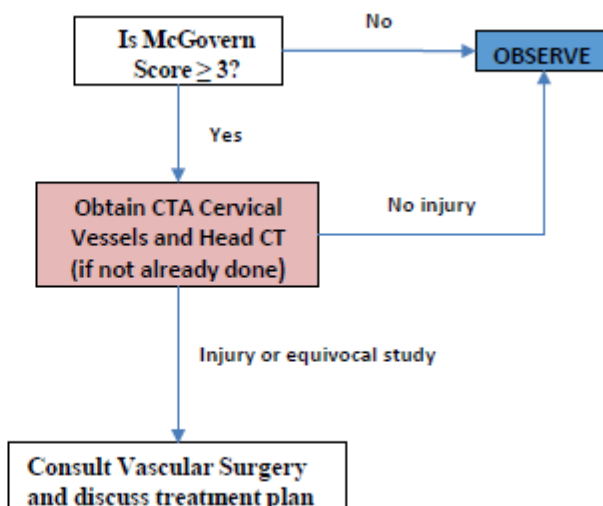
Blunt Neck Trauma/Blunt Cerebrovascular Injury Evaluation and Treatment Guideline

February 19, 2020; April 3, 2020

Blunt cerebrovascular injuries (BCVI) are rare. Most patients are symptomatic. According to Herbert, *et al.*, the McGovern score is the first BCVI screening tool to incorporate the mechanism of injury in screening criteria. This can allow physicians to minimize the use of radiation and determine which patient are high risk and which actually need angiographic imaging.

McGovern Score	
Criteria	Points
GCS score ≤ 8	1
Focal Neuro Deficit	2
Carotid Canal Fracture	2
Mechanism (MVC, HBC)	2
Petrous temporal bone fracture	3
Cerebral infarction on CT	3
Score ≥ 3 points signifies high risk for BCVI. The patient should undergo angiography.	

NOTE: If neuro decline, ACTIVATE STROKE ALERT



Reference Information:

Blunt Carotid Arterial Injury Grading Scale

- Grade I: luminal irregularity or dissection with $< 25\%$ luminal narrowing
- Grade II: dissection or intramural hematoma with $\geq 25\%$ luminal narrowing, intraluminal thrombus, or raised intimal flap
- Grade III: pseudoaneurysm
- Grade IV: occlusion
- Grade V: transection with free extravasation

Management Recommendations:

- Grade I -II: Aspirin
- Grade III – IV: Full anticoagulation with heparin preferred; consider ASA if full anti-coagulation contraindicated (possible stent or operative repair for grade III)
- Grade V: OR

References

1. Herbert JP; Venkataraman SS, *et al.* Pediatric blunt cerebrovascular injury: the McGovern screening score. J Neurosurg Pediatr. 2018; 21:639-49. DOI: 10.3171/2017.12.PEDS17498
2. Mallicote MU, Mubina AI, *et al.* Screening for blunt cerebrovascular injuries in pediatric trauma patients. J Pediatr Surg. 2019; 54:1861-5. DOI: 10.1016/j.jpedsurg.2019.04.014
3. Biff WL, Moore EE, *et al.* Blunt carotid arterial injuries: implications of a new grading scale. J Trauma. 1999; 47:845-53. DOI: 10.1097/00005373-199911000-00004

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NECK

Penetrating Neck Injury Evaluation and Treatment Algorithm University of Kentucky Hospital Trauma Center

Overview

- The neck is divided into zones:
Zone 1-Clavicles to cricoid cartilage Zone 2-
Cricoid to angles of the mandible
Zone 3-Angles of the mandible to skull base
- Management of patients with penetrating wounds to the neck has historically been determined by zone of injury.
- Because zones 1 and 3 are challenging to expose surgically, patients with injuries in zones 1 and/or 3 warrant thorough diagnostics because non-therapeutic surgery in these areas is both difficult and morbid.
- Zones 1 and 3 should be approached surgically only if an injury is felt to be present. However, zone 2 of the neck is easily exposed surgically.
- Controversy has existed as to whether patients with zone 2 injuries should undergo exhaustive diagnostics to exclude or characterize injuries in this area, or simply undergo neck exploration with limited or no preoperative evaluation of the esophagus and cervical vasculature (esophagography or esophagoscopy or both plus angiography).
- Evidence suggests that for patients with no clinical evidence of vascular injury (shown on the algorithm as “Mandatory Criteria for Neck Exploration”) then physical examination has 100% sensitivity, thus definitively excluding vascular injury without any angiography. This is most true if the injury is to Zone 2.
- The second major advance in the care of these patients has been the advent of helical CT angiography as an alternative to conventional catheter-based angiography, producing equivalent results.
- The anatomic detail provided by the neck CT may permit the clinician to exclude injury to the esophagus if the CT clearly shows a missile trajectory remote from the esophagus. However, the precise role that CT will play in excluding injury to the esophagus in these patients remains to be established.
- If the CT does not conclusively exclude injury to the esophagus, contrast esophagography and/or esophagoscopy (or both) should be performed.

Summary

1. CT scan of the neck including CT cervical angiography is the initial diagnostic of choice.
2. Some asymptomatic patients may avoid angiography entirely.
3. Esophageal injury must be definitively excluded, which may require esophagography or esophagoscopy or both.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NECK

Penetrating Neck Injury Evaluation and Treatment Algorithm University of Kentucky Hospital Trauma Center (continued)

References

References regarding the utility of CT/CTA for evaluating penetrating neck injury:

1. Gracias VH, Reilly PM, Philpott J, Klein WP, Lee SY, Singer M, Schwab CW. Computed tomography in the evaluation of penetrating neck trauma: a preliminary study. Arch Surg 2001;136: 1231-5.
2. Mazolewski PJ, Curry JD, Browder T, Fildes J. Computed tomographic scan can be used for surgical decision making in zone II penetrating neck injuries. J Trauma 2001;51:315-9.
3. Múnera F, Soto JA, Palacio DM, Castañeda J, Morales C, Sanabria A, Gutiérrez JE, García G. Penetrating neck injuries: helical CT angiography for initial evaluation. Radiology 2002; 224:366-72.
4. Ferguson E, Dennis JW, Vu JH, Frykberg ER. Redefining the role of arterial imaging in the management of penetrating zone 3 neck injuries. Vascular. May-jun 2005;13(3):158-63. [Medline].

References demonstrating the diagnostic accuracy of physical exam for vascular injuries in Zone II requiring intervention:

1. Nason RW, Assuras GN, Gray PR, Lipschitz J, Burns CM. Penetrating neck injuries: analysis of experience from a Canadian trauma centre. Can J Surg 2001;44:122-6.
2. Sriussadaporn S, Pak-Art R, Tharavej C, Sirichindakul B, Chiamanantthapong S. Selective management of penetrating neck injuries based on clinical presentations is safe and practical. Int Surg 2001;86:90-3.
3. Azuaje RE, Jacobson LE, Glover J, Gomez GA, Rodman GH Jr, Broadie TA, Simons CJ, Bjerke HS. Reliability of physical examination as a predictor of vascular injury after penetrating neck trauma. Am Surg 2003;69:804-7.
4. Osborn TM, Bell RB, Qaisi W, Long WB. Computed tomographic angiography as an aid to clinical decision making in the selective management of penetrating injuries to the neck: a reduction in the need for operative exploration. J Trauma. Jun 2008;64(6):1466-71. [Medline].

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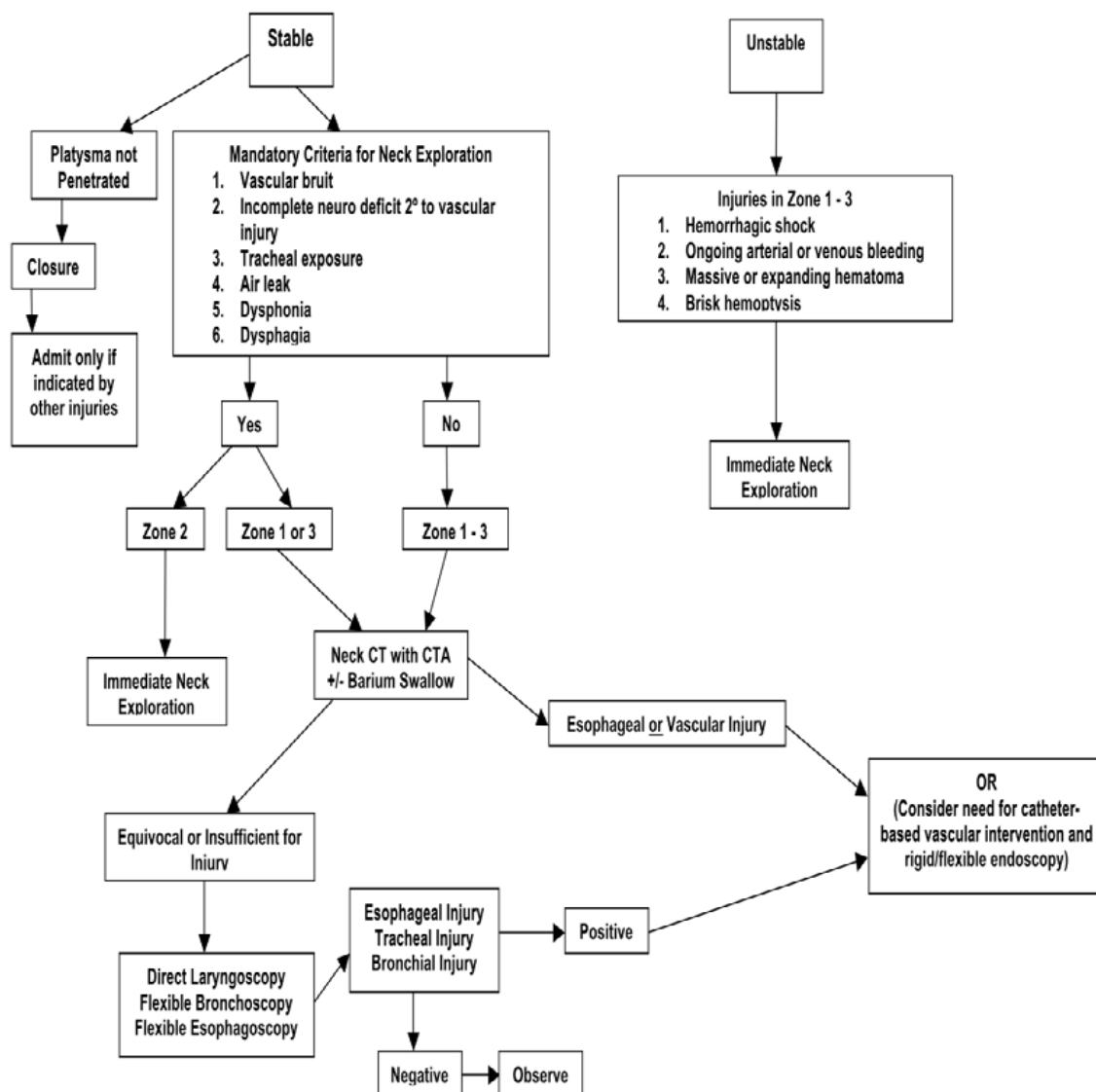
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NECK

Penetrating Neck Trauma Algorithm

PENETRATING NECK ALGORITHM



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Pediatric Thoracic Trauma

- **Thoracic trauma** is the second most common cause of death from trauma in children, with severe head injury being the most common.
- Anatomically and physiologically, children are different from adults.
- In infants and small children, the rib cage is very flexible, so significant compression can occur without associated rib fractures. Because of this, pulmonary contusion and pneumothorax are common.
- The flexible mediastinum of children allows for further shift with tension pneumothorax or hemothorax than in adults.
- Lower functional lung capacity and increased metabolic rate combined lead to more rapid decompensation when injuries are present.
- Most children with chest injuries require no more than observation or tube thoracostomy.
- If operative intervention is required, timely, effective assessment, identification, and management of injuries is directly related to positive outcomes.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Pediatric Chest Tube Placement

- Chest decompression may be necessary in patients with symptoms of pneumothorax, chylothorax, or hemothorax.
- Any chest decompression procedure is an invasive procedure and every attempt should be made to perform the procedure under sterile conditions.
- After first evacuating air with needle aspiration to stabilize clinical condition, preparations for chest tube insertion can be made.
- If infant is not in critical condition, prepare for chest tube insertion without needle evacuation.
- Give morphine 0.1 – 0.2 mg/kg for pain control and sedation.
- Place the patient in a supine position and restrain as necessary.
- Prep the anterior and lateral chest wall with betadine.
- Identify landmarks for chest tube placement.
- Chest tube should be placed at approximately the 4th intercostal space and anterior axillary line.
- Drape the area.
- If infant is not in extremely critical condition, infiltrate subcutaneous tissue and intercostal space with 1% Lidocaine.
- Using a #11 scalpel blade, make an incision through skin approximately ½ -1 cm in length at point where chest tube is to be inserted.
- Dissect through the tissue just over the rib by blunt dissection with curved hemostats.
- Avoid the underside of the adjacent rib; an artery and nerve run along it.
- Using extreme care, pressure should be applied on curved hemostats to create an entry into the pleural space.
- The hemostats should go over the top of the rib and care should be taken to avoid entering too far into the pleural cavity.
- A “pop” should be felt at this time followed by a hissing sound if the pneumothorax was under tension.
- **DO NOT** use a trochar.
- Open hemostats and insert chest tube into pleural space.
- Direct tube anteriorly and withdraw hemostat.
- Continue to advance tube but no further than the first black line approximately 3-5 cm.
- Attach chest tube to Heimlich valve which should fluctuate with each breath.
- If tube has been placed to evacuate fluid or blood, attach to suction with a collection device such as a Pleuro-Vac.
- Close the wound with a simple stitch using 3.0 silk suture.
- Tie an “air knot” 1cm above the skin stitch.
- Use enough throws to ensure that this knot will not slip.
- Tie the chest tube to the air knot, making the silk tight enough to dent the chest tube slightly.
- If needed, use additional stitches to approximate the edges of the wound.
- **AVOID PURSE STRING STITCHES** around the tube. They can cause an unsightly pucker type scar.
- Apply a dressing and tape the tube to the chest.
- Obtain chest x-ray, anterior and lateral, to establish placement of tube which should be anterior.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Resuscitative Thoracotomy

Purpose:

- Emergency Department resuscitative thoracotomy may be necessary to salvage patients who present in extremis and may otherwise die without aggressive therapy.
- Emergency Department thoracotomy is not indicated in the resuscitation of all trauma patients who present in extremis.
- The following reference is intended to be a guide and is not intended to be all-inclusive or exclusive.
- Additional patients not covered by these criteria who might benefit from Emergency Department thoracotomy will be rare and case-specific.
- The procedure is performed in conjunction with other resuscitative efforts and should not be employed in isolation.
- Under certain conditions, resuscitative efforts might best be accomplished in the Operating Room.
- An Emergency Department resuscitative thoracotomy should only be performed by general surgery PGY-3, or higher, level residents or attendings.

Indications:

- Penetrating thoracic trauma that arrive pulseless, with signs of life.
- Penetrating non-thoracic, non-cranial trauma that arrive pulseless, with signs of life
- Cardiac arrest in blunt chest or abdominal trauma after arrival in Emergency Department with an obtainable blood pressure.
- Suspected systemic air embolism.

Definitions:

- Signs of life: reactive pupils, spontaneous movement, or organized ECG activity.
- Aggressive fluid resuscitation: Lactated Ringers or Normal Saline IV bolus of 20 ml/kg x 2 and/or Packed RBC 10 ml/kg.

Procedure:

- Rapid bilateral antero-lateral Betadine or Chlorhexidine prep while thoracotomy tray opened. Thoracotomy trays are located in Trauma Bay Omni cells.
- Left antero-lateral thoracotomy incision located beneath nipple in males and in inferior breast fold in females at 5th intercostal space. Incision extends from left sternal border to anterior border of latissimus dorsi and chest entered along the superior aspect of fourth or fifth rib. Care must be taken to avoid injury to heart and lung. A right antero-lateral thoracotomy may be preferred for primary right chest wounds.
- Insert rib spreader with handle located toward table laterally.
- Examine pericardium, if tense hemopericardium present (pericardium distended with maroon discoloration) then proceed to step 7.
- If systemic air embolism is suspected or massive hemorrhage from lung parenchyma or hilum is present, then place Satinsky clamp across hilum medially.
- Retract left lung with left hand. Locate aorta by running right hand medially along posterior chest wall. Aorta located along lateral aspect of vertebral bodies and will be postero-lateral to esophagus. Dissect around aorta inferior to pulmonary hilum and apply aortic cross-clamp.
- Enter pericardium by longitudinally incising pericardium anterior and parallel to phrenic nerve. This is best accomplished by grasping pericardium with forceps and cutting with Metzenbaum scissors. Pericardial incision is carried inferiorly to diaphragmatic reflection and superiorly to level of superior pulmonary hilum. Care must be taken to avoid injury to left atrial appendage and phrenic nerve. This is best accomplished by lifting tip of scissors laterally as incision is made.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Resuscitative Thoracotomy (continued)

- Manually lift heart from pericardial sac.
- If hemopericardium present, then examine for cardiac perforation.
- Teflon pledgetted 3-0 prolene suture on a taper needle is present in thoracotomy suture pack for repair of cardiovascular wounds.
- If hemopericardium is not present, then begin open cardiac compression.
- Aortic cross-clamping, if not previously performed, is indicated if no hemodynamic response is noted.
- Additional exposure may be accomplished by extending thoracotomy incision across sternum into contralateral chest cavity.

Reference:

American College of Surgeons' Committee on Trauma: *Advanced Trauma Life Support*, ed.8, Chicago, 2008, The College.

Powell DW, Moore EE et al. Is emergency department resuscitative thoracotomy futile care for the critically injured patient requiring prehospital cardiopulmonary resuscitation? *J Am Coll Surg*. 2004 Aug; 199(2):211-5.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Emergency Department Trauma Thoracotomy Trays

TRAY #1

<u>QUANTITY</u>		<u>INSTRUMENT</u>
1	-	Finochietto – Buford Rib Spreader (10" Spread)
1	-	Tuffier Rib Spreader
1	-	Small Rib Spreader
1	-	6 ¾" Curved Mayo Scissors
2	-	9" Metzenbaum Scissors
1	-	#10 Disposable Blade
2	-	5 ½" Tissue Forceps
2	-	9" DeBakey Forceps
2	-	8" DeBakey Needle holders
4	-	Vanderbilt Clamps
1	-	Medium Satinsky Clamp
1	-	9 ½" DeBakey Aortic Clamp
8	-	5 ¼" Towel Clamps
1	-	Liston Bone Cutter Straight
1	-	Lebsche Sternal Chisel
1	-	Hammer
4	-	Surgeons Towels
1	-	Teflon Sheet 6" x 6"
10	-	Laparotomy Pads

SUTURE BAG

<u>QUANTITY</u>	<u>SUTURE</u>	<u>PRODUCT NUMBER</u>
2 pkgs	2-0 Silk (SH-CR) 30"	Ethicon #C01 6T
1 pkg	Cotton Umbilical Tape 30"	Ethicon #U-11 T 1
pkg	Teflon Felt Pledgets ¼"	Deknatel #K-705
1 pkg	Teflon Felt Pledgets"	Deknatel #L-705 6
pkgs	3-0 Prolene (SH) 36"	Ethicon #8522H
4 pkgs	3-0 Prolene (SH) with Teflon Pledgets	
4 pkgs	4-0 Prolene (RB-1) 36"	Ethicon #8557H
10 pkgs	2-0 Ticron (T-10) 30"	D & G #3174-51

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Cardiac Injury (BCI)

- Blunt cardiac injury is rarely seen in children with multisystem trauma.
- Myocardial contusions are the most common lesion.
- Wall disruption and valvular lesions may occur.
- A normal ECG does not rule out blunt cardiac injury.
- The major priority is identification of patients at risk for adverse events resulting from BCI and providing appropriate workup, monitoring and treatment.
- Patients with appropriate mechanism of injury and clinical evidence of cardiac dysfunction should cause high suspicion for BCI.
- Clinical signs of cardiac injury include chest pain, shortness of breath, muffled heart sounds, ECG changes, or abnormal ECHO.
- If rib fractures, flail segments, or pulmonary contusions are present on chest radiograph, there should be clinical suspicion of occult myocardial damage.
- BCI results from 5 possible mechanisms:
 - direct pre-cordial impact,
 - crush between sternum and spine,
 - deceleration or torsion causing a tear in the heart at a point of fixation,
 - hydraulic effect resulting in rupture from elevated intra-abdominal and caval pressure, and
 - blast injury.³
- Few clinical signs are diagnostic of BCI.
- Chest pain is the most common finding, but dyspnea, chest wall ecchymosis and rib fractures may also be present.
- Associated injuries include hemothorax, sternal fracture and great vessel injury.
- Clinical signs consistent with BCI include dysrhythmias, cardiac ischemia, low cardiac output and hypotension.⁴
- Diagnostic tests include ECG, echocardiography, and enzyme analysis. Controversy exists regarding the application of these tests. Frequency of diagnosis of BCI will be proportional to the aggressiveness with which it is sought. Appropriate workup commands achieving a balance between cost-effectiveness and information acquisition with attention to the clinical value of information gained in changing patient management.
- Guidelines for using diagnostic tests are as follows:
- Level 1 evidence supports obtaining an ECG in the emergency department for at-risk patients (described above).⁵ Using any ECG abnormality, including sinus tachycardia, bradycardia conduction delays and PAC's/PVC's, the diagnostic sensitivity of ECG is 100%.⁶
- Echocardiography is not effective as a screening tool and does not identify patients at risk for complications.⁷
- Transthoracic (TTE) or transesophageal echocardiography (TEE) should be obtained in patients with evidence of hemodynamic instability or in whom coincident coronary ischemia is suspected.
- CK and CKMB fraction analysis is **NOT** indicated in suspected BCI because associated skeletal and visceral injury creates serum CK abnormalities that contribute to an unacceptable false-positive and negative rate.⁸
- However, a body of evidence exists suggesting some value to cardiac troponin I (cTnI) or troponin T (cTnT).
- That evidence is as follows:
 - Diagnosis of BCI should not rely solely on cTnI or cTnT. ECG should be included.
 - Normal ECG and normal cTnI is 100% sensitive for BCI.⁹
 - Abnormal ECG and abnormal cTnI is 100% specific for BCI.⁹
 - cTnI is of little added benefit in patients with a markedly abnormal ECG (diagnosis is already made).¹⁰
 - Though the utility of cTnI in patients with normal ECG's is limited, cTnI obtained 4-6 hours after the injury in patients with sinus tachycardia or non-specific EKG changes or in older patients may give reassurance that the likelihood of BCI-related complications is low.¹⁰

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Cardiac Injury (BCI) (continued)

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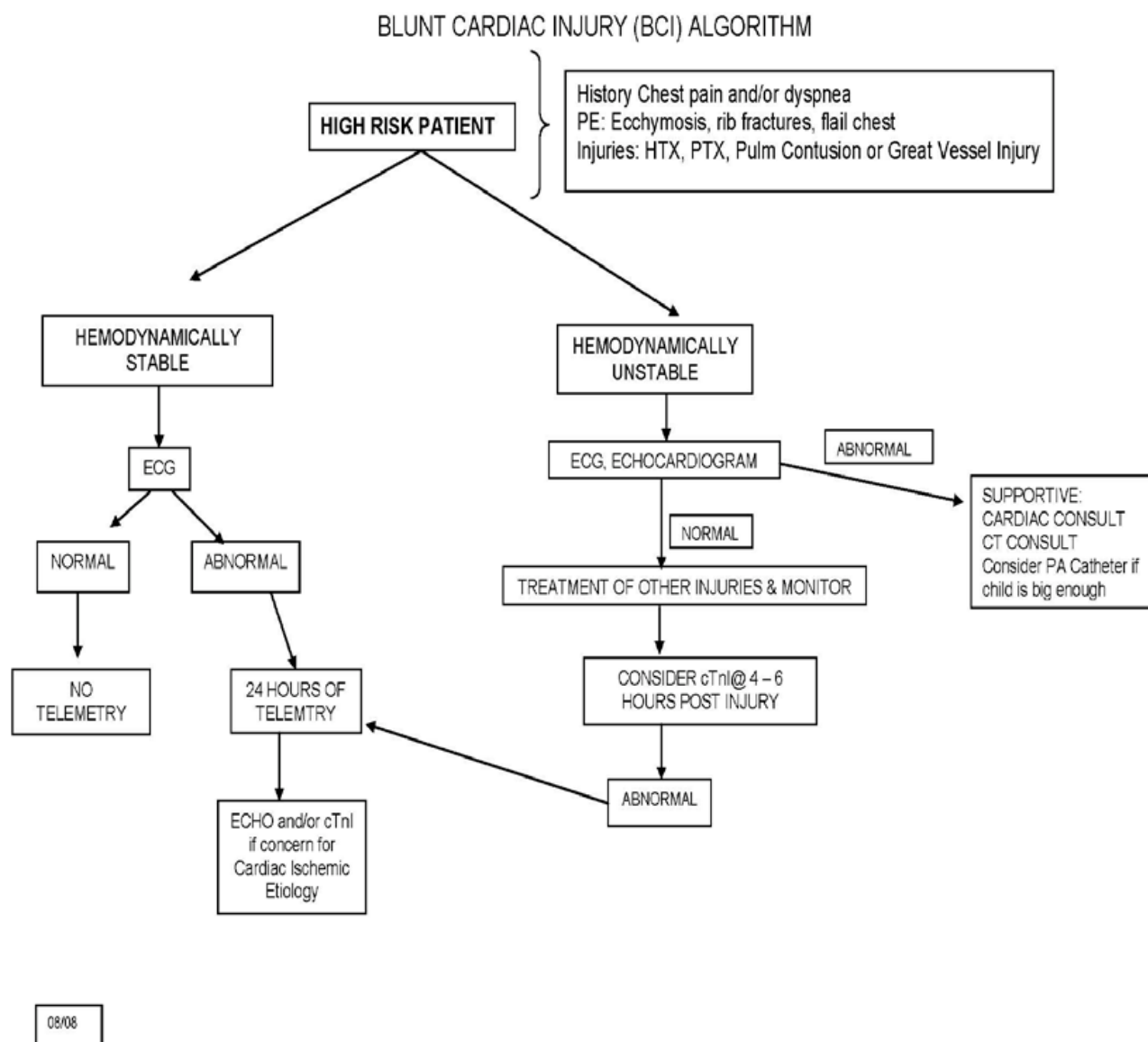
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt cardiac Injury (BCI) Algorithm



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Injury to the Thoracic Aorta

“THE DIAGNOSIS OF BLUNT INJURY TO THE THORACIC AORTA

What diagnostic test is best? Angiography, Transesophageal Echocardiography, or Computerized Axial Tomography?” Paul A. Kearney, M.D.F.A.C.S.

Background

- Blunt chest trauma with resulting aortic injury is a significant cause of death following high-speed motor vehicle collisions.¹⁻³
- The vast majority of these patients (80%-90%) expire at the collision scene.¹⁻³
- For the remaining 10%-20%, the mortality rate is high.
- Thirty percent expire within 6 hours and 40%-50% within 24 hours of injury.¹
- The recent multicenter trial by the American Association for the Surgery of Trauma reported an overall mortality of 31%, with 63% of the deaths attributable to aortic rupture.⁴
- Expeditious evaluation and timely surgical intervention are essential for patient survival.

Screening

- The goal of screening patients is to attain a zero nontherapeutic surgery rate without overlooking any significant aortic or arch vessel injury. Mechanism of injury, clinical exam, and the initial chest radiograph should reliably select patients who require further diagnostic evaluation.
- Chest radiographs demonstrating mediastinal hematoma have good sensitivity (93%) for aortic and arch vessel injury.⁵⁻⁷
- When combined with mechanism of injury, sensitivity rises to 98%.⁵
- More importantly the negative predictive value of a normal upright chest radiograph is almost 100%.
- A normal chest radiograph virtually excludes aortic and/or arch vessel injury.⁵⁻⁷
- Unfortunately, the specificity of an abnormal chest radiograph is only 10%-45%.
- Since most mediastinal hematoma originates from thoracic vascular structures other than the aorta, a definitive diagnostic test is required to establish the diagnosis of aortic injury.
- Computerized axial tomography (CT) of the chest can be used to screen patients for subsequent aortography.⁸⁻¹²
- CT is more sensitive than chest radiograph in detecting mediastinal hematoma.^{8,9}
- Screening thoracic CT is cost-effective and the negative predictive value of a normal study is 100%.⁸⁻¹⁰ Consequently, CT can be used to reduce the negative aortogram rate in patients with an abnormal chest radiograph.⁸⁻¹⁰

Diagnosis-Aortography

- Many authors advocate the liberal use of aortography based on mechanism of injury and chest radiographs.^{1,3,9,13}
- Using this approach, aortography yields positive results in only 10% of patients.
- Although angiography remains the “gold standard”, the procedure is invasive and time consuming, requires the use of intravenous contrast material and ionizing radiation, and can result in false negative or false positive results.^{3,9,13,14}
- Transporting an injured patient to the angiography suite is not without risk and interrupts the patient's ongoing evaluation, resuscitation, and treatment. On the other hand, when performed properly, aortography has a sensitivity and specificity of 99%.⁹
- Complications occur in less than 1% of patients.⁹
- Aortography remains the only study that provides detailed images of the entire thoracic aorta and the arch vessels.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Injury to the Thoracic Aorta (continued)

“THE DIAGNOSIS OF BLUNT INJURY TO THE THORACIC AORTA

What diagnostic test is best? Angiography, Transesophageal Echocardiography, or Computerized Axial Tomography?”? Paul A. Kearney, M.D.F.A.C.S.

Diagnosis-Transesophageal Echocardiography

- Transesophageal echocardiography (TEE) provides high-resolution real-time axial and longitudinal images of the aorta.¹⁴⁻¹⁸
- TEE can accurately demonstrate injury to the thoracic aorta.¹⁴⁻¹⁸
- In our hands, TEE was more sensitive (100%) and specific (98%) than aortography.^{14,15}
- TEE offers a number of advantages over aortography and CT scanning.
- The study can be performed at the bedside, eliminating transport risks.
- Concomitant diagnostic and therapeutic procedures can continue unhindered.
- TEE provides real-time images so that areas of interest can be examined repeatedly in different planes.
- Simultaneous evaluation of cardiac pathology and function can also be obtained.
- If urgent surgical intervention is indicated for other injuries, TEE can be performed in the operating room without delaying TEE or the surgical intervention.
- The study can be performed more rapidly than aortography making TEE ideal for the evaluation of the unstable trauma patient with a number of diagnostic and treatment priorities.
- Unfortunately, TEE requires the expertise of a well-trained and interested echocardiographer.
- Airway compromise, esophageal pathology, and unstable cervical spine fractures are contraindications to TEE.
- The depth and extent of injury are frequently difficult to determine particularly when atherosclerotic disease diminishes the sensitivity of the examination.
- In our experience, this has resulted in nontherapeutic thoracotomy.¹⁴
- There are blind spots related to the tracheal air column and the arch vessels simply cannot be imaged.

Diagnosis-Computerized Axial Tomography (CT)

- CT scanning technology is advancing rapidly. Perhaps the most promising technology for a fast, accurate, and less-invasive diagnostic test for detecting injuries to the thoracic aorta and arch vessels is the newer generation helical CT scanner.^{9,19-21}
- Rather than providing indirect evidence of aortic injury (detection of mediastinal hematoma), the helical scanners can provide direct evidence of aortic injury obviating the need for confirmatory tests.
- Sensitivity, specificity, and accuracy depend heavily on the technical skill of CT performance and the interpretative expertise.⁹
- Confirmatory angiography or TEE must be performed for indeterminate or equivocal CT scan results.
- Although CT can be performed more rapidly than angiography, transport of the injured patient to CT is not without risk and interrupts the patient's ongoing evaluation, resuscitation, and treatment.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Injury to the Thoracic Aorta (continued)

“THE DIAGNOSIS OF BLUNT INJURY TO THE THORACIC AORTA

What diagnostic test is best? Angiography, Transesophageal Echocardiography, or Computerized Axial Tomography?” Paul A. Kearney, M.D.F.A.C.S.

Summary

- Patients with suspected blunt injury to the thoracic aorta are a challenge for the trauma surgeon.
- Multisystem trauma, critical illness, and hemodynamic instability in this patient group result in diagnostic and treatment dilemmas.
- We employ a practical, evidence-based algorithm for the screening and diagnosis of injury to the thoracic aorta.
- Both mechanism of injury and an abnormal mediastinum on chest radiograph are required to trigger a diagnostic evaluation.
- Every attempt is made to obtain an “upright” AP chest radiograph to minimize distortion and magnification. However, this is not possible in all patients.
- Widening of the mediastinum alone is neither sensitive nor specific for mediastinal hematoma.^{5,6,7}
- Instead, we employ the criteria for mediastinal hematoma (abnormal mediastinal silhouette) as defined by Mirvis and Ayella.^{6,7}
- No further diagnostic evaluation is undertaken in patients with a normal chest radiograph, unless there are compelling physical findings that suggest aortic or arch vessel injury (i.e. Pulse deficit, unequal blood pressure/pulse measurement, unexplained hemodynamic instability, unexplained neurologic deficit).
- Thoracic CT scan using the aortic guideline is performed.^{9,19}
- A negative scan yields observation.
- A scan positive for aortic injury prompts surgical intervention or appropriate non-operative management.
- Patients with indeterminate scans undergo angiography or TEE to establish or exclude the diagnosis.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Injury to the Thoracic Aorta (continued)

“THE DIAGNOSIS OF BLUNT INJURY TO THE THORACIC AORTA

What diagnostic test is best? Angiography, Transesophageal Echocardiography, or Computerized Axial Tomography?” Paul A. Kearney, M.D.F.A.C.S.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Blunt Thoracic Injury with Suspected Injury to the Thoracic Aorta or Arch Vessels

- Patients who sustain blunt thoracic trauma are at risk for injury to the heart and great vessels.
- Patients should be selected for additional diagnostic studies based on mechanism of injury and evidence of mediastinal hematoma on chest radiograph.
- Every attempt is made to obtain an “upright” AP chest radiograph to minimize distortion and magnification. However, this is not possible in all patients.
- Widening of the mediastinum alone is neither sensitive nor specific for mediastinal hematoma (5,6,7).
- Instead, we employ the criteria for mediastinal hematoma (abnormal mediastinal silhouette) as defined by Mirvis and Ayella (6,7).
- No further diagnostic evaluation is undertaken in patients with a normal chest radiograph, unless there are compelling physical findings that suggest aortic or arch vessel injury (i.e. Pulse deficit, unequal blood pressure/pulse measurement, unexplained hemodynamic instability, unexplained neurologic deficit).

MECHANISM OF INJURY (History of significant deceleration)

- High-speed MVC (>30-40mph)
- Substantial vehicle deformity or associated fatalities
- Unrestrained and/or ejection from vehicle
- Pedestrian struck by vehicle
- Falls > 10 feet
- Hemodynamic instability

INJURY

- Sternal and/or scapular fracture
- Multiple rib fractures and/or flail chest

CHEST RADIOGRAPH (One or more of the following)

- Upper mediastinal widening
- Indistinct aortic contour
- Obscuration of the aortopulmonary window
- Widened left paraspinal stripe
- Deviation of the NG tube or trachea to the right
- Depression of the left mainstem bronchus
- Left apical cap (apical capping)

* Isolated fractures of the first and second ribs without evidence of mediastinal hematoma do not correlate with aortic or arch vessel injury and are not an indication for further imaging.

References:

1. Mirvis SE, Bidwell JK, Buddemeyer EU, et al. Imaging diagnosis of traumatic aortic rupture: A review and experience at a major trauma center. Invest Radiol 1987;22:187-190.
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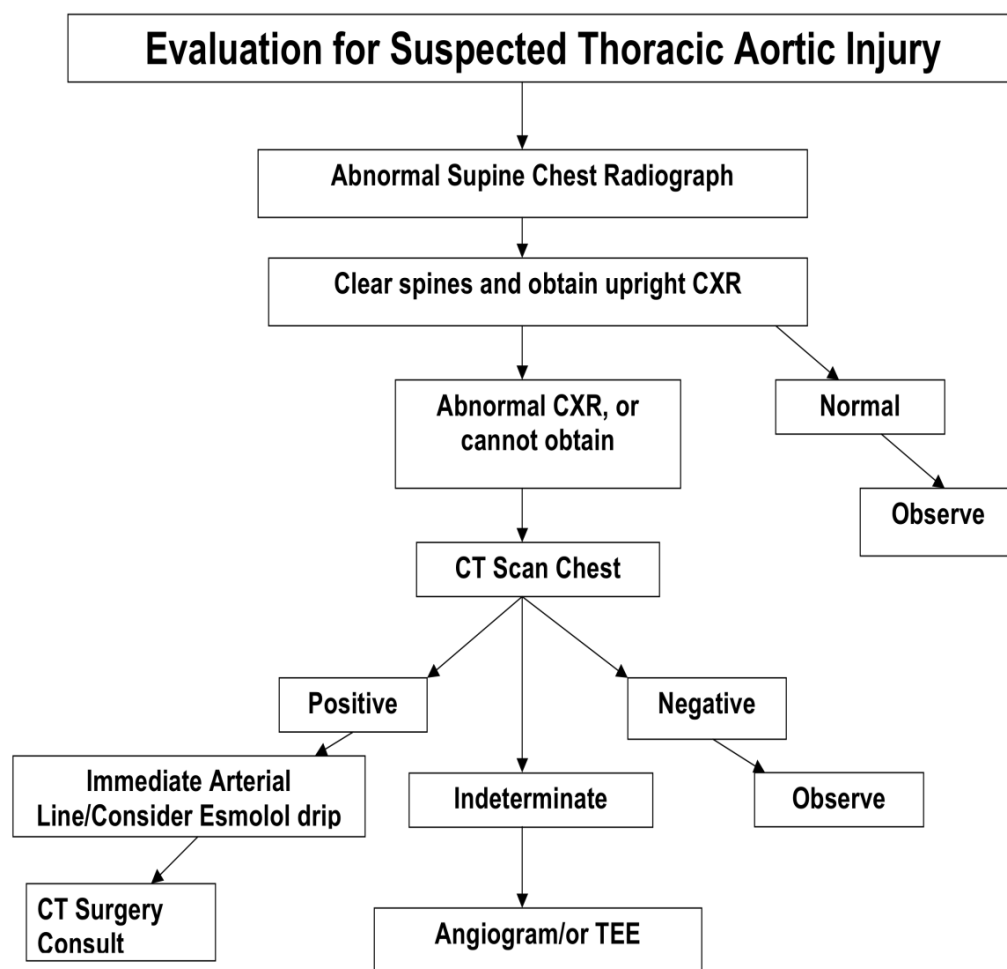
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Suspected Thoracic Aortic Injury Algorithm



****These injuries are rare in children**

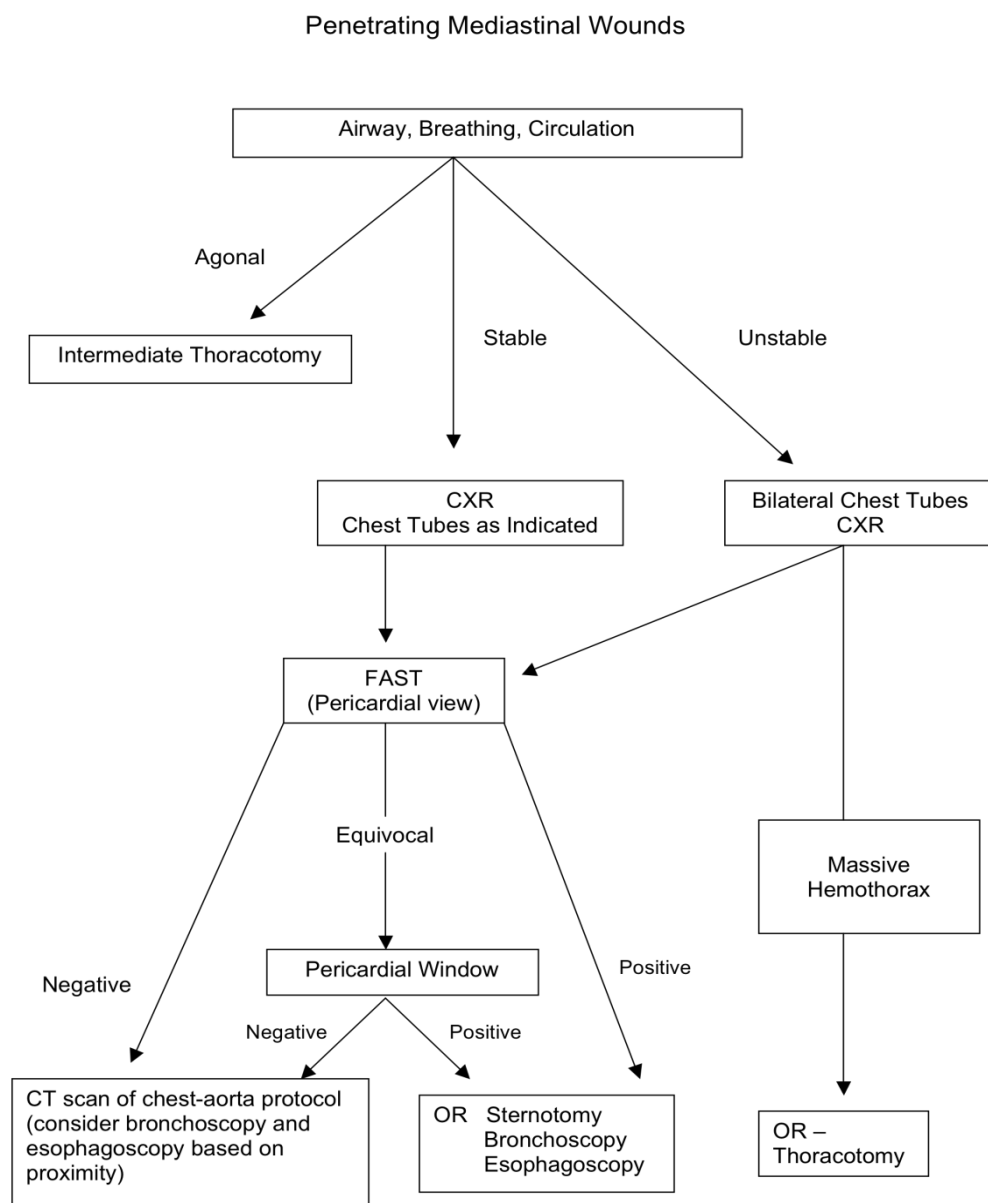
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Penetrating Mediastinal Wounds Algorithm



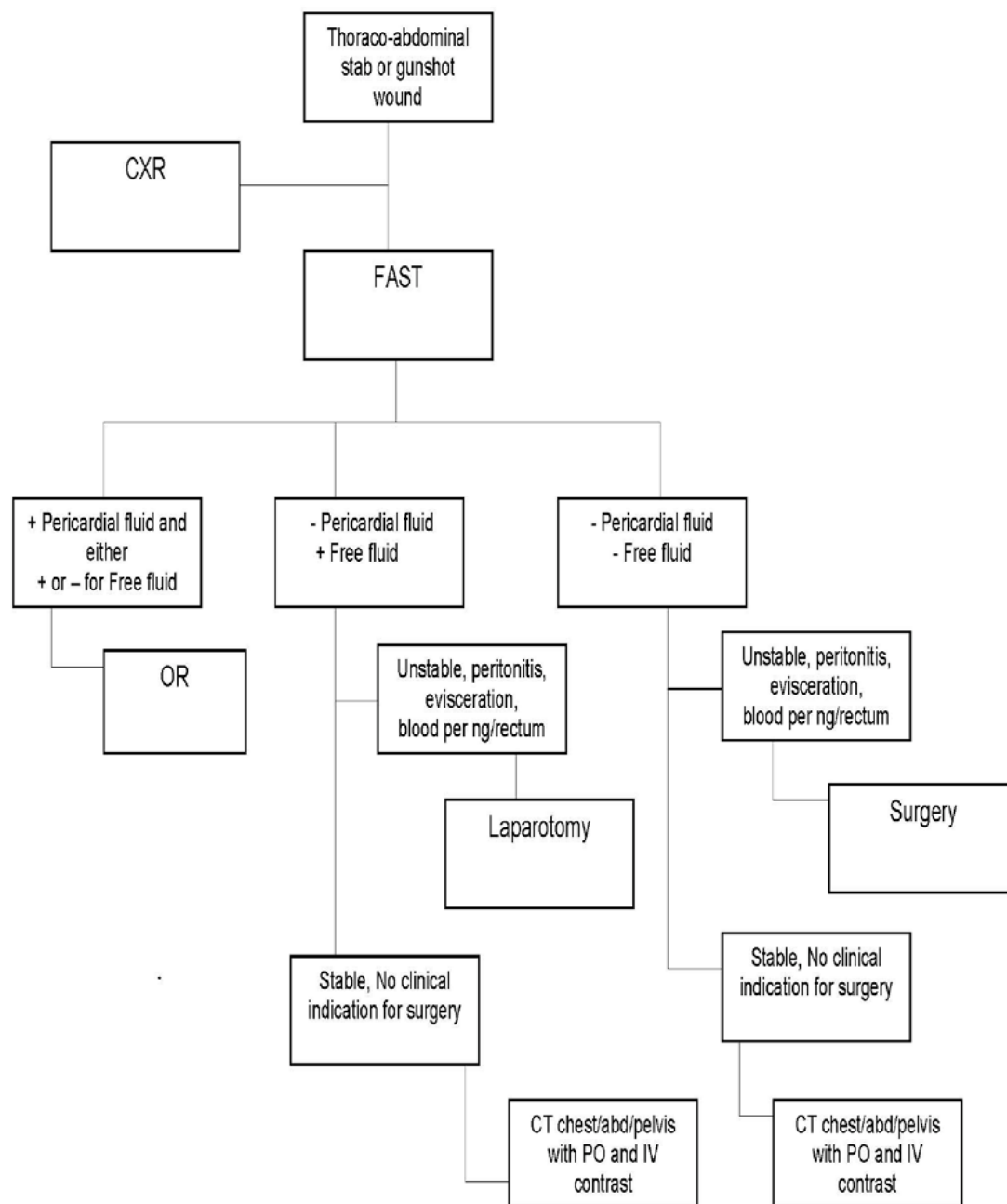
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: THORACIC

Thoraco-Abdominal Penetrating Wound Algorithm



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury

(Revised March 2016)

- **Scope**
 - **Condition** – Isolated spleen and/or liver lacerations incurred as the result of blunt abdominal trauma.
 - **Guideline categories** – Non-operative management, blunt trauma.
 - **Guideline objective** – To provide consistency in the evaluation and management of isolated solid organ injuries in our pediatric blunt trauma patients.
 - **Target Population** – Pediatric trauma patients with blunt abdominal trauma who are evaluated by the Pediatric Trauma Service.
- **Methodology**
 - The guideline developers performed literature searches of peer-reviewed medical journals. The major applicable articles were collected and reviewed.
 - We reviewed the clinical practice guidelines that have been shared by other ACS-Verified Level I Pediatric Trauma Hospitals.
- **Inclusion Criteria**
 - Single system abdominal trauma.
 - Multisystem injuries if patient has normal vital signs for age and is hemodynamically stable.
 - Stable vital signs after modest volume transfusions.
- **Exclusion Criteria**
 - Hemodynamic instability.
 - Multiple major trauma injuries when the spleen or liver injury does not direct the patient's plan of care.
 - Patients with systemic illnesses such as mononucleosis or blood clotting abnormalities.
- **Abdominal Trauma**
 - The vast majority of pediatric abdominal injuries are due to blunt trauma.
 - The spleen and liver are the abdominal organs which are most commonly injured in pediatric blunt abdominal trauma patients.
 - The non-operative treatment of isolated spleen and liver injuries in hemodynamically stable children is now standard practice.
 - Early recognition and standardized management are associated with excellent outcomes.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

Spleen Injuries

- The spleen is the most commonly injured abdominal organ in children.
- Injured patients may demonstrate left upper quadrant pain, left shoulder pain, abdominal distension, and ecchymosis/abrasions to the left upper quadrant.
- The American Association for the Surgery of Trauma (AAST) spleen injury scale:

AAST Spleen Injury Scale (1994 revision)

Grade*	Type of Injury	Description of injury
I	Hematoma Laceration	Subcapsular, <10% surface area Capsular tear, <1cm parenchymal depth
II	Hematoma Laceration	Subcapsular, 10%-50% surface area intraparenchymal, <5 cm in diameter Capsular tear, 1-3cm parenchymal depth that does not involve a trabecular vessel
III	Hematoma Laceration	Subcapsular, >50% surface area or expanding; ruptured subcapsular or parenchymal hematoma; intraparenchymal hematoma > 5 cm or expanding >3 cm parenchymal depth or involving trabecular vessels
IV	Laceration	Laceration involving segmental or hilar vessels producing major devascularization (>25% of spleen)
V	Laceration Vascular	Completely shattered spleen Hilar vascular injury with devascularized spleen

*Advance one grade for multiple injuries up to grade III.

Liver Injuries

- The liver is the second most commonly injured abdominal organ in children.
- The right lobe is injured slightly more frequently than the left.
- Injured patients may demonstrate abdominal pain and right shoulder pain.
- Elevated transaminases are highly suggestive of a liver injury.
- AAST liver injury scale:

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

Liver Injuries

- The liver is the second most commonly injured abdominal organ in children.
- The right lobe is injured slightly more frequently than the left.
- Injured patients may demonstrate abdominal pain and right shoulder pain.
- Elevated transaminases are highly suggestive of a liver injury.
- AAST liver injury scale:

II.	G	Type of Injury	Description of injury
I		Hematoma	Subcapsular, <10% surface area
		Laceration	Capsular tear, <1cm parenchymal depth
II		Hematoma	Subcapsular, 10% to 50% surface area intraparenchymal <10 cm in diameter
		Laceration	Capsular tear 1-3 parenchymal depth, <10 cm in length
III		Hematoma	Subcapsular, >50% surface area of ruptured subcapsular or parenchymal hematoma; intraparenchymal hematoma > 10 cm or expanding
		Laceration	>3 cm parenchymal depth
IV		Laceration	Parenchymal disruption involving 25% to 75% hepatic lobe or 1-3 Couinaud's segments
V		Laceration	Parenchymal disruption involving >75% of hepatic lobe or >3 Couinaud's segments within a single lobe
		Vascular	Juxtahepatic venous injuries; ie, retrohepatic vena cava/central major hepatic veins
VI		Vascular	Hepatic avulsion

*Advance one grade for multiple injuries up to grade III.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

Management Recommendations

Evaluation

- All trauma patients with blunt abdominal injuries should have a thorough physical examination.
- Serial abdominal exams should be performed by an experienced member of the Pediatric Trauma Service.
- **The decision not to operate is always a surgical decision.**

Imaging

- Abdominal CT scan with IV contrast.
- Radiologist reads CT and grades the injury using the AAST Injury Grading Scale.

Admission to the Children's Hospital

- Pediatric trauma patients with acute injuries should be admitted to the Pediatric Trauma Service.
- Hemodynamically stable patients can be admitted to the floor.
- Hemodynamically unstable patient or patients with multiple injuries should be admitted to the PICU.

Labs

- A baseline hemogram and a type & screen should be obtained as part of the initial evaluation.
- Serial hemoglobin & hematocrit (H&H) should be checked every 6 hours until they plateau.

Diet

- Acutely injured patients should be kept NPO at least until the first recheck of the H&H.
- Patients can be started on a clear liquid diet when deemed hemodynamically stable.
- Thereafter, the diet may be advanced as tolerated.

Activity

- Acutely injured patients should initially be kept on strict bed rest.
- Once the patient is hemodynamically stable, he or she may get out of bed and ambulate to the bathroom.
- The patient may be allowed to ambulate in the halls when his or her abdominal pain has subsided.

Discharge criteria

- Hemodynamically stable
- Adequate oral intake, and IV fluids discontinued
- Ambulatory, independent to bathroom
- Pain well controlled on oral medication

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

Discharge Instructions

- F/U 2 weeks in PDS clinic
- Diet as tolerated
- Activities
 - Quiet activities at home for the first 7 days. Avoid rough play with family, friends and pets.
 - Return to school after 1 week if no longer requiring narcotic pain medication
 - Avoid gym class, rough play, and contact sports until cleared by physician
- Analgesia
 - Tylenol for pain
 - Limit narcotic prescription to 5-7 days and encourage bowel regimen
- Contact Pediatric Trauma Surgery Office for
 - Fever over 101°F for 6 to 12 hours and does not resolve with a dose of Tylenol
 - Increasing pain, pallor, dizziness, difficulty breathing, vomiting, worsening shoulder pain, jaundice, gastrointestinal bleeding, or black tarry stools

Treatment	CT Grade of Solid Organ Injury (AAST Organ Injury Severity Scale)			
	I	II	III	IV
Labs	Hemoglobin & Hematocrit every 6 hours until plateau			
ICU Admit	If hemodynamically unstable			
OOB to chair & bathroom	When hemodynamically stable			
Floor activities	When hemodynamically stable and pain free			
Hospital stay* (post-injury day)	1	2	3	4
Activity restriction** (weeks)	3	4	5	6
Follow-up visits (weeks)	2 weeks in PDS clinic			
*Consider early discharge if hemodynamically stable, tolerating regular diet, and without abdominal pain.				
**Return to school and normal childhood activities after one week. No rough play or contact sports until cleared by a physician.				

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

Failure of Non-Operative Management

Indications for operative treatment:

- Ongoing hemorrhage
- Hemodynamic instability refractory to volume resuscitation
- Blood loss >1/2 patient's blood volume (40 mL/kg)
- Major multiple trauma when the spleen or liver injury does not primarily direct the patient's plan of care

Operative management

- Most spleen and liver injuries are amenable to simple methods of hemostasis
- Manual compression, direct suture, topical hemostatic agents

Interventional radiology

- The role of IR in the management of pediatric solid organ is evolving.
- IR consult may be requested at discretion of the Attending Pediatric Trauma surgeon.

Post-splenectomy sepsis prophylaxis

- **Vaccinations**
 - Ideally, immunizations should be given 2 weeks after emergent splenectomy. Parents should be instructed to follow-up with their pediatrician about these vaccines.
 - Pneumococcal 0.5cc SC/IM
 - Meningococcal 0.5cc SC/IM
 - Haemophilus influenza type b (Hib) 0.5cc SC/IM
- **Antibiotic prophylaxis**
 - Patients must be started on antimicrobial prophylaxis upon discharge.
 - It is recommended that antimicrobial prophylaxis be given at least until 5 years of age and for at least 1 year following splenectomy.
 - Oral Penicillin VK 125 mg PO BID for children younger than 3 years
250 mg PO BID for children 3 years and older
 - If allergic to penicillin, erythromycin 250 mg BID can be given

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Spleen and Liver: Blunt Injury (continued)

(Revised March 2016)

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Pediatric Abdominal Examination

The Role of CT scan, FAST (ultrasound), & DPL

Introduction

- Most abdominal injuries can be managed non-operatively.
- Awareness of anatomic patterns and physiologic responses has made it possible for most children with abdominal solid organ injury to be managed without surgical intervention.
- Screening capabilities of Computed tomography (CT) scanners, the focused abdominal sonograph for trauma, and less invasive percutaneous, angiographic, and endoscopic interventions allow non-surgeon members of the trauma team to play an active role in pediatric trauma care.
- The goal of the abdominal evaluation of multi-system trauma patients is the safe, accurate, and timely determination of the presence or absence of intra-abdominal injury, particularly those requiring surgical intervention.
- Physical examination can be unreliable in polytrauma patients, particularly when the abdominal examination and/or the level of consciousness have been altered by central system trauma, distracting pain, or alcohol/drugs.
- CT scan and abdominal sonography (FAST) are the most frequently employed diagnostic studies used for the abdominal examination of pediatric trauma patients.
- Each study has advantages and disadvantages.

CT Scans

- Abdominopelvic CT scanning for blunt abdominal trauma has a sensitivity of 96%, a specificity of 98%, and an accuracy of 97%.
- CT is organ specific, allowing the identification and grading of injured organs and the quantification of intraperitoneal fluid or blood. This allows for non-operative management of stable patients, thereby reducing the rate of non-therapeutic laparotomy.
- Extra peritoneal injuries (thorax, retroperitoneum, and pelvis) can be identified and graded.
- Sensitivity and specificity depend on quality of the scan and skill of the interpreter.
- Contrast aspiration and allergy may occur.
- Bowel and pancreatic injuries may be missed.
- CT is time consuming, and transport to the scanner interrupts other diagnostic and therapeutic interventions.
- CT is limited to hemodynamically stable patients.
- CT scan is the mainstay in the abdominal evaluation of the stable blunt trauma victim.
- CT has a role to play in the evaluation of penetrating torso injury.

FAST

- Use of ultrasound or FAST (Focused Assessment with Sonography in Trauma) in abdominal trauma is noninvasive, rapid (3 to 5 minutes), and relatively inexpensive.
- Performed at the bedside
- Does not interfere with other diagnostic and therapeutic interventions, and can be repeated as needed.
- The primary goal of FAST is to detect free fluid in Morison's Pouch, the pelvis, peri-splenic region and the pericardium.
- Extra peritoneal structures can be imaged (thorax, pericardium, retroperitoneum).
- The relative lack of subcutaneous tissue in children, compared to an adult, makes this study easy to perform and limits the child's exposure to radiation.
- The technique can be easily learned and performed by the treating surgeon. Sensitivity, specificity, and accuracy of US clearly improve with experience of the examiner.
- US are much less sensitive for specific organ injury (liver, spleen, etc.).
- US can miss bowel injuries.
- Sensitivity, specificity, and accuracy also depend on image quality and experience of the examiner.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Pediatric Abdominal Examination

The Role of CT scan, FAST (ultrasound), & DPL (continued)

DPL

- DPL (Diagnostic Peritoneal Lavage) procedure is of limited use in pediatric patients.
- DPL is rapid (< 30 minutes) and safe (< 2% complication rate) with well-established standards for surgical intervention on the basis of cell counts or the aspiration of free blood after lavage.
- The procedure is invasive and can be performed at the bedside while other diagnostic and therapeutic interventions proceed.
- DPL cannot evaluate extraperitoneal structures (thorax, retroperitoneum, and pelvis) and must be supplemented with other diagnostic procedures.
- DPL is not organ specific and the presence of pelvic fractures also diminishes specificity.
- A positive study may lead to a non-therapeutic laparotomy for a trivial injury.
- DPL has been replaced by FAST in the unstable patient and has a limited role in the stable patient.

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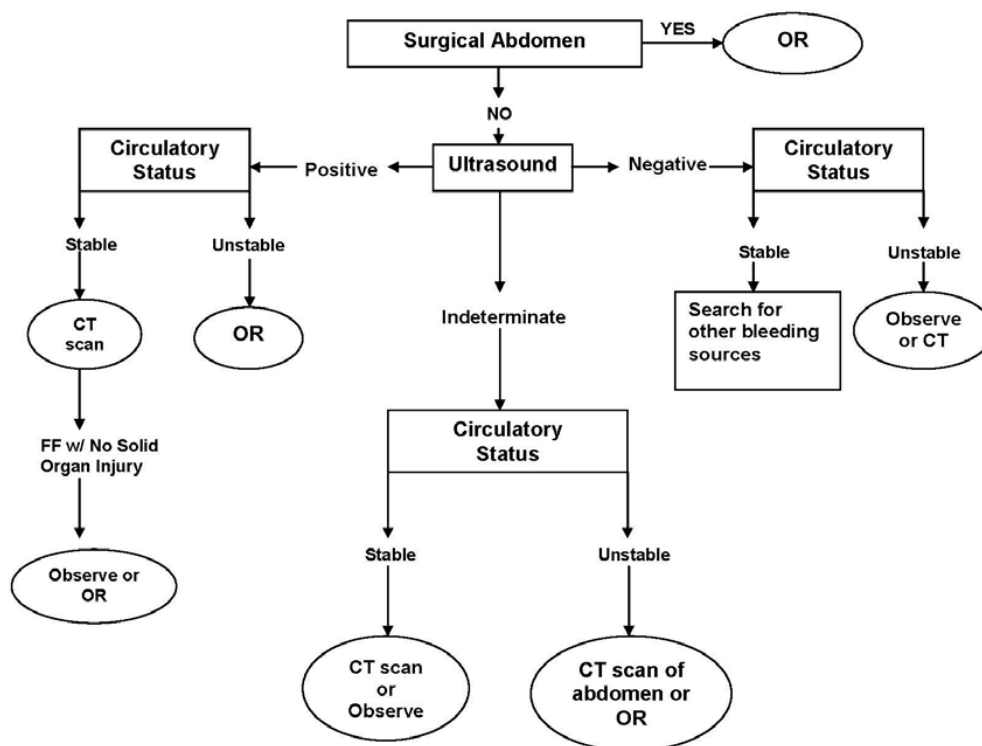
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Blunt Abdominal Trauma Algorithm

Pediatric Blunt Abdominal Trauma Algorithm



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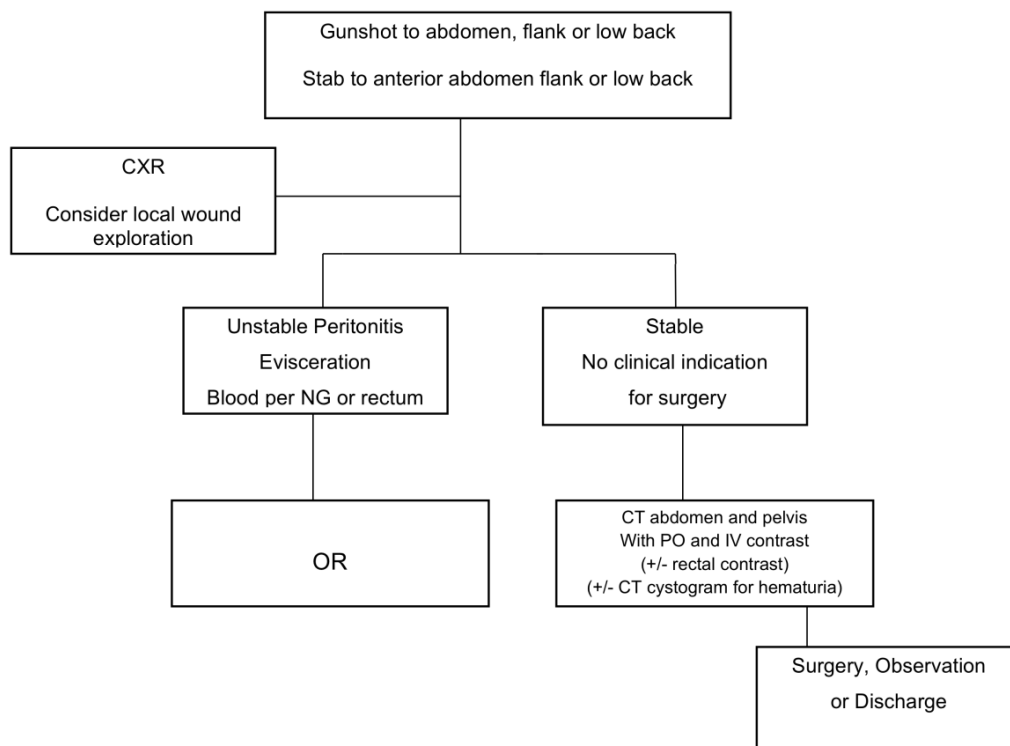
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Penetrating Abdominal Trauma Algorithm (Includes Flank & Low Back)

Penetrating Abdominal Trauma



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Genitourinary Trauma

Reviewed/Revised: November 2019, March 2020

Introduction

- Trauma to the genitourinary (GU) tract is present in approximately 10% of all injuries.
- Children are at increased risk of renal injury from blunt trauma due to decreased perirenal fat, weaker abdominal musculature, and less ossified thoracic cage.

Urethral Trauma

- Urethral injuries are usually due to blunt trauma and are associated with pelvic fracture or straddle-type injuries.
- Injury of the bulbar urethra is most common. Injuries of the female urethra are rare.
- Physical findings of blood at the meatus, perineal hematoma or extensive laceration, a high riding prostate, or a large hematoma found on rectal exam mandates a retrograde urethrogram **prior to insertion of a Foley catheter**.
- Management requires suprapubic urinary diversion or endoscopic assisted placement of Foley catheter.

Bladder Trauma

- Bladder rupture can result from penetrating or blunt trauma.
- Extraperitoneal bladder rupture is commonly associated with pelvic fracture.
- Intraperitoneal bladder rupture is a result of blunt lower abdominal force on a full bladder.
- Classic physical findings of bladder rupture include suprapubic pain, hematuria, and inability to void.
- If bladder trauma suspected, a CT cystogram should be obtained.
- Extraperitoneal bladder rupture is generally managed with Foley catheter drainage, and intraperitoneal bladder rupture requires immediate surgical intervention.

Kidney and Ureter Trauma

- The kidneys are the most commonly injured organs in the GU tract.
- **Gross hematuria** mandates further workup of the GU tract – prompt surgical consultation is warranted.
- If renal trauma suspected, CT abdomen and pelvis with IV contrast and delayed images (excretory phase) should be obtained.
- Management depends on hemodynamic stability of the patient and grade of injury.
- Most renal injuries do not require an operation in a stable patient.
- Incidental intra-operative finding of perinephric hematoma should be explored if the mechanism is penetrating trauma.
- Most ureteral injuries are penetrating. They require surgical intervention.
- Ureteral reconstruction includes debridement of devitalized tissue and tension-free anastomosis with absorbable suture in a spatulated fashion over a double J stent.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Genitourinary Trauma (continued)

Reviewed/Revised: November 2019, March 2020

AAST Kidney Injury Scoring Scale

<http://www.aast.org/library/traumatools/injuryscoringscales.aspx>

Kidney injury scale		
Grade*	Type of injury	Description of injury
I	Contusion	Microscopic or gross hematuria, urologic studies normal
	Hematoma	Subcapsular, nonexpanding without parenchymal laceration
II	Hematoma	Nonexpanding perirenal hematoma confirmed to renal retroperitoneum
	Laceration	<1.0 cm parenchymal depth of renal cortex without urinary extravagation
III	Laceration	<1.0 cm parenchymal depth of renal cortex without collecting system rupture or urinary extravagation
IV	Laceration	Parenchymal laceration extending through renal cortex, medulla, and collecting system
	Vascular	Main renal artery or vein injury with contained hemorrhage
V	Laceration	Completely shattered kidney
	Vascular	Avulsion of renal hilum which devascularizes kidney

*Advance one grade for bilateral injuries up to grade III

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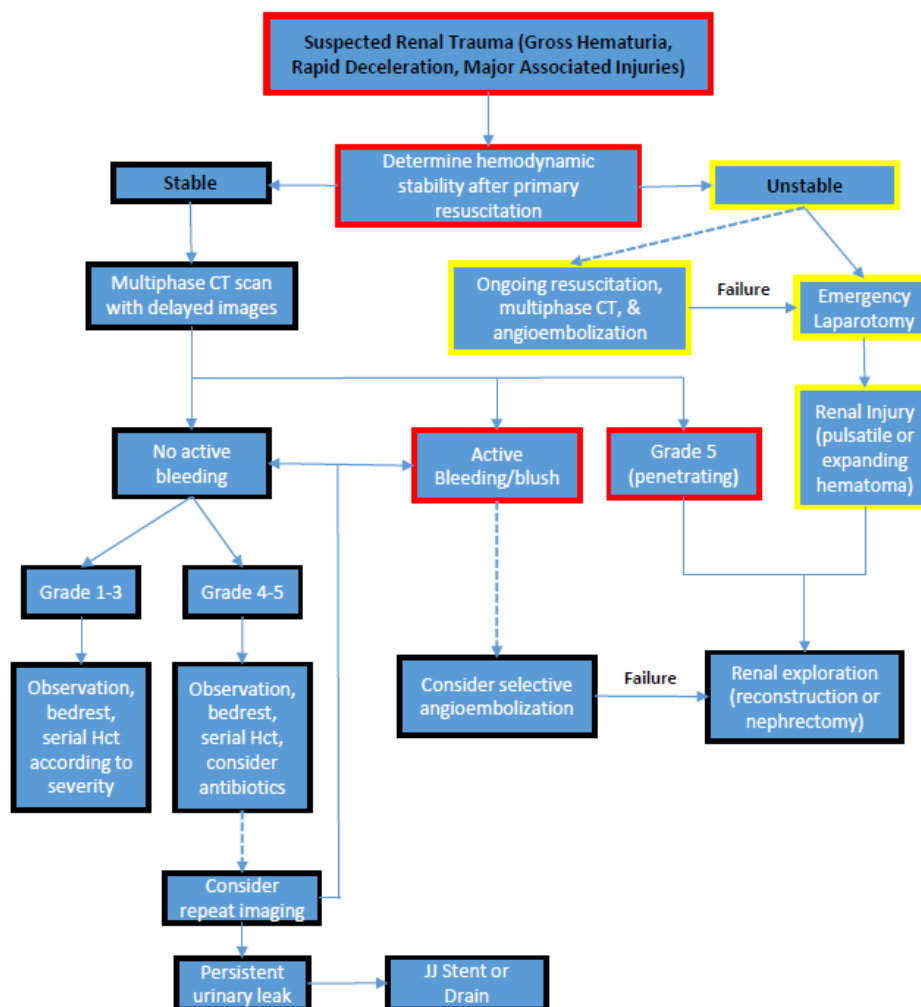
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Genitourinary Trauma (continued)

Reviewed/Revised: November 2019, March 2020



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Genitourinary Trauma (continued)

Reviewed/Revised: November 2019, March 2020

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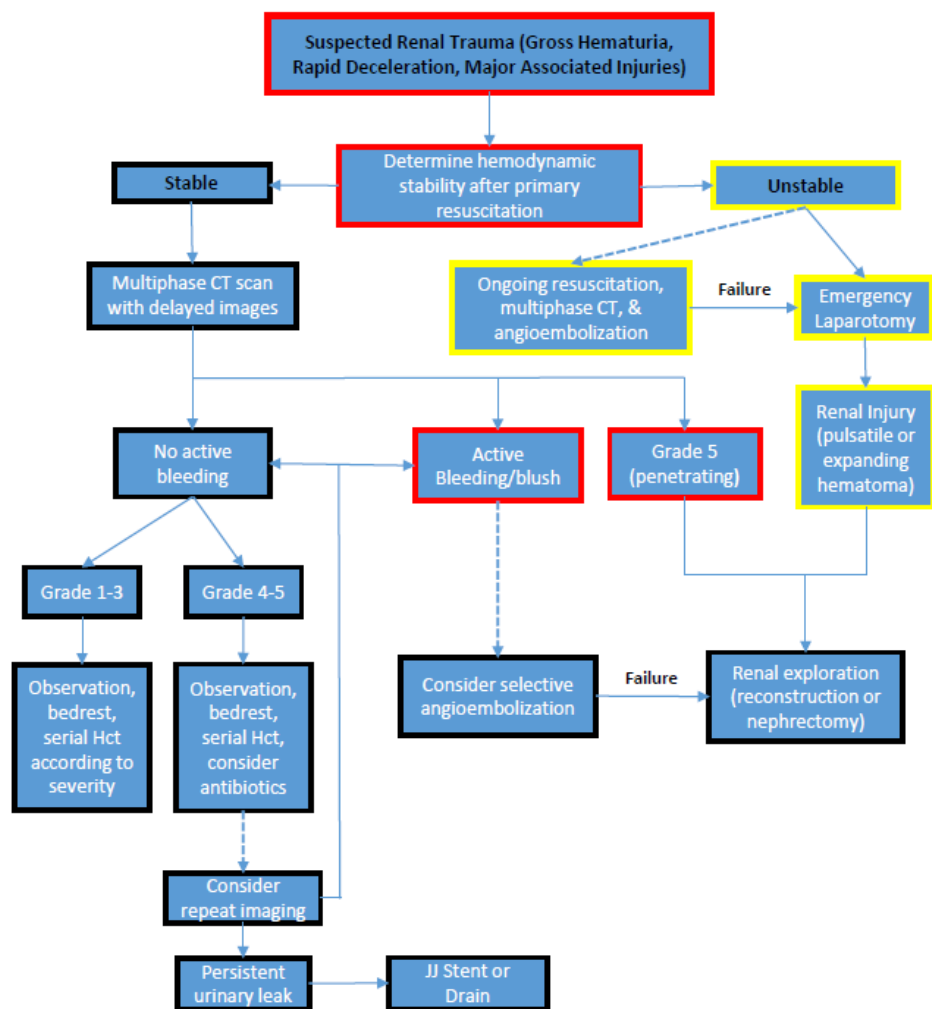
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Genitourinary Trauma Algorithm

Reviewed/Revised: November 2019, March 2020



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Abdominal Compartment Syndrome

Intra-Abdominal Hypertension and Abdominal Compartment Syndrome Assessment and Monitoring

- The abdominal cavity can be considered a single cavity and change in the volume of contents will elevate abdominal pressures.
- Abdominal Compartment Syndrome (ACS) is a condition in which the increased pressure in the anatomic space results in organ dysfunction.
- Undetected increases in intra-abdominal pressure (IAP) can be life threatening.
- Identification of patients at risk is essential to prevent hemodynamic and respiratory compromise from undetected ACS.
- ACS is preceded by intra-abdominal hypertension (IAH) which is defined as an IAP of ≥ 12 mmHg.
- Organ dysfunction also may precede development of ACS.
- Abdominal Compartment Syndrome (ACS) is defined by Intra-abdominal pressure (IAP) ≥ 20 mmHg (with or without an Abdominal Perfusion Pressure < 60 mmHg) in a minimum of three standardized measurements taken four to six hours apart plus at least one new end-organ failure.
- Experience with abdominal decompression in children is limited, but in some cases this has been shown to decrease oxygen requirements and high airway pressures, so it may be a consideration.

Etiology of Increased IAP :

Acute

- Intra-abdominal Hemorrhage
 - Post resuscitation visceral hemorrhage
 - Hypothermic or consumptive coagulopathic bleeding
 - Rupture of abdominal aortic or visceral artery aneurysm
 - Post traumatic intra-abdominal hemorrhage
- Retroperitoneal Hemorrhage
 - Blunt trauma (i.e., pelvic fracture, kidney laceration)
 - Hemorrhagic Pancreatitis
 - Ruptured abdominal aortic aneurysm
- Accumulation of Fluid/Visceral Swelling
 - Septic shock
 - Peritonitis (i.e., perforated viscus, postoperative abscess)
 - Paralytic ileus
 - Bowel obstruction
 - Mesenteric venous thrombosis
 - Mesenteric ischemia/reperfusion
 - Pancreatitis
- Other
 - Tension pneumoperitoneum
 - Intra-abdominal packing

Chronic

- Ascites
- Pregnancy
- Large abdominal tumor/ovarian mass

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Abdominal Compartment Syndrome (continued)

Physiological Consequences

Cardiopulmonary Effects

- Pulmonary
 - Decreased compliance (see higher peak airway pressures)
 - Increased inspiratory pressure
 - Hypercarbia (decreased ventilation)
 - Hypovolemia (compresses SVC, decreasing preload = decreased CO)
 - Respiratory Acidosis (decreased FRC and TV = decreased ventilation)
 - Increased pulmonary vascular resistance (increases pulmonary shunt and increases work on heart to generate same CO)
- Cardiac
 - Decreased ventricular compliance (requires increased preload for same CO)
 - Increased CVP, PWP, PAP (Falsely elevated when euvolemic!)
 - Diminished venous return (compresses SVC/IVC)
 - Tachycardia (decreased preload, need increased HR to keep same CO)
 - Decreased cardiac output (seen when compensatory mechanisms fail)
 - Increased SVR
 - Venous stasis may increase risk of DVT/PE

Renal Effects

- Increased IAP compresses the inferior vena cava and renal veins.
- Direct extrinsic pressure on the kidney creates a circumferential constriction.
- The combination of direct trauma, hypoperfusion and venous backpressure can create an intra-renal compartment syndrome. As a consequence, urine output diminishes.

Neurological Effects

- By increasing ITP, increased IAP impedes venous outflow from the cerebral circulation.
- Increased ICP
- Decreased CPP

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Abdominal Compartment Syndrome (continued)

Gastrointestinal/Hepatic/Wound Healing

- Increased abdominal pressure reduces blood flow to the abdominal viscera.
- Decreased celiac and portal blood flow
- Decreased mucosal blood flow
- Decreased fascial blood flow- (increases risk of wound infection and dehiscence)
- Increased bacterial translocation

Multiple Compartment Syndrome

- $\uparrow$ IAP $\rightarrow$ $\uparrow$ ITP $\rightarrow$ $\uparrow$ CVP $\rightarrow$ $\uparrow$ ICP $\rightarrow$ $\downarrow$ CPP
- When ICP remains elevated despite maximal medical maneuvers, give consideration to MCS which may require decompressive laparotomy and/or decompressive craniotomy.

Grades of Intra-abdominal Hypertension (mmHg)

- Normal = 0 – 11
- Grade I = 12 – 15 (IAH)
- Grade II = 16 - 20
- Grade III = 21 – 25
- Grade IV = >25

- Please notify physician if IAP is > 12 , or per physician's orders.
- **Profound physiologic derangements** can occur with IAH reinforcing the need to recognize and treat IAH early before ACS develops.
- **Abdominal Perfusion Pressure (APP) = MAP – IAP**
- **Goal APP > 60 mmHg**

Procedure for Measuring Abdominal Pressures

- Supplies Needed
 - 500 ml Normal Saline
 - Transducer/pressure tubing
 - 60 ml syringe
 - 18 gauge needle
 - Providone-iodine swab
 - Alcohol Swab
 - Kelly
 - 4 x 4 gauze

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ABDOMINAL

Abdominal Compartment Syndrome (continued)

Procedure for Measuring Abdominal Pressures

- Guidelines
 - Place patient in supine position
 - Clamp tubing of the indwelling urinary catheter distal to sampling port with Kelly clamp, using 4 x 4 gauze to protect tubing.
 - Clean sampling port with providone-iodine/alcohol
 - Attach IAP monitoring device to Foley
 - Insert 18-gauge needle into sampling port of bladder drainage system.
 - Attach 60-ml syringe to 3-way stopcock and withdraw 60 ml of normal saline from the flush system.
 - Instill 60 ml of normal saline into the bladder
 - Using the proximal stopcock, level and zero the transducer at the symphysis pubis. Please mark position if you are going to be doing more intra-abdominal pressure measurements.
 - Measure abdominal pressure at end expiration (please note that the intra-abdominal pressure waveform is a relatively flat line with excursion that corresponds to respiratory cycle.
 - Record reading on flow sheet
 - Subtract NS instilled into the bladder from urine output
 - Label and date flush bag/transducer used for intra-abdominal pressure readings if repeated measurements are ordered. Please ensure that the flush bag/transducer is labeled for intra-abdominal pressure readings only.

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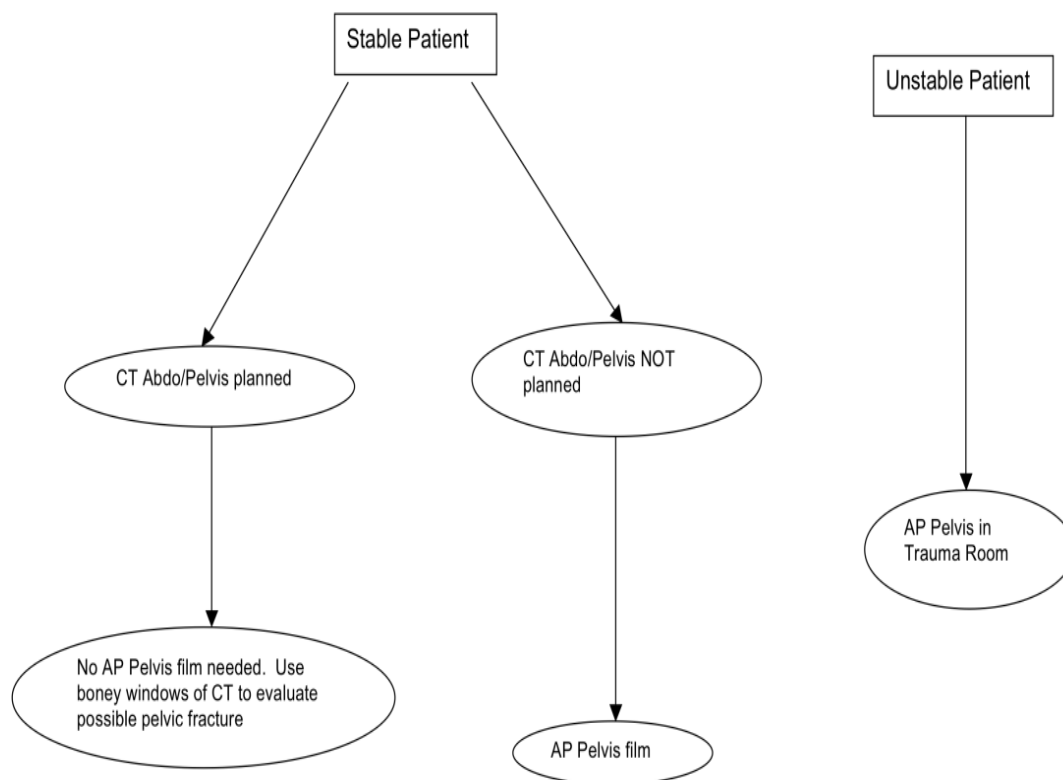
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: PELVIS

Pelvic Fracture Radiologic Evaluation Algorithm

EMERGENCY DEPARTMENT RADIOLOGICAL ALGORITHM TO EVALUATE POSSIBLE PELVIC BONE FRACTURE IN BLUNT TRAUMA VICTIMS

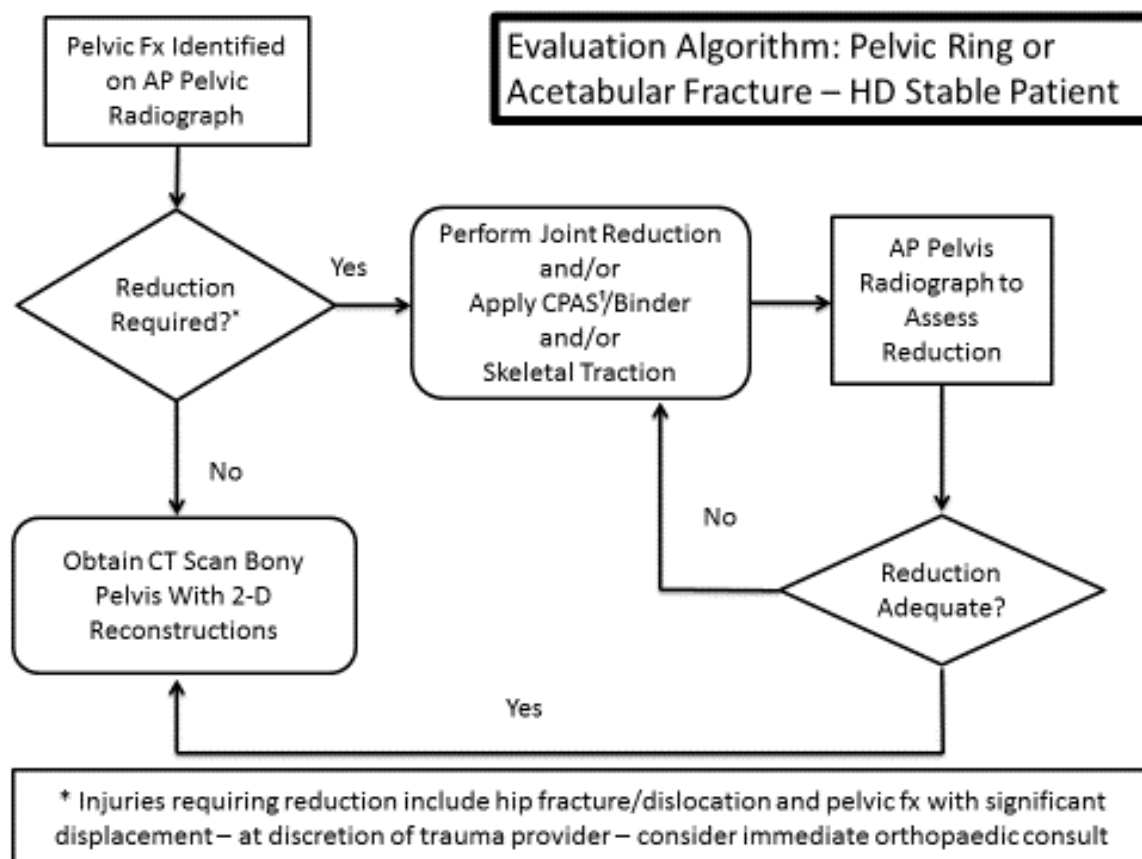


10/08

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO/PELVIS

Stable Pelvic Fracture Evaluation



¹circumferential pelvic antishock sheeting

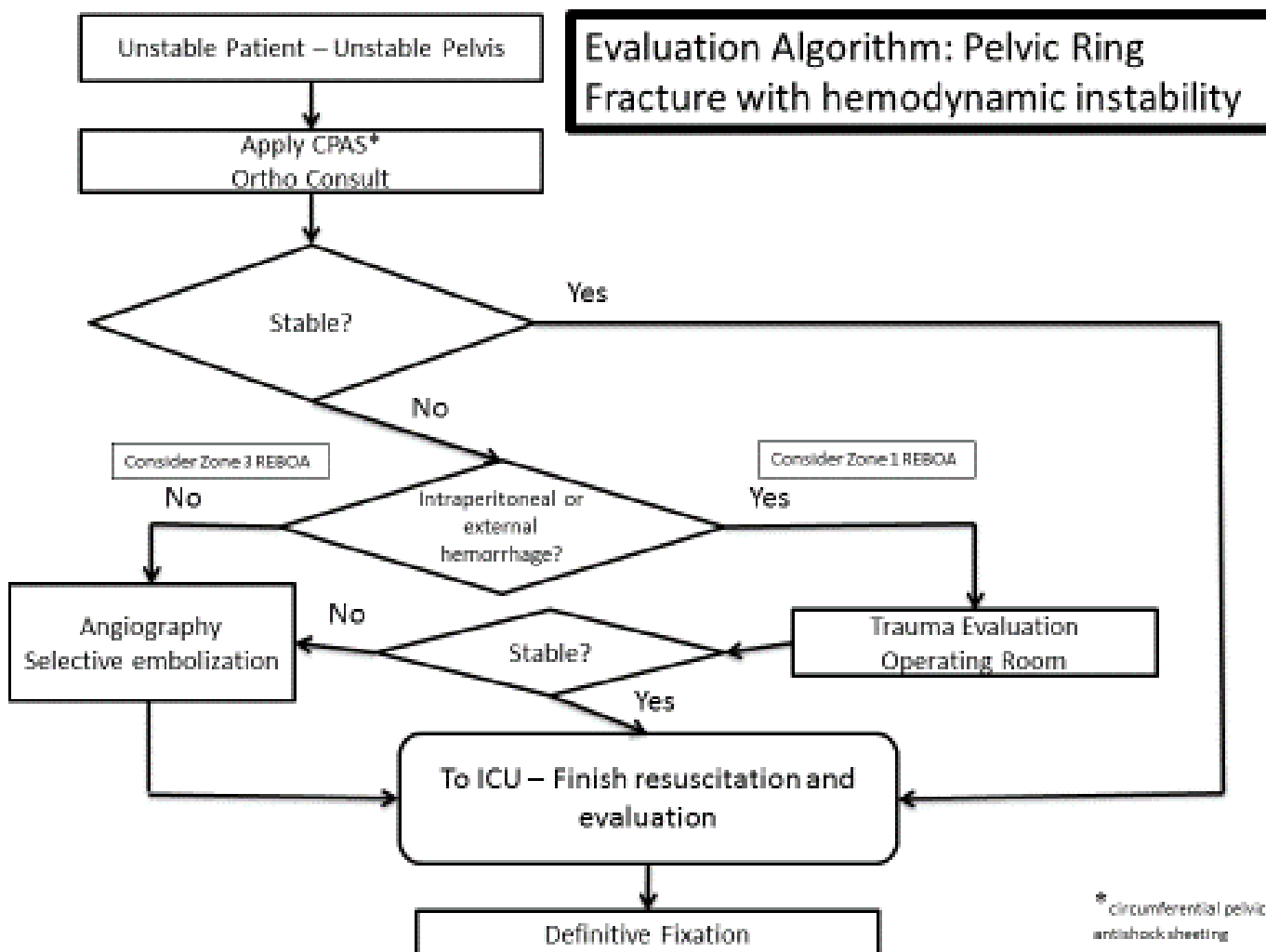
June 2017

Refer to Guideline for Admitting Service of Pediatric Fractures

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO/PELVIS

Unstable Pelvic Fracture Evaluation



June 2017

Refer to [Guideline for Admitting Service of Pediatric Fractures](#)

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO/EXTREMITY INJURIES

Care of Extremity Fractures

Assessment

- Document time, mechanism of injury and any associated injuries.
- Note obvious swelling, deformity, tissue integrity, dislocation, or inability to bear weight or move affected extremity. Compare to uninjured extremity.
- Assess and document quality and severity of pain using age appropriate tools
 - children over 12yrs of age use the 5 P's.
 - Pain
 - Pressure
 - Pulses
 - Paresthesia
 - Paralysis
 - Pallor
 - children 0-11yrs, use the 3 A's
 - Analgesia
 - Anxiety
 - Agitation
- Perform serial assessments for compartment syndrome (CS). Assess for:
 - Taut, firm extremity compartments
 - Pain unrelieved by narcotics – often the first noted indication of CS
 - Extreme pain elicited by passive stretch
 - Paresthesia
 - Loss of pulse is the LAST clinical finding in developing CS
- Determine relevant history, including previous fracture or injury; possible pregnancy status in females; tetanus immunization status; last meal; medications received.
- Document treatment PTA – self-treatment, at outlying facility, or by pre-hospital personnel.

Interventions:

- Remove jewelry and/or constrictive clothing as soon as possible.
- RICE – rest, ice, compression, elevation.
- Irrigate, clean, and dress open wounds (See [Antibiotic & Prophylaxis](#) guideline) – give Td as indicated
- Splint injured site to prevent further injury or allow patient to hold extremity in position of comfort
- Reevaluate neurovascular status proximal to injury before and after any interventions.

Diagnostics

- Image affected limb with joints above and below the injury site
- Consult orthopaedics for abnormality noted on imaging or concern for compartment syndrome.
- Consider CTA, angiogram, and/or vascular consult if vascular injury suspected. Hard signs are:
 - expanding/pulsatile hematoma
 - presence of bruit/thrill
 - absence of Doppler signals
 - signs of limb ischemia
 - API < 0.9
- Refer to [Mangled Extremity Guideline](#) or [Extremity Vascular Injury Guideline](#) for further care of these injuries.

Revised July 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO/EXTREMITY INJURIES

Care of Extremity Fractures (continued)

Pain management recommendations for the orthopaedically injured child

- Acetaminophen 15mg/kg/dose q 4h PRN
- Ibuprofen 10 mg/kg/dose q 6h PRN pain
- Acetaminophen/Hydrocodone, Hydrocodone 0.1 - 0.2 mg/kg/dose q 4h PRN pain
- Oxycodone 0.1 – 0.2 mg/kg/dose q 4h PRN pain
- Morphine Inj. 0.05 to 0.2 mg/kg/dose q 2h PRN pain
- Diazepam 0.05 – 0.1 mg/kg/dose q 6h PRN spasms

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Revised July 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO/EXTREMITY INJURIES

ED Management of Pediatric Orthopaedically Injured Patients with Anticipated Discharge from ED

- The purpose of this guideline is to help guide the decision for Orthopaedic Surgery consult, specifically for children who are expected to be discharged with Pediatric Orthopaedic follow-up.

Independent ER Management

- Non-displaced clavicle fractures
- Acromioclavicular separation
- Glenohumeral dislocation – with axillary lateral x-ray demonstrating anatomic reduction
- Non-displaced forearm fracture
- Distal radius buckle fracture
- Non-displaced metacarpal or phalanx fractures of the hand
- Soft tissue injury without bone or joint involvement
- Transient synovitis of the hip – $\leq 1/4$ Kocher criteria (per septic hip algorithm)
- Toddler fracture (non-displaced, spiral tibia fracture)
- Non-displaced metatarsal or phalanx fractures of the foot

Orthopedic Surgery Consult Recommended

General rules:

- Any fracture involving the growth plate
- Any displaced fracture
- Any elbow injury
- Any musculoskeletal infection

Specific Injuries:

- Displaced clavicle fractures
- Proximal humerus fractures
- Humeral shaft fractures
- Any elbow injury – normal x-ray w/ effusion, elbow dislocation, medial epicondyle fx, lateral condyle fx, supracondylar humerus fx, radial neck fx, olecranon fx
- Displaced forearm fractures
- Displaced metacarpal or phalanx fractures of the hand
- Septic hip – $\geq 2/4$ Kocher criteria (per septic hip algorithm)
- Proximal femur fractures
- Femoral shaft fractures
- Distal femur fractures
- Intra-articular fractures of the knee – tibial spine fx, proximal tibial epiphyseal fx, tibial tubercle fx
- Tibia shaft fractures
- Physeal fractures of the ankle
- Displaced metatarsal or phalanx fractures of the foot

This list was compiled by representatives of Orthopaedics (peds & trauma) and Emergency Medicine (chairman & pediatric).

Developed: 2014 Revised: 5/2016 Reviewed 2017.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO ADMISSIONS

Guideline for Admitting Service of Pediatric Fractures

- The following applies only to the single system injury children who have been cleared to go to the floor by the Emergency Medicine (EM) attending or the Pediatric Surgery (PDS) team.
- Any multisystem injury (defined as the need for more than one consult service to care for injuries) will be admitted to Pediatric Surgery (PDS).
- Any patient admitted to PICU or PCU will be admitted to the PDS or ICU appropriate teams.

Injury	Admitting Service	Svc post tertiary exam
Femur fx, any mechanism	PDS	ORF
Femur fx w/ concern for NAT (NAT=nonaccidental trauma)	PDS	PEDS (ORF if abuse w/u complete)
Pelvic fx	PDS	ORF
Pelvic fx w/ concern for NAT (NAT=nonaccidental trauma)	PDS	PEDS (ORF if abuse w/u complete)
All other isolated fractures	ORF	ORF
All other isolated fractures w/ concern for NAT (NAT=nonaccidental trauma)	PEDS w/ PDS consult	PEDS (ORF if abuse w/u complete)

- PDS will need to assess any patient requiring an abuse workup. This can be a consult on patients admitted to pediatrics.
- Orthopaedic surgery is expected to follow the patient through completion of discharge plans including plans for equipment, follow-up, and PT/OT to ensure timely discharge of patients and to prevent confusion related to sign offs.

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Developed: 8/2015

Reviewed: 2017

Antimicrobial prophylaxis should be administered in the trauma bay **as soon as possible**. Doses and frequencies listed detail/describe open fracture prophylaxis while awaiting closure. For antibiotic dosing prior to any operative incision and intraoperative dosing frequency, see surgical prophylaxis guidelines on CareWeb. Contact pharmacy with dosing questions.

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CSF Leak	NONE Pneumococcal Vaccination Prior to Discharge [‡]	NONE Pneumococcal Vaccination Prior to Discharge [‡]
Antibiotics should be continued <u>until</u> soft-tissue coverage, and no longer than 48 hours total		

Patients with isolated hand injuries or GSW to foot or transabdominal pathway into Hip/Pelvis/Spine should receive routine grade I-III antibiotic prophylaxis as described above.

[‡] PLUS chlorhexidine rinses PO QID for mandible fractures

[‡] In obese patients (>125% IBW) utilize dosing body weight (DBW) for gentamicin; DBW = Ideal BW + 0.4 (Actual BW-Ideal BW)

[‡] See Table III

Table IA: Gustilo Open Fracture Classification

Gustilo Grade	Definition
I	Clean wound <1 cm in length
II	Wound > 1 cm without extensive soft-tissue damage, flaps, avulsions
III	Extensive soft-tissue laceration, damage, or loss or an open segmental fracture.
IIIA	Adequate periosteal coverage despite the extensive soft-tissue laceration or damage
IIIB	Extensive soft-tissue loss and periosteal stripping and bone damage. Usually associated with massive contamination. Will often need further soft-tissue coverage procedure (i.e. free or rotational flap)
IIIC	Arterial injury requiring repair, irrespective of degree of soft-tissue injury

Table II: Tetanus Recommendations

History of Tetanus Vaccination	Last Dose	Wound Type	DTaP, Tdap or Td [†]	TIG [‡]
Unknown or <3 doses		Clean, minor wound	Yes	No
		All other wounds*	Yes	Yes
≥3 doses	<5 years prior	All wounds	No	No
	5-10 years prior	Clean, minor wounds	No	No
	5-10 years prior	All other wounds*	Yes	No
	≥10 years prior	All wounds	Yes	No

Abbreviations: DTaP = Diphtheria and Tetanus toxoids and acellular pertussis vaccine; Tdap = tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis; Td = tetanus and diphtheria toxoids; TIG = Tetanus immune globulin

*Such as, but not limited to, wounds contaminated with dirt, feces, soil, and saliva; puncture wounds; avulsions; and wounds resulting from missiles, crushing, burns, and frostbite.

[†] DTaP is recommended for children <7 years of age. Tdap is preferred to Td for persons aged 11 years or older who have not previously received Tdap. Persons aged 7 years or older who are not fully immunized against pertussis, tetanus, or diphtheria should receive one dose of Tdap (preferably the first) for wound management and as part of the catch-up series; if additional tetanus toxoid-containing doses are required, either Td or Tdap vaccine can be used.

[‡] People with HIV infection or severe immunodeficiency who have contaminated wounds (including minor wounds) should also receive TIG, regardless of their history of tetanus immunizations.

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Table III: Pneumococcal Vaccination in Patients with CSF Leak

Age 2–5 years	
Any Incomplete[#] Series with:	
≥3 PCV13 doses	Administer 1 dose PCV13 now [‡] [‡] Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose
<3 PCV13 doses	Administer 1 dose of PCV13 now [‡] Administer 2nd dose of PCV13 in ≥8 weeks [‡] Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose
No history of PPSV23	Administer 1 dose of PPSV23 in ≥8 weeks from PCV13 dose
Age 6–18 years	
No history of either PCV13 or PPSV23	Administer 1 dose of PCV13 now [‡] Administer 1 dose of PPSV23 in ≥8 weeks [‡] Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose
Any PCV13 but no PPSV23	Administer 1 dose of PPSV23 now [‡] [‡] Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose
PPSV23 but no PCV13	Administer 1 dose of PCV13 now [‡] [‡] Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose

[#]*Incomplete series* = Not having received all doses in either the recommended series or an age-appropriate catch-up series. See Tables 8, 9, and 11 in the in the ACIP pneumococcal vaccine recommendations (<https://www.cdc.gov/mmwr/pdf/rr/rr5911.pdfpdf icon>) for complete schedule details.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHOPAEDIC INJURIES

General Guidelines for Orthopaedic Injuries/Emergencies

- ALL pelvic and acetabular fractures are treated at UKHealthCare (see pelvic fracture algorithms – stable & unstable).
- Long bone fractures, isolated or multiply injured, are taken to the operating room for initial or definitive fixation within 24 hours of presentation to UKHealthCare unless not medically cleared. Refer to “Determining Extent of Orthopaedic Intervention in Acute Multi-System Trauma” for damage control priorities.
- Open fractures are evaluated for proper management and are taken to operating room for surgical irrigation & debridement within 24 hours of presentation to UKHealthCare unless not medically cleared. Initial washout may be performed at bedside as needed. Refer to “Antibiotic & Prophylaxis” guideline and “Determining Extent of Orthopaedic Intervention in Acute Multi-System Trauma” for priorities of care in the multi-injured patient.
- Open fractures receive the first dose of antibiotics within 60 minutes of arrival at UKHealthCare. See “Antibiotic & Prophylaxis” for identification of appropriate antibiotics and dosage.
- Supracondylar humerus fractures, Gartland type III or greater, are taken for repair at least 18 hours from time of injury.

The Orthopaedic Trauma team is readily available after consultation requested by the surgical team leader for multiply injured patients with these time-sensitive injuries:

- Fractures with neurovascular compromise
- Pelvic fractures with hemodynamic instability
- Joint dislocations

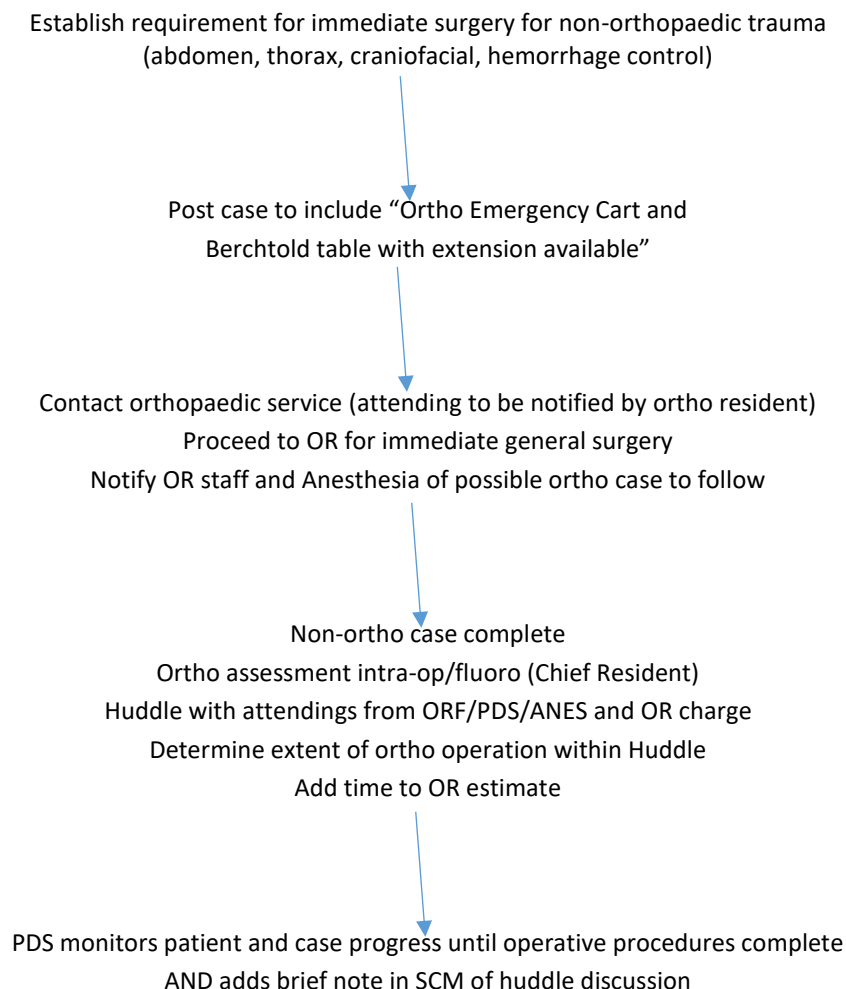
NOTE: Physician coverage for the KCH is available through the online “Provider On Call” system. Additionally, specifics are posted to inpatient areas with name & photo, pager, and other information (see example provided).

June 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO INJURY GUIDE

Determining Extent of Same Setting Orthopaedic Surgery in Patients with Combined Emergency Non-Orthopaedic & Orthopaedic Injuries Guideline



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Established 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO GENERAL CARE

Mobility In The Injured Patient

Principles of movement in the injured patient

Weight bear status (WB) defines whether or not **and** how much weight the child can put on his or her affected extremity:

- NWB – non-weight bearing, or no weight
- WBAT – weight bear as tolerated or whatever the child will tolerate
- PWB, TTWB, or TDWB – partial weight bear, toe touch weight bear, touch down weight bear – put only some weight on the extremity (used in older children who are able to understand)

Below are images of EXAMPLES of some supportive devices:



- Movement within these devices is based upon weight bear status as the device provides the necessary support when appropriately applied.
- EXPECT initial discomfort with movement – **however**, the discomfort SHOULD NOT outweigh the need and benefits of movement.
- Proper premedication when used can be very effective in positioning.

Developed July 2017

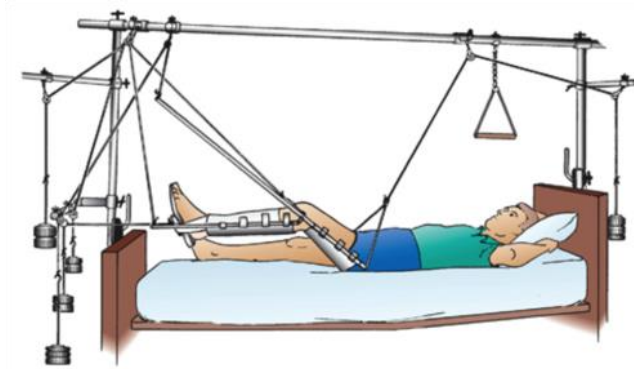
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO GENERAL CARE

Skeletal Traction

Skeletal Traction



Purpose: application of pull to injured extremity while counter force (patient weight) pulls in opposite direction to reduce or re-align fracture

Primarily used short term and in older or larger children.

Care of the Child in Skeletal Traction

Educate patient & family – purpose of traction, ability & methods of movement in traction

- Encourage use of Trapeze to assist with movement
- Child can raise himself/herself off bed with trapeze, provided other traumatic injuries are not contraindications (e.g. upper extremity fractures)

Pain Control – provide pain control measures, keeping in mind that the fracture, while in better alignment with traction, still requires fixation for proper healing.

Muscle spasms – anticipate muscle spasms as the traction weights pull/hold the fracture in proper alignment.

Documentation – neurovascular assessments, skin & pin site assessments, and (per hospital policy) number of pounds of traction

Turning & Positioning

- Nurses can support weights gently during patient movement, ensuring smooth movement and free hanging of weights (e.g. preventing weights getting caught in foot board of bed)
- Bed sheets can be changed top to bottom, with child's assistance lifting off bed (rather than moving side-to-side)
- Trendelenburg positioning can be used to help keep child from sliding to foot of bed
- Lower extremity with traction may be positioned on the bed (with pillow) provided weights are free hanging and extremity is aligned with the pull **AND** the heel of the foot is elevated off the bed to prevent skin breakdown
- If child's feet are against footboard, child needs to be moved up in bed
- For larger or multiply injured children, consider using Tortoise positioning system

Developed July 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ORTHO GENERAL CARE

Pin Care

Pin Care: Yes or No?

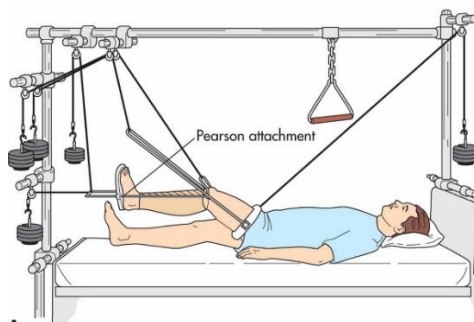
PRN only



External Fixator**



Halo traction



Skeletal traction

PRN Pin Care

with gauze in place:

- Remove 4X4's from pins
- Gently cleanse skin
- Replace Chlor-prep soaked 4X4's

without gauze:

- Gently cleanse skin
- Using Chlor-prep (Hibiclens)
- Do NOT aggressively push skin away from pin sites



**for Ilizarov devices, see specific pin care instructions.

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Developed July 2017

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: Mangled Extremity/Amputation

Mangled Extremity/Traumatic Amputation Guideline

Upper Extremity

- Definition of Mangled Upper Extremity – any major injury to the upper extremity from the shoulder down, which involves 3 of the following 4 components: Skin & Soft Tissue, Bone, Major Nerves, & Major Vessels.
- For any mangled upper extremity/traumatic amputation, the ED or Pediatric Trauma Surgery will contact Hand Surgery on-call as first consult.
- The Hand Surgeon will consult Orthopedic Trauma, Vascular Surgery, and/or Plastic Surgery as dictated by the nature of the injury.

****The Hand Service should also be consulted for any significant injury or infection of the soft tissue distal to the elbow, or bones/joints distal to the radiocarpal joint.**

Lower Extremity

- Definition of Mangled Lower Extremity – any major injury to the lower extremity from the hip down, which involves 3 of the following 4 components: Skin & Soft Tissue, Bone, Major Nerves, & Major Vessels.
- For any mangled lower extremity/traumatic amputation, the ED or Pediatric Trauma Surgery will contact Orthopedic Surgery on-call as first consult.
- The Orthopedic Trauma Surgeon will consult Vascular Surgery, and/or Plastic Surgery as dictated by the nature of the injury.

****Lower extremity injuries involving fracture, however severe, should prompt consultation of the Orthopedic Trauma Service.**

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Revised: March 2018

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: VASCULAR INJURIES

Traumatic Peripheral Vascular Injury (Reviewed UK Vascular Service May 2017)

- Any injured extremity should be thoroughly evaluated for a possible vascular injury.
- The presence of obvious arterial injury from a blunt and/or penetrating mechanism rarely requires imaging and should not delay emergent operative exploration.
- The presence of “hard signs” strongly supports vascular injury and typically necessitates emergent repair. These “hard signs” are:
 - Bruit/Thrill
 - Active/Pulsatile hemorrhage
 - Pulsatile/Expanding hematoma
 - Signs of limb ischemia and or compartment syndrome including the 6 "P's" pain, pallor, pressure, paresthasias, pulse deficit, paralysis, and pain on passive extension of the compartment (**pain on passive extension is the earliest and most sensitive physical finding**)
 - Diminished or absent pulses with + Doppler signals (*this is not a sensitive prognostic finding, as up to 30% of patients with major vascular injuries requiring repair have normal pulses or Doppler signals distal to the injury due to collateral flow*) [1]
- The Arterial Perfusion Index, API, is a validated tool for screening for peripheral vascular injury[2].
- This is performed by placing a blood pressure cuff above the ankle or on the bicep of the limb of concern.
- The systolic pressure is determined with a Doppler probe at the dorsalis pedis or brachial artery.
- Repeat this procedure on the ipsilateral uninjured limb.
- The API is calculated by dividing the systolic pressure in the injured limb by the systolic pressure in the uninjured limb.
- An API < 0.9 has a sensitivity of 95% and specificity of 97% for a major arterial extremity injury.
- In a study on blunt orthopedic extremity injuries the negative predictive value is 100% for an API > 0.9 to exclude an arterial injury.[3-5]
- The purpose of these algorithms is to diagnose the occult injury early before irreversible tissue ischemia is present.
- In patients where the “hard” signs are **NOT** present it is imperative to maintain a high suspicion of peripheral vascular injury in the injured extremity [2, 6, 7].
- If “hard signs” are not present but peripheral vascular injury is suspected then expedient consultation with Vascular Surgery is indicated and the use of imaging, per Vascular Surgery, should be liberal to avoid missed injuries.

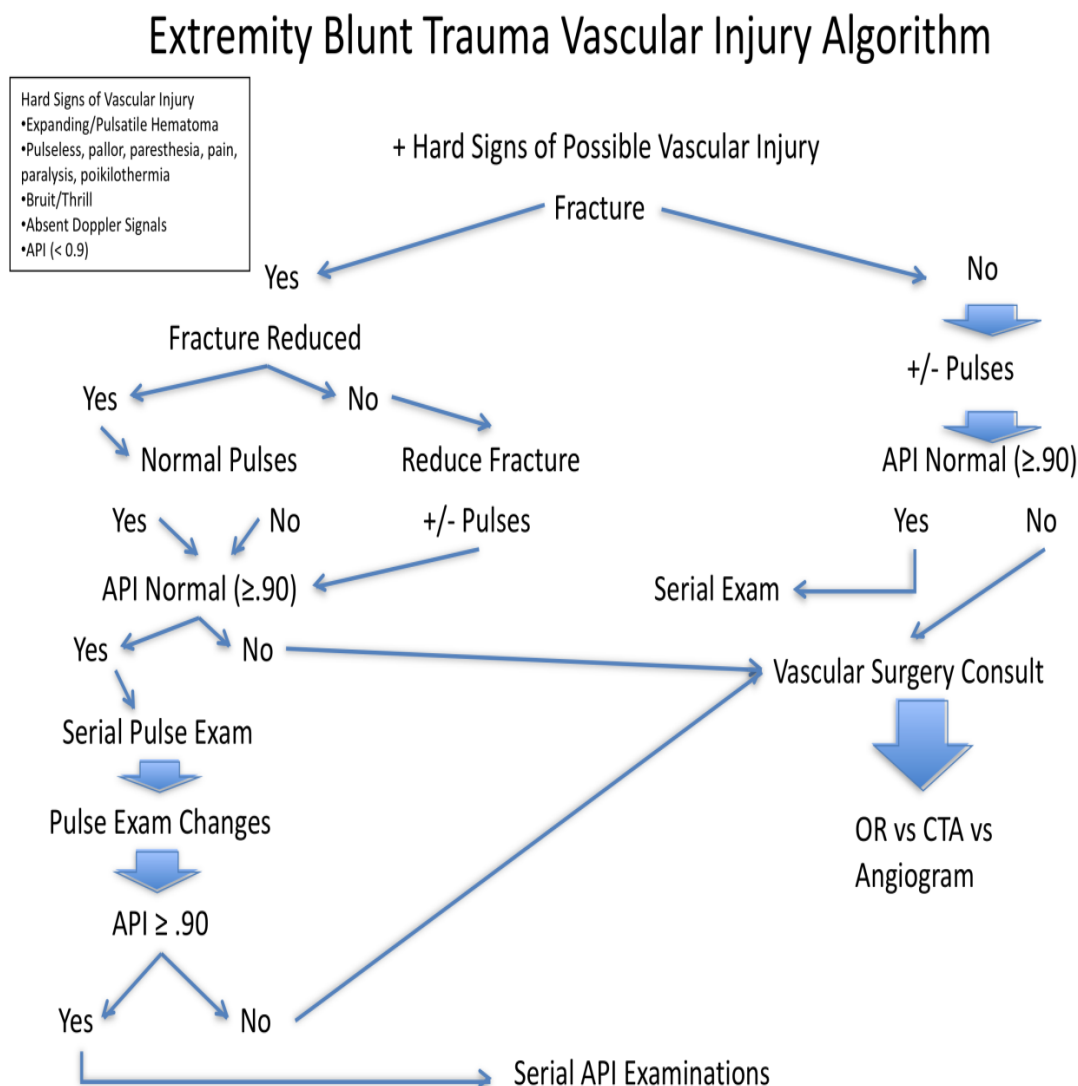
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: VASCULAR INJURIES

Extremity Blunt Trauma Vascular Injury Algorithm (Reviewed UK Vascular Service May 2017)

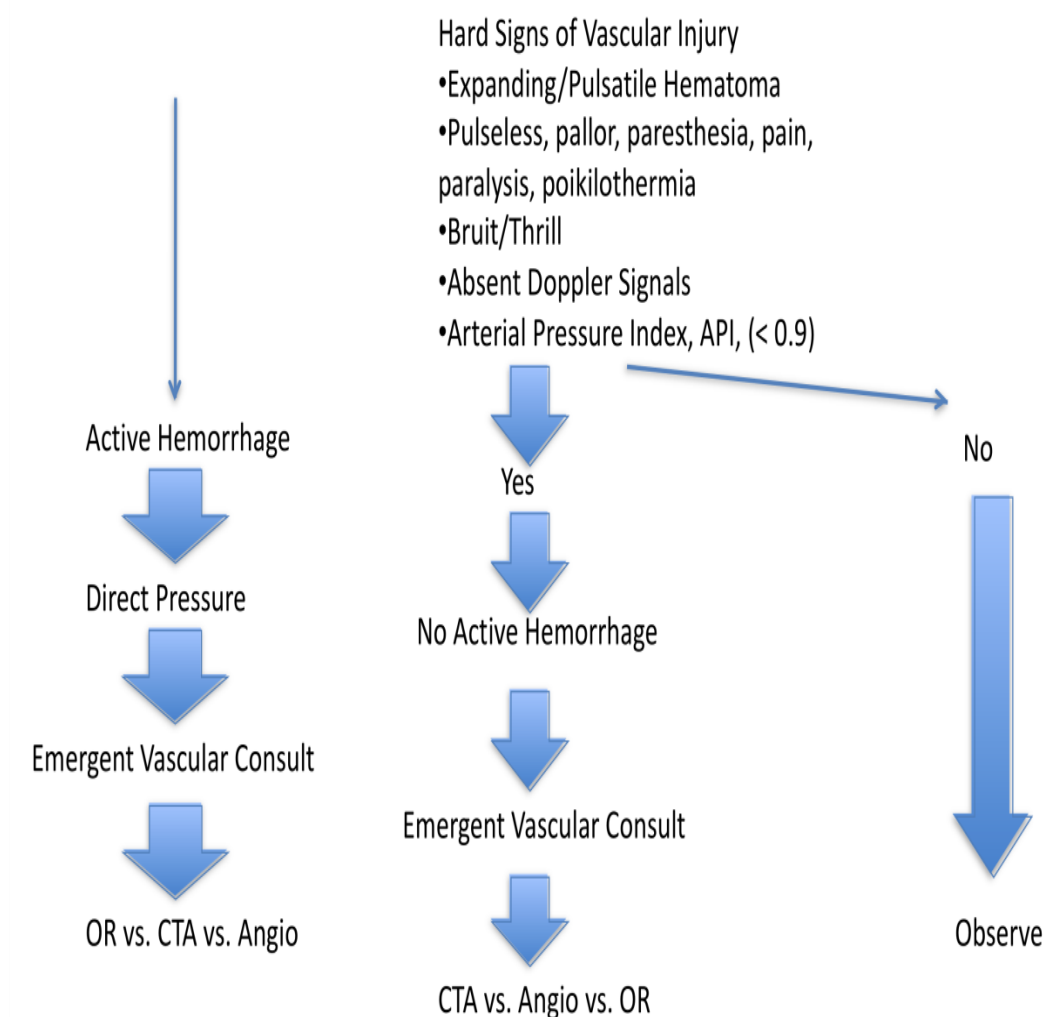


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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: VASCULAR INJURIES

Penetrating Extremity Vascular Injury Algorithm *(Reviewed UK Vascular Service May 2017)*

Penetrating Extremity Vascular Injury



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: BURNS

Evaluation of Burn Patients (Revision: May 2017; May 2019; May 2020, Dec 2020, June 2021)

Referral, Transfer and Admission

- Any pediatric patient with the following should be evaluated for potential benefit from referral and/or transfer to a specialty pediatric burn center (**Shriners Children's Ohio**):
 - burns involving face, hands, feet, genitalia, perineum, or major joints
 - partial thickness burns $\geq 10\%$ TBSA
 - any third degree burns
 - electrical burns, including lightning chemical burns
- **Pediatric patients with $\geq 20\%$ TBSA burns or concern for inhalation injury will be stabilized and then transferred to Shriners Children's Ohio.**
 - Any child with concomitant traumatic injury which poses IMMEDIATE RISK will be stabilized prior to burn center transfer.
 - Any child who cannot be transferred due to weather must be stabilized and appropriately treated until the child can be safely transferred.
- Pediatric patients with $< 20\%$ TBSA burns requiring hospital admission may be admitted to the Pediatric Trauma Service. Pediatric Surgery will manage overall care. Plastic surgery will be consulted for wound management.

Introduction

- 75% of burns are $\leq 10\%$ Total Body Surface Area (%TBSA).
- 60% of burn victims are children ≤ 5 years of age.
- 90% of burns can be managed on an outpatient basis.
- Many pediatric burn injuries (6 - 20%) are secondary to child abuse.

Stabilization

- Avoid hypothermia – administer warmed fluids, increase ambient temperature and cover patient with warm blankets.
- Assume carbon monoxide poisoning and assess for cyanide toxicity if history of altered mental status or confinement in a burning environment.
- Maintain airway - supraglottic airway is extremely susceptible to edema/obstruction from superheated air.
Consider intubation if $> 40\%$ TBSA or clinical signs of inhalation injury:
 - Hoarse voice or cry, stridor, drooling, difficulty speaking, respiratory distress, obvious swelling of the oropharynx.
 - Obtunded patient with absent airway reflexes (no cough/no gag).
 - Facial burns/singeing of eyebrows and nasal hair do not necessarily indicate inhalation injury.
- Standardized ETT securement for patients with severe facial burns:
 - **Shriners Children's Ohio** recommends tape from ear to ear and use surgical staples to secure the tape to the skin.

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Circulation – Initial fluid resuscitation

- **Bolus ONLY** for hypotension [$SBP = 70 + 2 \times \text{age (years)}$]. Tachycardia is poor marker of resuscitation in burn patient.
- If TBSA >10%, the following fluids should be started during primary survey:

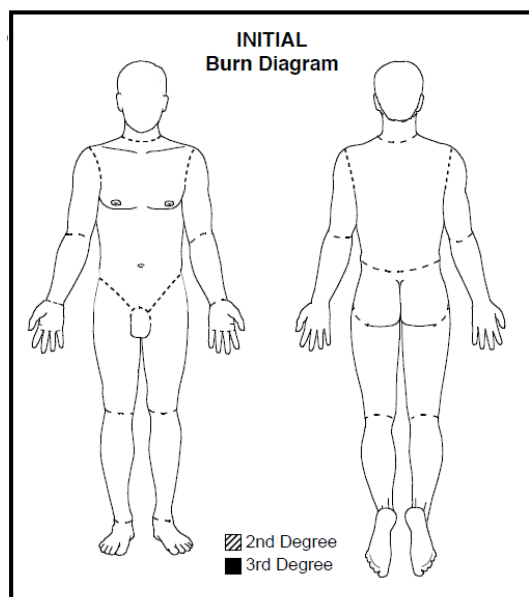
Age/Weight of Child	Initial fluid rate & type (prior to burn size calculation)
≤5 years old & <10kg	125mL/hr of D5LR
≤5 years old & >10kg	125mL/hr of LR
6-14 years old	250mL/hr of LR
>14 years old	500mL/hr of LR

- Calculate burn TBSA and convert resuscitation fluids as instructed below under “Burn Fluid Resuscitation”.

Assessment

- Assess for associated injuries.
- History → time and mechanism of injury, enclosed fire or toxic chemicals involved.
- Assess burn and estimate Total Surface Body Area (%TBSA) using the “Burn Estimate and Diagram Form.”
 - **First-degree** (sunburns) – characterized by erythema, pain, and absence of blisters. **First-degree burns are not counted in TBSA calculation.**
 - **Second-degree** (superficial partial thickness or deep partial thickness) – characterized by red or blanching appearance with swelling and blister formation. Surface may have weeping, wet appearance and painfully hypersensitive.
 - **Third-degree** (full thickness) – skin appears dark and leathery; may also appear translucent, mottled, or waxy white; surface is painless* and generally dry, but may be moist. (*Pain presents at the subcutaneous tissue surrounding or within the burn area).

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Area	Birth 1 yr.	1-4 yrs.	5-9 yrs.	10-14 yrs.	15 yrs.	Adult	Initial TBSA	
							2°	3°
Head	19	17	13	11	9	7		
Neck	2	2	2	2	2	2		
Ant. Trunk	13	13	13	13	13	13		
Post. Trunk	13	13	13	13	13	13		
R. Buttock	2.5	2.5	2.5	2.5	2.5	2.5		
L. Buttock	2.5	2.5	2.5	2.5	2.5	2.5		
Genitalia	1	1	1	1	1	1		
R.U. Arm	4	4	4	4	4	4		
L.U. Arm	4	4	4	4	4	4		
R.L. Arm	3	3	3	3	3	3		
L.L. Arm	3	3	3	3	3	3		
R. Hand	2.5	2.5	2.5	2.5	2.5	2.5		
L. Hand	2.5	2.5	2.5	2.5	2.5	2.5		
R. Thigh	5.5	6.5	8	8.5	9	9.5		
L. Thigh	5.5	6.5	8	8.5	9	9.5		
R. Leg	5	5	5.5	6	6.5	7		
L. Leg	5	5	5.5	6	6.5	7		
R. Foot	3.5	3.5	3.5	3.5	3.5	3.5		
L. Foot	3.5	3.5	3.5	3.5	3.5	3.5		
Total:								

Burn Fluid Resuscitation

- Bolus ONLY for hypotension.
- Replace volume based on estimate of TBSA.
 - If < 20% TBSA, burn fluid resuscitation is not indicated. Discontinue IVF or decrease to MIVF unless otherwise indicated or patient unable to take PO.
 - If > 20% TBSA, start 2-bag resuscitation per the following:
 - Initiate two large bore IVs; overlying burns should not prevent IV placement (upper extremities preferred)
 - **Bag 1:** Follow the Pediatric Burn Formula $[3\text{ml} \times \% \text{ TBSA burned} \times \text{weight (kg)}] / 16 = \text{ml/hr LR}$ (never add dextrose to Bag 1 fluids).
 - **Bag 2** if < 30 kg: Calculate MIVF of D5LR.

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- Place urinary catheter. Initial output should not count towards hourly output. Titrate Bag 1 rate hourly according to individual patient's response:
 - < 15 years and < 50 kg: goal UOP = 1.0 ml/kg/hour
 - If UOP < 1ml/kg/hr, increase fluids by 10%
 - If UOP > 1ml/kg/hr, decrease fluids by 10%
 - > 15 years or > 50 kg: goal UOP = 30 ml/hr
 - If UOP < 30ml/hr, increase fluids by 10%
 - If UOP > 30ml/hr, decrease fluids by 10%
- Place feeding tube and start enteral feeds within 12 hours if medically stable and unable to take PO. Feeds held for sedation per KCH policies. Wean Bag 2 fluids as enteral feeds are increased.
- Monitor finger stick glucoses every 1-2 hours until stable.
- Continue resuscitation fluids until enteral intake is sufficient to maintain adequate UOP and patient has had full thickness burns excised. Patients will have an ongoing inflammatory response, capillary leak and relative intravascular hypovolemia.

Basic Wound Care

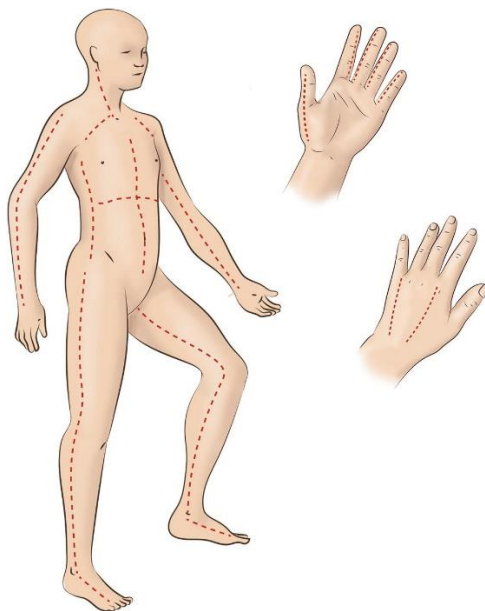
- Remove affected clothing if not already done and anything wet/cold.
- Cover burns with dry dressing.
- Patients for whom transfer is directed by Shriners Hospital: no need to wash, debride or apply topical medications.
- Patients for whom outpatient referral is directed by Shriners Hospital:
 - Wash burn with soap and water if visibly dirty. One wipe is sufficient; no need to debride loose skin.
 - Apply antibiotic ointment to non-adherent dressing. DO NOT USE SILVER SULFADIAZINE.
 - Place dressing directly on burn and wrap lightly with gauze, maintaining full range of motion.
- **Wound Care for those not transferred or referred:**
 - Superficial partial thickness: Mepilex AG, wrap with kerlix or stretch net
 - Deep partial thickness: bacitracin with Adaptic or Vaseline gauze; Dermanet for larger areas; pad with 10 ply burn gauze, wrap with kerlix/stretch net. Transition to Mepilex AG prior to discharge home.
 - Full thickness burn: can use Silvadene for older for than 2 months of age.

Circumferential burns

- Extremities:
 - Remove rings and bracelets and assess distal circulation.
 - Check pulses with Doppler.
 - Observe for cyanosis, impaired capillary refill, or progressive neurological signs (i.e. paresthesia and deep tissue pain).
 - Limb Escharotomy:
 - Relieve compromised distal circulation in circumferentially burned limb by escharotomy, which can be done with sedation or local anesthesia in lieu of general anesthesia, due to insensitive full thickness burn.
 - Incision must extend across entire length of eschar in lateral and/or medial line of limb including fingers and joints.
 - Incision should be deep enough to allow cut edges of eschar to separate.
 - Escharotomy should be performed through burned skin when possible.

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- Thorax/Abdomen/Neck
 - Circumferential burns of thorax can impair respiratory excursion.
 - Circumferential burns of abdomen can cause abdominal compartment syndrome.
 - Circumferential burns of neck can cause airway compromise.
 - Thoracic Escharotomy:
 - Bilateral, mid-axillary escharotomy incisions should be considered.
 - Escharotomy should be performed through burned skin when possible.



Special Burn Considerations

Chemical burns

- Brush dry powder off before irrigation.
- Irrigate burns with water for at least 20 to 30 minutes; alkali burns require longer irrigation.
- Alkali burns to the eye require continuous irrigation during first 8 hours – consult Ophthalmology.

Electrical burns - frequently more serious than they appear on the surface

- Initial care as above including EKG monitoring and urinary catheter placement.
- Full spinal immobilization.
- Observe for myoglobinuria (due to rhabdomyolysis); consult with burn center regarding alkalization of fluids.
- Increase IVF rate to ensure UOP 1–1.5 mL/kg/hr in children weighing less than 30 kg or 100 ml/hr in adults. Once urine is clear of pigmentation, titrate IVF down to ensure standard UOP as described above.

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Explosive Injuries

- Any patient involved in an explosion has mechanism for traumatic injuries.
- Even if child was not thrown, a flash flame explosion can have enough energy to cause mild TBI (concussion).

Other Pediatric Considerations

- Consult Child Life for all burn patients.
- Consult Pediatric Forensic Medicine Service and/or DCBS if injury due to suspected abuse, neglect, or safety hazards in the home.
- Consider Pediatric Psychiatry or Psychology consult to help child cope with hospitalization, body image changes, and/or acute stress disorders.
- Assess tetanus immunization status and provide immunization per AAP Red Book recommendations:
<https://redbook.solutions.aap.org/chapter.aspx?sectionId=189640195&bookId=2205&resultClick=24#192303674>

Referral, Transfer and Admission

- Any pediatric patient with the following should be evaluated for potential benefit from referral and/or transfer to a specialty pediatric burn center (**Shriners Children's Ohio**):
 - burns involving face, hands, feet, genitalia, perineum, or major joints
 - partial thickness burns $\geq 10\%$ TBSA
 - any third degree burns
 - electrical burns, including lightning chemical burns
- **Pediatric patients with $\geq 20\%$ TBSA burns or concern for inhalation injury will be stabilized and then transferred to Shriners Children's Ohio – (Dayton, OH).**
 - Any child with concomitant traumatic injury which poses IMMEDIATE RISK will be stabilized prior to burn center transfer.
 - Any child who cannot be transferred due to weather must be stabilized and appropriately treated until the child can be safely transferred.
- Pediatric patients with $< 20\%$ TBSA burns requiring hospital admission will be admitted to the Pediatric Trauma Service. Pediatric Surgery will manage overall care. Plastic surgery will be consulted for wound management.

Resources:

- **Shriners Children's Ohio** is available 24/7 to assist with management of burn patients. Call 866-947-7840 and ask to speak with the Charge Nurse.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ANEMIA

Anemia Management

Pediatric patients commonly suffer from anemia on admission to the Pediatric Intensive Care Unit (PICU). Additionally, anemia frequently develops or worsens during the course of a patient's PICU stay. Anemia can cause prolonged PICU stays, increase the costs of healthcare, and lead to other negative patient outcomes. If transfusion is required, the patient may be exposed to potential harmful effects of transfused blood products.

Prevention

- Minimize phlebotomy (avoid daily lab draws, Use pediatric sized specimen tubes to limit the amount of blood required for the tests, obtain all blood needed at once, use specimen tubes that can be used for multiple labs, add tests to blood already in the lab when appropriate)
- Adequate nutritional support

Transfusion of PRBCs

- Patients may require 10 ml/kg of PRBCs during their initial resuscitation after trauma. This may be repeated.
- If the patient remains hemodynamically unstable after 2 boluses of PRBCs, institute the Massive Transfusion Guideline until hemorrhage control is achieved. See Massive Transfusion Guideline.
- The goal after receiving a transfusion is a Hgb ≥ 7 g/dL (Hct ≥ 21).
- If clinically indicated to monitor response and further bleeding, blood counts should be collected 30 minutes after the end of the infusion.
- Routine post-transfusion hematocrits are not mandatory. The most blood-economical manner of monitoring hematocrit is with blood gas panels, which require only 1cc of blood (venous or arterial).

Adjunctive Medications

- Consider the addition of these medications when treating anemic patients:
 - **Folic acid**
 - **Cyanocobalamin (vitamin B12)**
 - **Ferrous sulfate**

SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: MTP

Massive Transfusion Clinical Practice Guidelines for Pediatric Trauma Patients

Review Date:

Revision Date: 2/10/2017; 7/19/2018; 10/22/2018; 11/16/2018

Objectives

- To outline the process for rapid preparation, issue, and infusion of blood components of patients in extremis due to massive hemorrhage.
- To provide goal directed therapy through the use of thromboelastography (TEG).
- In the absence of goal directed therapy, a ratio of 1:1:1 component therapy is desired.

Target Population

- Pediatric trauma patients with hemorrhagic shock who are undergoing resuscitation in the Emergency Department, Operating Room, and Pediatric Intensive Care Unit.

Pediatric Massive Transfusion

- The actual or anticipated rapid transfusion of blood products and other intravenous fluids to pediatric trauma patients to replace greater than the patient's estimated blood volume within a 24-hour period.
- The need for transfusion equal to half of the patient's estimated blood volume at one time, such as within one hour.
- Estimates of a child's blood volume vary by age:

Age	Blood Volume
Newborn to 1 month	90 mL/kg
1 - 12 months	80 mL/kg
1 – 14 years	80 mL/kg
>14 years	75 mL/kg

Clinical Criteria

- Exsanguinating shock with associated acidosis
- Transfusion of crystalloid > 40 mL/kg or pRBC > 20 mL/kg with continued hemodynamic instability
- Estimated blood loss > 20 – 40 mL/kg
- High energy chest, abdominal or pelvic trauma
- Aortic or other vascular injury
- Severe CHI in a young child
- Long bone injury with multisystem injuries
- Known coagulation disorders

Laboratory Criteria

- Admission hematocrit < 30 g/dL
- Base deficit > 8
- pH < 7.1
- INR > 1.5
- PT > 18 seconds
- PTT > 60 seconds

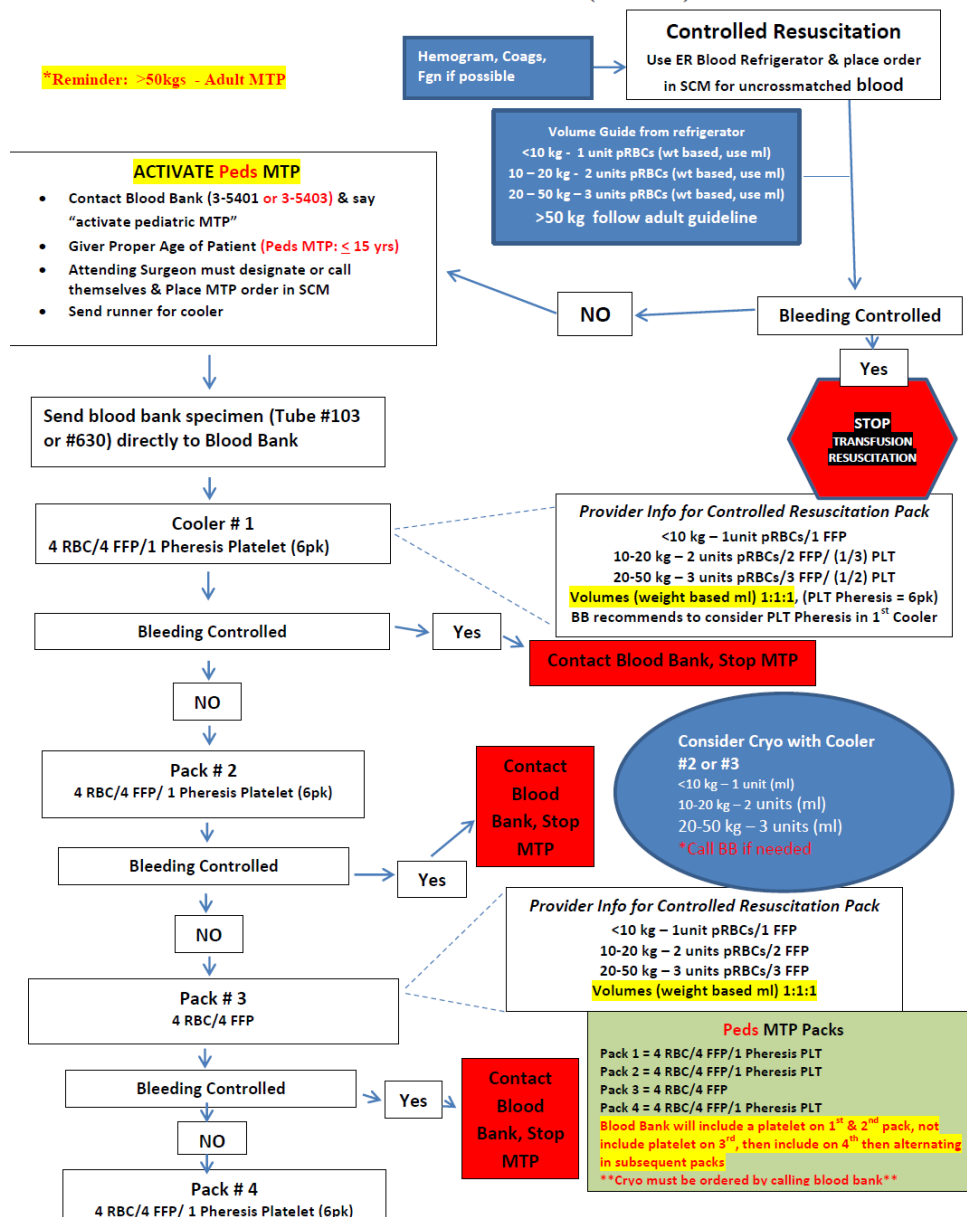
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: MTP

Massive Transfusion Clinical Practice Guidelines for Pediatric Trauma Patients (continued)

MTP Decision Algorithm

KCH Pediatric Level I Trauma Massive Transfusion (**Peds** MTP) Activation Guide



Transfusion Notes

- Type and Screen plus Crossmatch should be obtained ASAP and send directly to the blood bank (not the main lab) by the tube system using blood bank tube station numbers 103 and 630
- All blood products should be administered through filters.
- Warmers should be used whenever possible.
- Give O positive pRBCs to male children and O negative pRBCs to female children until ABO/Rh is obtained.
- Carefully document the blood products and volumes in the patient care record.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: MTP

Massive Transfusion Clinical Practice Guidelines for Pediatric Trauma Patients (continued)

Thromboelastography (TEG)

- In severely injured trauma patients, TEG can be utilized to guide the administration of blood products with the goal of improving patient mortality, reducing the amount of blood required for transfusion, and preventing unnecessary transfusions.
- Refer to TEG Clinical Practice Guideline for additional information.
<http://uktraumaprotocol.blogspot.com/2013/03/teg.html>

Selected References

1. Christiaans SC, Duhachek-Stapelman AL, Russell RT, et al. Coagulopathy after severe pediatric trauma. *Shock* 2014; 41:476-90.
2. Hendrickson JE, Shaz BH, Pereira G, et al. Implementation of a pediatric trauma massive transfusion protocol: one institution's experience. *Transfusion* 2012; 52: 1228-36.
3. "Pediatric Trauma." Advanced Trauma Life Support (ATLS): Ninth Edition. ATLS Subcommittee; American College of Surgeons' Committee on Trauma.
4. Horst J, Leonard JC, Vogel A, et al. A survey of US and Canadian hospitals' paediatric massive transfusion protocol policies. *Transfus Med* 2016; 26:49-56.
5. Hwu RS, Spinella PC, Keller MS, et al. The effect of massive transfusion protocol implementation on pediatric trauma care. *Transfusion*. 2016; 56:2712-19.

Volumes of Blood Products:

1. **PRBCs** ≈ **360 mL** (200-220 mL RBCs, 40 mL plasma, 100-120 mL additive solution)
2. **THAWED PLASMA** is kept refrigerated up to 5 days and contains **250 mL** of Factors II, VII, IX, X, and fibrinogen.
3. Factors V and VIII may be somewhat decreased.
4. **PLATELETS** contain ~ 3.0×10^{11} platelets in **250 mL** (equivalent of 6 pack pooled platelets)
5. 1 unit of **CRYO** contains ≥ 80 IU Factor VIII, ≥ 150 mg fibrinogen, as well as significant amounts of Factor XIII and vWF in **10-15 mL**

Revised: Approved IDTQA 11/16/2018

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee Non-Accidental Trauma Clinical Practice Guideline

Revisions: February 19, 2016; July 15, 2016; July 2017

Section I

A. Recognizing Child Physical Abuse

Child physical abuse is non-accidental physical injury as a result of punching, beating, kicking, biting, shaking, throwing, stabbing, choking, hitting, burning, or otherwise harming a child that is inflicted by a parent, caregiver, or other person who has responsibility for the child. Such injury is considered abuse regardless of whether the caregiver intended to hurt the child.

B. Risk Factors for Physical Abuse

- Parent or Caregiver Factors
 - Personality characteristics
 - Psychological well-being (untreated/inadequately treated mental illness)
 - History of maltreatment
 - Substance abuse
 - Attitudes and knowledge
 - Immaturity
- Family Factors
 - Non-biological parental male living in the home
 - Marital conflict/domestic violence
 - Lower economic status
 - High stress level/lack of social support
- Child Factors
 - Age ≤ 3 years have the highest risk
 - Disability (physical/cognitive/emotional)
 - Prematurity
 - Long-awaited child
- The absence of risk factors is not the absence of risk!!!

C. Bruising and Other Skin Trauma

- **TEN/4 rule** for body region/age raising clinical suspicion for abuse.
 - to perform complete child abuse work-up for age.
- Bite marks
 - Measurement of diameter can be helpful determination of adult vs. child but can still be difficult.
 - Inflicted adult bite marks are very worrisome – typically indicate a more sadistic abuser.
 - Swabbing fresh bite marks can help identify perpetrator by DNA analysis. This can be accomplished by moistening a swab, swipe over bite mark several times, allow to dry and put in evidence collection kit.
 - Photography of bite marks (and all inflicted injuries) is very important – include scale
 - Most common perpetrator is TODDLER

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee

Non-Accidental Trauma Clinical Practice Guideline (continued)

Revisions: February 19, 2016; July 15, 2016; July 2017

D. Burns

- Characteristics of inflicted burns:
 - Sharply demarcated edges
 - Paucity of splash marks
 - Burns in the genital area
 - Predominance of second and third degree burns
- The appearance of a burn can change rapidly (in a matter of hours). Photo documentation should be done IMMEDIATELY if the child is otherwise stable.
- Abuse work-up should be done and would consider skeletal survey especially in the 2-3 year age group

E. Abusive Head Trauma (AHT)

- **#1 cause of death in child physical abuse**
- Global brain injury caused by rotational forces
- Involves shaking, impact, or both
- Subdural hematomas, +/- retinal hemorrhage, bruising, fractures
- “Shaken baby syndrome”
 - Often triggered by crying
 - Typically, not a one-time event
 - Frequently, these children present with NO history of trauma
 - Beware of vomiting without diarrhea – this is a commonly missed presentation for abusive head injury.
 - AHT and young age are significant risk factors for **seizures** in children with TBI and up to 77% of victims of AHT have been report to have either clinical or subclinical seizures.
 - Beware of rapid increase in head circumference.
 1. Consider head CT in any child who has crossed two major percentile lines in head circumference over a short time (a few months).
 2. Head ultrasound is known to miss subdural especially as child gets older and fontanelle gets smaller.

F. Abdominal Trauma

- Any abrasion or bruise on the abdominal area should prompt labs
- If labs are abnormal, then imaging should be obtained
 - In cases of abdominal trauma, liver enzymes > 80 has been associated with forensically significant abdominal trauma (i.e., liver contusion) that may not necessarily require surgical intervention.
 - Hollow organ injury could still be a concern with otherwise normal labs, so if significant abdominal bruising or concerning abdominal exam, would still obtain imaging despite normal screening labs.
 - FAST scan/ultrasound is not sufficient.

G. Skeletal Trauma

- When evaluating skeletal injuries, the physician should consider the location of the fracture, the mechanism given, the child's chronologic age and developmental status.
- Specificity of radiographic findings in infants and toddlers (highest specificity applies in infants).

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee Non-Accidental Trauma Clinical Practice Guideline (continued)

Revisions: February 19, 2016; July 15, 2016; July 2017

High specificity	Moderate specificity	Low specificity, but common
Metaphyseal corner fractures Rib fractures, especially posteromedial Scapular fractures Spinous process fractures Sternal fractures	Multiple fractures, especially bilateral Fractures of different ages Epiphyseal separations Vertebral body fractures and subluxations Digital fractures Complex skull fractures	Subperiosteal new bone formation Clavicular fractures Long-bone shaft fractures Linear skull fractures

- Additional fracture information:
 - **Buckle fractures** in any child under 9 months of age are of concern due to a lack of mobility and underdeveloped protective reflexes that might lead to an “accidental” mechanism such as falling on an outstretched arm.
 - **Spiral fractures** can occur from seemingly innocuous trauma such as tripping while running and are not always indicative of abuse.
 - **Transverse fractures** – the mechanism should reflect the specific type and magnitude of forces required to cause this specific fracture morphology.
 - **Metaphyseal corner fractures** are highly suspicious for abuse. Often associated with soft tissue injury as well.

Section II

A. Admission to the Children's Hospital

- If the child's condition/presentation/injuries meet **Pediatric Trauma Activation Criteria** (Trauma Alert or Trauma Alert Red), then the patient is to be evaluated in the trauma resuscitation area of the emergency room, and the **ED Pediatric Trauma Process Guidelines** must be followed.
- Children with acute injuries should be admitted to the **Pediatric Trauma Service**.
- Children without acute injuries may be admitted to the **Pediatric Hospitalist Service**.

B. Documentation

- Questions to Ask
 - Who is/are the primary caregivers?
 - When did the caregivers first notice symptoms/bruises?
 - Does the history change with changing information given to the caregiver?
 - Do different witnesses give different accounts?
 - What did they do after they noticed these symptoms?
 - When was the last time the child was acting normally?
 - In young infants it can be difficult to tell
 - Tracking, cooing, smiling, eating without vomiting
 - Have there been any accidents?
 - Are there other children at home or not living with the family?
 - Has the child had any injuries before?
 - Is there a history of SIDS or any other unexpected death of a young child?

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee Non-Accidental Trauma Clinical Practice Guideline (continued)

- b. What to Document
 - Detailed physical examination documentation with appropriate drawings
 - Be sure the child is completely unclothed during examination and the lights are on in the room
 - Document location, size, and shape of all bruising or unusual markings
 - A careful and well documented history
 - Use quotes whenever possible
 - Document detailed descriptions of the mechanisms of injury or injuries with inclusion of the progression of symptoms
 - Also make note of what the caregiver did after these events/symptoms
 - Photo Documentation
 - Photo-documentation as soon as possible is becoming the standard of care.
 - The **pediatric forensic medicine team** or **hospital social workers** are typically available to take photographs and appropriately store the images.
 - Informed consent is not required in open investigations of child abuse
 - Photograph injuries:
 - Prior to treatment
 - Turn on the lights in the room
 - Document the patient's name, DOB, MRN, etc. (this can be accomplished by the first picture being of the patient's sticker).
 - Should have patient's face then a full body photo (with appropriately covered genitals).
 - Each injury should be photographed at least 2 times: 1 establishes location of injury on the body and the second is closer up to obtain detail and measurement.
- C. Use a ruler or measurement device to give perspective. A paper measuring tape used for measuring head circumference with a patient label attached is a great way to accomplish this.
- D. **Work-up for Child Physical Abuse**
 1. **≤ 12 months**
 - Skeletal survey
 - Consult Orthopedic Surgery if abnormal findings identified
 - Follow-up skeletal survey should be performed within 10 to 14 days-to be arrange by Forensics
 - Trauma labs
 - CBC, CMP, PT, PTT, lipase, and urinalysis
 - The urinalysis is looking for blood and preferably a bag specimen
 - Consider toxicology screen
 - Consider CK and platelet function analysis if extensive bruising
 - CT of head without IV contrast or MRI
 - If head/neck/ear/face bruising or swelling (however if present for >3 days, admission for MRI)
 - If signs or symptoms of neurological impairment present
 - If CT head obtained at outlying hospital, speak with radiologist for an official UK over-read.
 - Must speak with radiologist and need to have images and report (if available).

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee

Non-Accidental Trauma Clinical Practice Guideline (continued)

- If radiologist agrees an over-read is needed, then either radiologist or requesting physician can call imaging records desk at 3-2131 to get the study assigned an accession number.
- Immediate Neurology consult if abnormal finding identified
- Needs to be placed on continuous EEG for at least 24 hours or until evaluated by Neurology-if continuous EEG not available, needs, at minimum a routine EEG.
- Antiepileptic drugs to be given for positive findings of seizure activity or for positive EEG
- Admit to PICU if GCS <13 **or** findings identified on head imaging
- Immediate Forensic Medicine consult if findings identified on head imaging
- MRI of head
 - If evidence of trauma on physical exam
 - If head CT is abnormal at all, a MRI should be obtained
 - This often requires sedation and can be performed after the child is admitted
- Dilated fundoscopic exam
- CT of abdomen/pelvis with IV contrast – if trauma labs abnormal
- Consult Pediatric Forensic Medicine Team for any of the following:
 - Any skin findings
 - Positive trauma labs
 - Positive skeletal survey
 - Positive head imaging
 - To arrange follow-up skeletal survey
 - Any questions/concerns regarding family or work-up
- Consult Pediatric Surgery for any of the following:
 - Positive trauma labs
 - Bruising on abdomen/trunk
 - Bilious vomiting

2. 13 to 24 months

- Skeletal survey
 1. Consult Orthopedic Surgery if abnormal findings identified
 2. Follow-up skeletal survey should be performed within 10 to 14 days
- Trauma labs
 1. CBC, CMP, PT, PTT, lipase, and urinalysis
 2. The urinalysis is looking for blood and preferably a bag specimen
 3. Consider toxicology screen
 4. Consider CK and platelet function analysis if extensive bruising
- CT of head without IV contrast
 1. If head/neck/ear/face bruising or swelling
 2. If signs or symptoms of neurological impairment present
 3. Consult Neurosurgery if abnormal finding identified
 4. Immediate Neurology consult if abnormal findings identified
 - Consider immediate EEG per Neurology, but recommend EEG during admission
 - Antiepileptic drugs if seizure activity present or EEG abnormal
 - Consider admission to PICU
 5. If head CT is abnormal at all, a MRI should be obtained

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee Non-Accidental Trauma Clinical Practice Guideline (continued)

- Dilated fundoscopic exam – if brain injury present
- CT of abdomen/pelvis with IV contrast – if trauma labs elevated

- Consult Pediatric Forensic Medicine Team for any of the following:
 1. Any skin findings
 2. Positive trauma labs
 3. Positive skeletal survey
 4. Positive head imaging
 5. To arrange follow-up skeletal survey
 6. Any questions/concerns regarding family or work-up
- Consult Pediatric Surgery for any of the following:
 1. Positive trauma labs
 2. Bruising on abdomen/trunk
 3. Bilious vomiting

- 3. 2 to 5 years**
 - Skeletal survey
 1. If severe trauma
 2. If child is non-verbal, unresponsive, or developmentally delayed
 3. Consult Orthopedic Surgery if abnormal findings identified
 4. Follow-up skeletal survey should be performed within 10 to 14 days
 - Trauma labs
 1. CBC, CMP, PT, PTT, lipase, and urinalysis
 2. The urinalysis is looking for blood and can be a bag specimen
 3. Consider toxicology screen
 4. Consider CK and platelet function analysis if extensive bruising
 - CT of head without IV contrast
 1. If signs or symptoms of neurological impairment present
 2. Consider for head/neck/ear/face bruising or swelling
 3. Consult Neurosurgery if abnormal finding identified
 4. Consult Neurology
 - Consider EEG
 - Antiepileptic drugs if seizure activity present or EEG abnormal
 5. If head CT is abnormal at all, a MRI should be obtained
 - Dilated fundoscopic exam – if brain injury present
 - CT of abdomen/pelvis with IV contrast – if trauma labs abnormal
 - Consult Pediatric Forensic Medicine Team for any of the following:
 1. Any skin findings
 2. Positive trauma labs
 3. Positive skeletal survey
 4. Positive head imaging
 5. To arrange follow-up skeletal survey
 6. Any questions/concerns regarding family or work-up

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee

Non-Accidental Trauma Clinical Practice Guideline (continued)

- Consult Pediatric Surgery for any of the following:
 1. Positive trauma labs
 2. Bruising on abdomen/trunk
 3. Bilious vomiting

4. Follow-up skeletal survey

- Should be performed 10 to 14 days after the initial skeletal survey.
- The hospital social worker will schedule this appointment prior to discharge.
- The patient must follow up in Pediatric Forensics Clinic to be examined and to have study radiographs completed.

Suggested Guidelines for Evaluation of Suspected Child Physical Abuse

	Head CT or MRI	Skeletal Survey	Labs*	Abdominal CT	Fundoscopy Exam	Social Work Consult
0-1 year	Yes	Yes	Yes	If abdominal bruising or elevated transaminases	Yes	Yes
1-2 years	Consider [†]	Yes	Yes	If abdominal bruising or elevated transaminases	If positive neuroimaging or evidence of head trauma [†]	Yes
2-5 years	No	If extensive injury or developmental delay	Yes	If elevated transaminases or suggested by physical exam	Not typically recommended	Yes
Above 5 years	No	No	No	If suggested by physical exam	Not typically recommended	Yes

* Labs include CBC, CMP, Lipase, PT, PTT and UA [†] Indicated if decreased mental status, skull fracture, or head injury

E. Reporting

a. All healthcare providers are mandated reporters.

- b. First call should always be to the hospital social worker (in-house or on call)
 - The social worker will interview the family
 - They can help facilitate communication with DCBS and the police
 - Pager: 330-0331
 - Phone: 218-5587

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee

Non-Accidental Trauma Clinical Practice Guideline (continued)

- c. In Fayette County:
 - Protection and Permanency
Phone(859)245-5258
- d. Child Protection Hot Line
 - They will send to appropriate county or give you the phone number.
 - Phone: (800) 752-6200
- e. If outside the county, may go to the following address:
 - https://apps.chfs.ky.gov/Office_Phone/index.aspx
- f. University of Kentucky Pediatric Forensic Medicine Team
 - Phone: (859) 218-6727
 - Call UK•MDs at 1-800-888-5533 and ask for clinician on call
- g. Remember: the DCBS report must be made in the county where the event occurred!!

Section III

Kentucky Unified Juvenile Code

A. KRS Chapters 600 to 645

- a. Laws that declare a child's right to be free from abuse and neglect.
- b. Requires the reporting of neglect, physical, sexual or emotional abuse, and dependency of children.
- c. Requires that reports will be assessed and investigated.
- d. Requires that social services will be provided to victims of children maltreatment.

B. KRS 620.030(1)

- a. The law states that it is the duty of everyone who has reasonable cause to believe that a child is dependent, abused or neglected to report this information.
- b. *Any person who knows or has reasonable cause to believe that a child is dependent, neglected or abused shall immediately cause an oral or written report to be made to a local law enforcement agency or the Kentucky State Police; the Cabinet or its designated representative; the commonwealth's attorney or the county attorney; by telephone or otherwise...*

C. KRS 620.040 (5)(b)

- a. Medical personnel have the right to hold a child whom they feel is in imminent danger.
- b. *If a child who is in the hospital or under the immediate care of a physician appears to be in imminent danger if he is returned to the persons having custody of him, the physician or hospital administrator may hold a child without court order, provided that a request is made to the court for an emergency custody order at the earliest practicable time, not to exceed seventy-two (72) hours.*

D. KRS 620.050(1)

- a. Immunity from prosecution is given to any person making a report or assisting legal authorities or the child protection program in making an assessment.
- b. *Anyone acting upon reasonable cause in the making of a report or acting under KRS 620.030 to KRS 620.050 in good faith shall have immunity from any liability, civil or criminal, that might otherwise be incurred or imposed. Any such participant shall have the same immunity with respect to participation in any judicial proceeding or resulting from such report or action.*

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UK HealthCare Pediatric Trauma Service Child Protection Team Subcommittee

Non-Accidental Trauma Clinical Practice Guideline (continued)

- E. KRS 620.990(1)**
- a. The failure to report or falsely reporting child abuse or neglect can result in criminal charges.
 - b. *Any person intentionally violating the provisions of this chapter shall be guilty of a Class B misdemeanor. A class B misdemeanor carries a penalty of up to 90 days in jail and/or a fine of up to \$250.*
- F. KRS 214.185(4)**
- a. allows a professional to provide medical, dental, and other health services to minors without a parent or guardian's consent if (1) requiring consent would lead to a delay in or the denial of treatment and (2) the risk to the minor's life requires treatment without delay.
- G. KRS 214.185(7)**
- a. notes that except as otherwise provided in KRS 214.185, CHFS or a custodian or guardian of a minor will not be financially responsible for services rendered to a minor without consent unless such services were essential to preserve the minor's health.

UK HealthCare Reference Contact: In the event any UK HealthCare employee has a question regarding legal or risk management they may contact:

Paula J. Holbrook, RN, BHS, JD, CPHRM
Associate General Counsel, University of Kentucky
Clinical Risk Manager, Assoc. Director, Risk Management
800 Rose Street, Room HG 608
Lexington, Kentucky 40536
859-257-6212 859-257-2498 facsimile

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

DETERMINING A NEAR FATALITY TIPSHEET

“Near fatality” is a term used in child protective services. It is **NOT** a medical term.

- Child Abuse Prevention & Treatment Act Section 106.B.4 defines near fatality as “an act that, as certified by a physician, places the child in serious or critical condition”.
- KRS 600.020(38) mirrors the federal definition as law in Kentucky.
- The Department of Community Based Services maintains the same definition of near fatality in their standards of practice.

Box #1

THE QUESTION TO ASK THE REPORTING SOURCE:

Is abuse or neglect by a caregiver suspected?

Box #2

THE QUESTION TO ASK THE PHYSICIAN TREATING THE CHILD:

Is the child or has the child been in serious or critical condition as a result of suspected abuse or neglect?

- If this is answered **YES** the case is automatically a **NEAR FATALITY**

If the answer to Box #2 is unclear, the following questions are to be asked of the physician treating the child:

Box #3

- **Have life-saving measures been performed?**
 - Use of Narcan, CPR, intubated, etc.
- **Will this child be admitted to the ICU (Intensive Care Unit) as a DIRECT RESULT of an alleged abusive injury or neglectful situation?**
 - The condition of the child admitted to the ICU must be considered. If the child is admitted for observation only, the condition may not meet that of a near fatality.
- **Will this child be emergently transferred to a referral/specialty hospital as a result of the alleged abusive injury or neglectful situation?**
 - The condition of the child should be the criteria for transfer. If the child is transferred solely because a medical facility is not available locally, it may not be a near fatality.
- **Is there or has there been a substantial risk of death as related to the child's condition?**

- If **YES** is answered to any question in Box #3, designate the case as a **NEAR FATALITY**. This will allow the investigation to determine whether the child was in serious or critical condition. Consultation with forensic professionals may be needed.
- If the service region needs additional assistance in making the determination, please contact the Child Fatality Program at (502) 564-6852.

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These guidelines are not intended as a directive or to present a definitive statement of the applicable standard of patient care. They are offered as an approach for quality assurance and risk management and are subject to (1) revision as warranted by the continuing evaluation technology and practice; (2) the circumstance; and (3) the patient's willingness to follow the recommended treatment.

SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

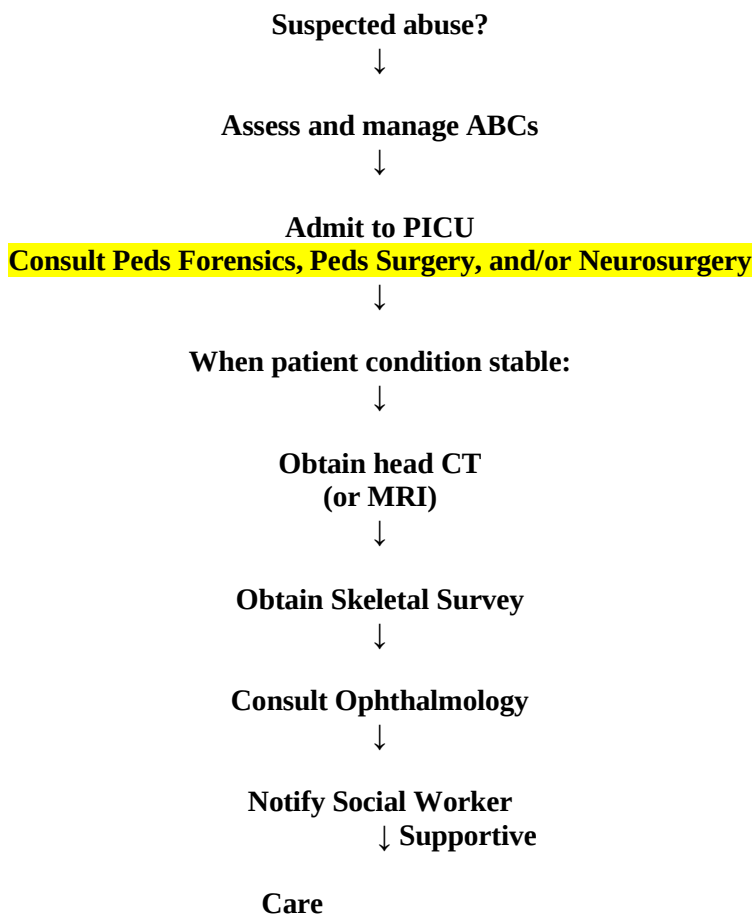
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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NAT

Altered Infant/Child



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Child Abuse Tip Sheet

Basic Principles: Does injury fit mechanism? Does developmental ability fit mechanism?

Presentation	Call PFM prior to	Work-up tips
Bruises		
Any bruising anywhere in child less than 4 months of age (or pre-mobile infant)	yes	Pediatric trauma survey, head imaging (CT or MRI), ophthalmology, CBC, coags, CMP, Lipase, urine, photographs
< 4 years of age and:		
- Face (excluding forehead), ears, mouth, neck and poorly explained	yes	head imaging for those less than 1 year of age and consider for >1 year, Pediatric trauma survey for < 2 years, trauma labs: CBC, coags, CMP, Lipase, urine, and photos
- Trunk or abdomen and poorly explained	yes	
- Pattern (belt, loop, parallel linear)	yes	
Burns		PHOTOGRAPHS
Any non-ambulating child (<1 year)	yes	Pediatric trauma survey, head imaging (CT or MRI), trauma labs. PHOTOS
Concerning pattern:		
- Genitalia	yes	Age appropriate work-up (pediatric trauma survey for < 2 years, trauma labs)
- Circumferential	yes	
- Hot object	yes	
- Classic Immersion pattern	yes	
Fractures		
Poorly Explained (child <4)	consider	Age appropriate work-up, consider bone labs (CA, PO4, 25 OH Vit D, PTH,
Any fracture 0-6 months	yes	Trauma survey, head imaging (CT or MRI), trauma labs, bone labs
Any fracture 6 months-1 year	yes	Trauma survey, trauma labs, consider head imaging
Any fracture 1yr-2 years	if concern	if poorly explained: trauma survey, trauma labs
Other		
sibling evaluation or R/O NAT	Not if w/u neg	<1 year: thorough skin assessment, pediatric trauma survey, trauma labs, consider head imaging. <i>Twins are special case-head imaging on both if one with injury.</i> 1y-2 yrs: thorough skin assessment, trauma survey, CBC, coags,
Neglect: question of adherence, supervision, delay of care, near drowning, ingestion,	No	consider urine tox
Known serious abuse	Yes	urine tox screen, appropriate trauma work-up for age
Intracranial bleed with or without fracture (excluding MVA, obvious accidental or known	Yes	MRI brain, pediatric trauma survey, trauma labs

***Not Intended to replace full guidelines, ED Pediatric Trauma Process Guidelines must be followed.**

-Children with acute injuries should be admitted to the **Pediatric Trauma Service**.

-Children without acute injuries may be admitted to the **Pediatric Hospitalist Service**.

*Hospital social worker should be involved in **all** above cases

*Contact Clinician on call for Pediatric Forensic Medicine through UK MD (859-257-1000), available 24/7 for questions/concerns

*Remember it is your duty to report if you have concern regarding abuse or neglect. KY CPS hotline: 1-877-KYSAFE1 (1-877-597-2331).

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FEVER

Fever Management with Tylenol/Ibuprofen

Procedure

- Medication Administration:
 - Administer Tylenol calculated at 10-15 mg/Kg po/pr administered as first-line medication intervention to treat fever ≥ 101.5 degrees F. If more than 3-4 hours since last dose, repeat appropriate Tylenol dose.
- If above criteria has been done < 3-4 hours, may give Ibuprofen 10 mg/Kg for fever greater ≥ 101.5 under following conditions:
 - No drug sensitivity to Ibuprofen or ASA
 - No history of GI ulcers or bleeding problems

General

- If Tylenol or Ibuprofen were administered prior to arrival to the hospital, discuss the dose administered with the caregiver to ensure that appropriate dose was given.
- Consult attending M.D. for Ibuprofen administration, if child is less than 3 months old.

Hyperthermia Management

- The patient with a temperature >104 degrees, hypotension, and decreased LOC will require emergency treatment:
 - Assess patient's temperature if possible, otherwise assess by symptoms.
 - Maintain ABC's, cardiac monitoring, and seizure precautions.
 - Attempt to lower body temperature by removing clothes, pouring tepid fluid over body, and allowing to evaporate or placing ice to groin and axilla if available.
 - Monitor core temperature if possible.
 - Cool IV Fluids should be administered at a high infusion rate.
 - High flow oxygen should be given.
 - NG lavage with iced saline and foley irrigation with iced saline is indicated if temp >106 degrees
 - Notify receiving hospital of need for cooling blanket if necessary.
 - Document assessments, actions, and results.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FEVER

Hypothermia Management

- **Information**

- Any individual will become hypothermic given the proper circumstances.
- There are specific high risk groups: infants, elderly, alcoholics, trauma victims, outdoor people, and those with CNS dysfunction.

- **Procedure**

- **Mild Hypothermia >90 degrees Fahrenheit (32C)**
 - Handle patient gently
 - Insulate patient from cold
 - Warm humidified oxygen
 - Warm IV fluids, active external re-warming
- **Moderate Hypothermia <90 degrees Fahrenheit (vital signs present)**
 - Handle patient gently
 - Insulate from cold
 - Warm humidified oxygen
 - Warm IV fluids, active re-warming of truncal areas only
 - **DO NOT** administer medication until temperature > 90 degrees Fahrenheit
- **Severe Hypothermia (vital signs absent)**
 - Assess pulse, respirations, and cardiac rhythm for 1-2 minutes
 - If no pulse or respirations and asystole, begin CPR
 - If VF defib up to 3 shocks (2-4 Joules/Kg)
 - Check rectal core temperature
 - Intubate and administer warm humidified oxygen
 - Administer warm IV fluids
 - NGF lavage with warm saline and foley irrigation with warm saline

***Note:** Discretion should be used when determining whether or not to transport a patient whose core temperature is <90 degrees Fahrenheit.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NEAR DROWNING

Near Drowning Management

- **Information**
 - Injury due to submersion is usually the sole problem seen in the near-drowning victim.
 - It is important to assess the near drowning victim for other associated injuries.
 - Trauma to the head, neck, or other organs may have occurred if the patient fell or was diving into the water.
 - A history of drug or alcohol ingestion should be sought, as well as a history of myocardial disease, hypoglycemia, and loss of consciousness, seizures, or other serious illness.
 - The patient should be assessed and treatment priorities assigned.
- **Instructions**
 - If the patient is in arrest, treat according to the Pediatric Advanced Life Support guidelines.
 - If the patient is not in full arrest, the patient should be managed according to the airway, breathing, and circulation pathway.
 - Airway
 - Airway secured along with cervical spine and maintained
 - Breathing
 - Evaluate breathing and manage according to breathing management guidelines.
 - ALL patients suffering from Near Drowning should receive 100% oxygen
 - Patients with spontaneous breathing and are alert should be supplemented with a non-rebreather.
 - Apneic patients should be provided with source of mechanical ventilator support.
 - 100% oxygen should be continued throughout transport period.
 - If oxygenation remains poor, judicious application of PEEP should be considered.
 - Bronchospasm may occasionally occur after near drowning and if so should be treated with bronchodilator
 - Pulmonary Edema accompanying near drowning is best treated with PEEP or CPAP.
 - If signs and symptoms of increasing intracranial pressure are present from a concurrent head injury, then controlled hyperventilation to obtain a pCO₂ of 30-35 mmHG should be employed.
 - Circulation
 - Impaired circulation due to cardiac problems should be treated according to PALS guidelines.
 - If circulation dysfunction is secondary to hypovolemia, resuscitation with crystalloid is indicated.
 - Normally, maintenance fluids are required
 - Patients who are comatose or cerebral edema is suspected, IV maintenance fluids should be utilized.
 - Hypothermic patients may become hypovolemic through inhibition of ADH secretion.
 - Illness and injuries should be sought and treated according to protocol
 - CPR should continue until patient is normothermic.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: VTE

Venous Thromboembolism Prophylaxis for Pediatric Trauma Patients

Objectives

- To assess the risk of venous thromboembolism (VTE) in pediatric trauma patients.
- To identify pediatric trauma patients who should receive VTE prophylaxis.

Pediatric Venous Thromboembolism

- VTE events are rare in pediatric trauma. The incidence of VTE is less than 1% in the general pediatric trauma population.
- Although rare, VTEs are potentially serious events.
- Pharmacologic prophylaxis against VTE is a reasonable strategy in critically injured adolescent trauma patients.
- The effectiveness of prophylaxis compared to its potential harms has not been proven in this population.
- Attending physician discretion should always guide complex risk assessment.

VTE Prophylaxis

- Begin chemical prophylaxis if following criteria met:
 - Age $\geq$ 13 years
 - Non-weight bearing pelvic fracture, femur fracture, or bilateral lower extremity fractures
- Consider VTE prophylaxis if:
 - Major polytrauma
 - Spinal cord injury
 - PICU admission $\geq$ 36 hours
 - Morbid obesity
 - Personal history of DVT or PE
- Preferred medication is low-molecular-weight heparin (LMWH, enoxaparin).
 - Start LMWH at 0.5 mg/kg Q 12H (up to 30mg SQ Q 12H)
 - If weight > 60 kg and BMI > 40, start LMWH 40mg SQ Q12H
- Continue LMWH until orthopedic follow-up (no US needed).

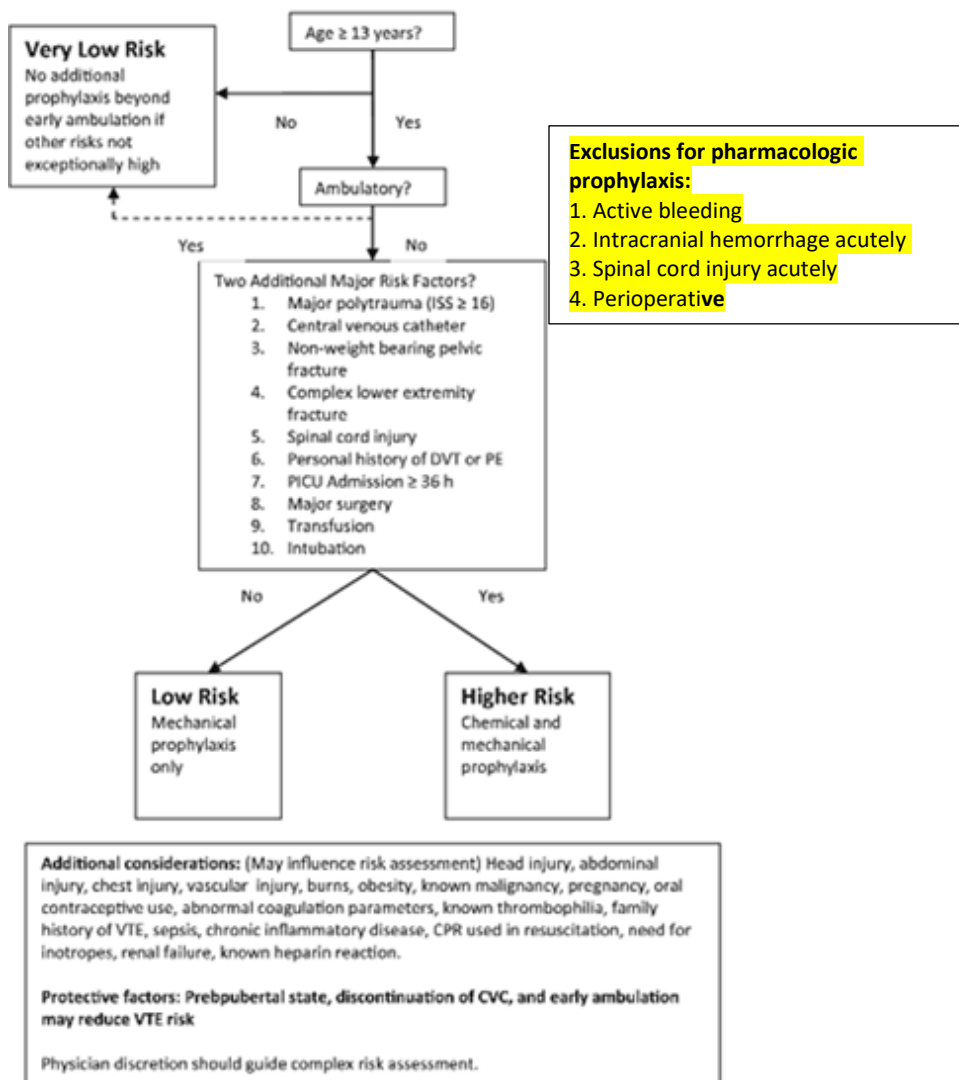
Exclusions for Pharmacologic Prophylaxis:

- Active bleeding
- Intracranial hemorrhage acutely (36-48h) and per neurosurgery.
- Spinal cord injury acutely and per spine service.
- Perioperative:
 - General surgery – day of operation
 - Neurosurgery – day of and 3 days after
 - Orthopedic surgery – day of operation
- Use SCDs if patient has contra-indication to LMWH.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: VTE continued

Venous Thromboembolism Prophylaxis for Pediatric Trauma Patients



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: Blood & Bleeding Control

Anticoagulant Therapy, Anticoagulant Reversal Agent, Transfusions, Blood Products

- The University of Kentucky HealthCare Enterprise maintains protocols, guidelines, policies and procedures in digital format and are accessible via the Care Web
- These policies and procedures
- The following site via the Care Web accesses the UKHC Anticoagulation Stewardship guidelines that are enterprise standards: <http://careweb/ICISdocs/anticoagulation.html>
- Other sites related to anticoagulant therapy, transfusion, blood products, and bleeding management:
 - <http://careweb/ICISdocs/Pediatric%20Heparin%20Protocol.pdf>
 - <http://careweb/ICISdocs/anticoagulation.html>
 - <http://careweb/ICISdocs/VTE%20updated%20reference%2011.25.pdf>
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 - https://ukhealthcare.mc.uky.edu/policies/enterprise/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/enterprise/Enterprise/A08-125%20Blood%20or%20Blood%20Product%20Transfusions.docx&action=default&DefaultItemOpen=1
 - <http://careweb/ICISdocs/anticoagulation.html>
- UK HealthCare Policy A14-035, “Heparin Continuous Infusion” is accessible via the Care Web: https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A14-035_Heparin_Continuous_Infusion.docx
- The following link is to the Nursing guideline on VTE Prevention. This is found on the Care Web through Hospital Policy and Procedure: [https://ukhealthcare.mc.uky.edu/policies/nursing/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/nursing/Nursing%20Guidelines/Venous%20Thrombosis%20Embolic%20\(VTE\)%20Guideline.docx&action=default](https://ukhealthcare.mc.uky.edu/policies/nursing/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/nursing/Nursing%20Guidelines/Venous%20Thrombosis%20Embolic%20(VTE)%20Guideline.docx&action=default)

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: SNAKEBITE

Crotalidae Snakebite Guideline

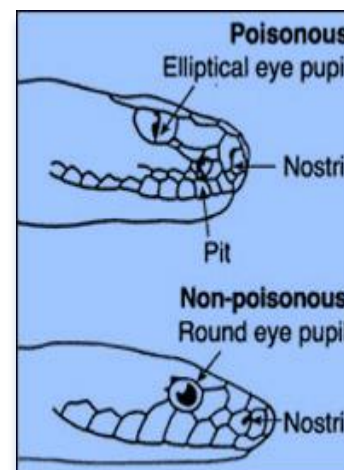
Description

The majority of snakebites in North America are caused by the Crotalidae family, including Copperheads and Rattlesnakes.

- Mortality is rare, less than 1%.
- The mainstay of therapy to attenuate morbidity is antivenom.

Guideline

- Assess and secure airway. Apply oxygen if respiratory distress noted.
- Assess hemodynamic status: Give 20 cc/kg NS if tachycardic or poor perfusion.
- Follow PALS guidelines if further resuscitation necessary.
- Assess area of bite for fang marks, erythema, and ecchymosis.
- The area should be accurately measured and recorded.
- Assess neurovascular status. Do not apply ice or tourniquets.
- Remove jewelry and clothing. Cleanse wound area – leave unbandaged.
- Position or splint limb in gravity neutral position.
- Attempt to identify snake species as well as time of envenomation.
- Antivenom is more effective within 6 hours of envenomation.
- Monitor affected extremities for signs of compartment syndrome.
- 5(P's) Pain out of proportion to what is expected, Paresthesia, Pallor, Paralysis, Pulselessness (pulselessness is a very late sign of compartment syndrome).
- Initial lab work should include CBC with diff, Platelet count, PT/PTT, Fibrin Split Products (FSP).
- Consider CMP, Type and Cross, UA, Stool Hemocult, and ABG if in respiratory distress.
- Administer Crofab for moderate to severe envenomations per pediatric dosing Guide



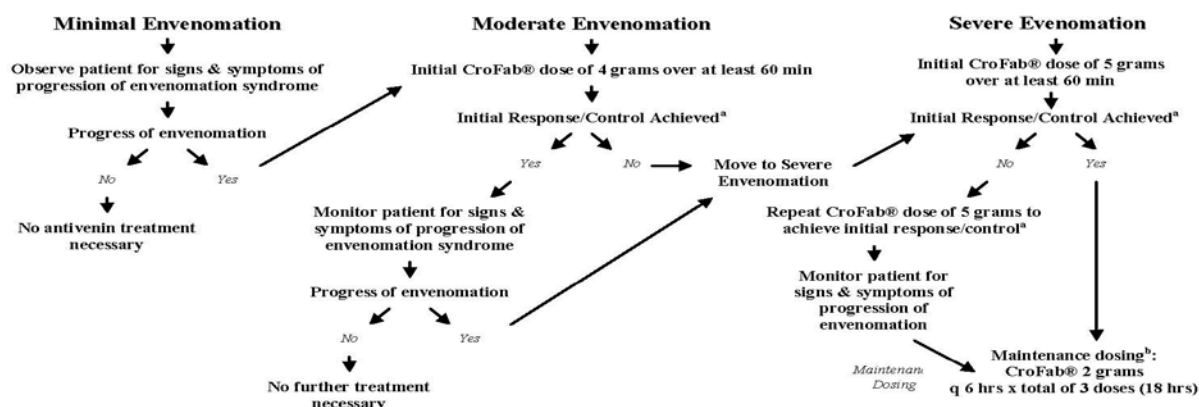
Type of Signs or Symptoms	Severity of Envenomation		
	Minimal	Moderate	Severe
Local	Swelling, erythema, or ecchymosis confined to site of the bite	Progression of swelling, erythema, or ecchymosis beyond site of the bite	Rapid swelling, erythema, or ecchymosis involving the entire body part
Systemic	No systemic signs or symptoms	Non-life threatening signs and symptoms (N/V, mild hypotension, perioral paresthesias, myokymia)	Markedly severe signs and symptoms [hypotension (SBP < 80 mmHg), altered sensorium, tachycardia, tachypnea, and resp distress]*
Coagulation	No coagulation abnormalities or other lab abnormalities	Mild abnl coagulation profile without significant bleeding	Abnl coag profile with bleeding [↓ INR, APTT, fibrinogen; plt count <20,000 per mm ³]
Snakebite Severity Score (SSS)*	0-3	4-7	8-20

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: SNAKEBITE

Treatment Algorithm for Snakebites

Treatment algorithm:



^a Initial response or control: Initial control is cessation of progression of local effects, systemic effects, and coagulopathy from envenomation. Patients should be monitored up to one hour following FabAV dosing to assess initial response/control.

Clinical response: Pre-treatment signs and symptoms of envenomation were arrested or improved after treatment.

Partial response: Signs and symptoms of envenomation worsened, but at a slower rate than expected after treatment.

Non-response: Patient's condition was not favorably affected by treatment.

^b Patients with documented rattlesnake envenomations should be evaluated for scheduled maintenance dosing to prevent recurrence of envenomation.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FOREIGN BODY REMOVAL

Foreign Body Removal

Esophageal

Presentation

- Child (6 mo to 5 yo) noted to swallow foreign body (coin, toy, marble, battery) by parents or family. Initially may have choking or gagging. Can have persistent drooling.
- **NOTE:** if the object is a battery, it needs **immediate** removal – see below.

Physical Exam

- Examine oropharynx, look for excessive drooling, palpate for abdominal pain.

Radiology

- CXR PA/LAT including upper abdomen. Most objects are visible on plain radiographs. Can be located at thoracic inlet, mid esophagus or stomach.

Management

- If object is lodged in upper or mid esophagus will need endoscopic removal, either rigid or flexible. If intragastric most objects will pass and removal not necessary and can have repeat films in 3-4 weeks by PCP to see if passed. No need to sift through stool by parents.

Admit orders

- Admit as observation
- Repeat films if done at outside or greater than 4 hours old
- NPO
- Maintenance IVF
- Consent for endoscopic foreign body removal

Surgical procedure

- After anesthesia, place shoulder roll. Place either rigid or flexible scope through oropharynx into esophagus. Search circumferentially for object, may use grasping forceps or net to carefully remove object. Once object removed re-scope to see if any mucosal injury.

Post-op

- CXR in recovery looking for mediastinal air: if +, then NPO and watch
- If CXR clear may start liquids and advance as tolerated. May d/c home.

Discharge Instructions

- Follow-up with PCP, if intragastric foreign body then film in 3-4 weeks to see if object passed.
- NO Peds surg follow-up needed.

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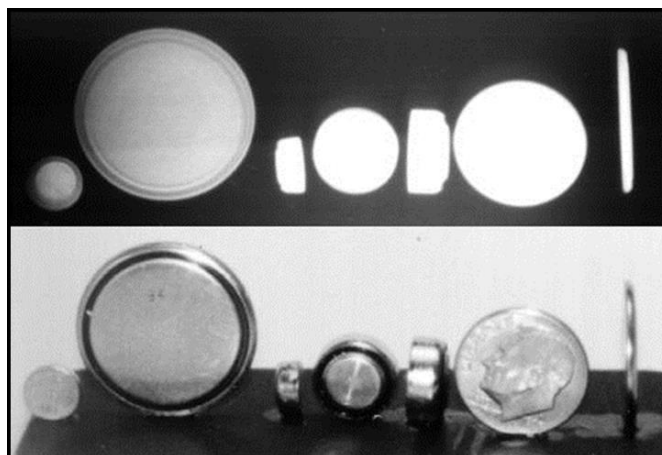
SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FOREIGN BODY REMOVAL

Foreign Body Removal (continued)

Special Situations

Disk Batteries

- Esophageal batteries should be removed immediately.
- Distal batteries can be managed with observation and weekly radiographs to ensure passage.
- Lithium cell ingestions associated with most severe outcomes.
- Higher risk of perforation, erosion, fistula, stenosis if lodged in esophagus.
 - Electricity flows between both battery poles through contact of tightly surrounding esophageal walls may cause liquefaction necrosis and perforation.
 - Leakage of contents: acidic environment may erode seal of battery and release heavy metals and cause necrosis of membranes.
 - Physical compression of adjacent tissues leading to pressure necrosis.
- CXR findings:
 - Beware of the “coin fake out”
 - Look for “halo sign” of disk battery.
 - On lateral, battery shows step off appearance of edges, while coin has sharp edges.



Magnets

- Multiple magnets can attract across multiple loops of bowel causing ischemia, pressure necrosis, perforation, fistula formation, or volvulus.
- Single magnets shouldn't cause problems.
- Difficult to tell whether single or multiple magnets have been ingested.
- Suspect magnet ingestion if metallic object fails to progress.
- Any suspected magnet ingestions should be removed.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FOREIGN BODY REMOVAL

Foreign Body Removal (continued)

Airway

NOTE: Most airway foreign bodies are managed by the Peds ENT service. The PDS service will manage these patients if Peds ENT is not available.

Presentation

- Child (6 mo to 5 yo) noted to aspirate foreign body or found to have spontaneous coughing and no foreign body witnessed.
- Initially may have choking or gagging, cyanosis or new onset wheezing.

Physical Exam

- On auscultation may have absent breath sounds or wheezing

Radiology

- PA/Lat CXR: object may be radio-opaque. Look for hyperinflation on one side, mediastinal shift, infiltrate or lobar collapse

Management

- To OR for rigid or flexible bronchoscopy and foreign body removal.

Admit orders

- Admit
- Repeat films if done at outside
- NPO
- Maintenance IVF
- Consent for bronchoscopy and foreign body removal

Surgical procedure

- After anesthesia, place shoulder roll. Place either rigid or flexible scope through oropharynx into trachea. Search circumferentially for object, may use grasping forceps or Fogarty catheter to carefully remove object. Once object removed re-scope to see if any mucosal injury or mucus plugging beyond object.

Post-op

- CXR in recovery, if lungs clear may advance diet.
- If lobar collapse or infiltrate, admit overnight and repeat film in AM.

Discharge Instructions

- Follow-up with PCP, may need repeat CXR in 4 weeks if lobar collapse to ensure adequate re-inflation.
- NO Peds surg follow-up needed.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: FOREIGN BODY REMOVAL

Foreign Body Removal (continued)

OR Equipment needed for Foreign Body Removal



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ANIMAL BITES

Animal Bites

- Animal bites are considered reportable cases under Kentucky state law.
- Communicable disease is any illness capable of being transmitted to a susceptible host. Animal bites are included.
- Physicians are to report name, age, sex, and precise location of the person bitten to victim's local health department within 12 hours of first seeing the patient (KRS 258.065).
- *UK HealthCare Enterprise* Policy provides information on "Reporting Communicable Diseases"
https://ukhealthcare.mc.uky.edu/policies/enterprise/_layouts/15/WopiFrame.aspx?sourcedoc=/policies/enterprise/Enterprise/A03-060%20Reporting%20Communicable%20Diseases.docx&action=default&DefaultItemOpen=1
- The Emergency Department Policy outlines the immediate information that is required for documentation and reporting.
<https://ukhealthcare.mc.uky.edu/policies/departamental/Emergency%20Department/Archived%20Policies/ED08-10-Guidelines%20for%20the%20Care%20of%20Animal%20and%20Human%20Bites.pdf?Web=1>
- The Emergency Department maintains a form that is to be completed for all animal bite cases and reported to health department. Forms are typically specific to the Health Department of county of residence.

LEXINGTON-FAYETTE COUNTY HEALTH DEPARTMENT

Division of Environmental Health & Protection
650 Newtown Pike, 2nd Floor
Lexington, KY 40508
Phone: (859) 231-9791 — Fax: (859) 231-9459

PROCEDURE FOR REPORTING ANIMAL BITES

• Under KRS 258.065 all incidents involving persons bitten by animals must be reported to the Health Department within twelve (12) hours after the physician's first attendance. The information that is to be reported is at the bottom of this page and in the adjacent column. If the person bitten does not know the name and address of the animal owner, you may instruct them to find out and call the Health Department with the information at: (859) 231-9791 from 8:00 AM to 4:30 PM, Monday through Friday.

• Do not mail this form to the Health Department, fax the information as soon as possible to (859) 231-9459. If you are unable to fax the report call the Health Department at: (859) 231-9791 from 8:00 AM to 4:30 PM, Monday through Friday.

ANIMAL BITE INFORMATION

DATE OF BITE: _____

CIRCUMSTANCES OF BITE: _____

PART OF BODY BITTEN: _____

Attending Physician's Name: _____

Reported by:

☐ Doctor

☐ Hospital

☐ Emergency Room

☐ Other

If unable to FAX: HEALTH DEPARTMENT CALLED ON: ☐ AM ☐ PM

DATE: _____

LHD-407 (08/2016)

PERSON BITTEN:

NAME: _____

(If a child) AGE: _____ PARENT'S NAME: _____

ADDRESS: _____

DAY TELEPHONE NO. () _____

ANIMAL OWNER:

NAME: _____

ADDRESS: _____

DAY TELEPHONE: () _____

ANIMAL DESCRIPTION:

TYPE: _____

BREED: _____

SIZE: _____ COLOR: _____

SEX: ☐ MALE ☐ FEMALE

ANIMAL'S NAME: _____

STRAY ANIMAL ☐

COMMENTS: _____

- Information pertinent to the case for documentation and reporting:
 - Date, time, and circumstances of bite wound
 - Location wound (s), complete description
 - Tetanus/Immunization status
 - If Animal: breed, domestic or wild, owner information, rabies vaccination history
- Rabies Vaccination Guide found here:
<https://ukhealthcare.mc.uky.edu/policies/departamental/Emergency%20Department/ED08-17%20Guide%20for%20Recommended%20Rabies%20Prophylaxis.docx?Web=1>
- Patients are at significant risk for infections related to the bite.
- Overall wound care and medication therapy is high priority.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: REFERENCE LINKS

UK HealthCare Enterprise and Kentucky Children's Hospital Reference Links

- All enterprise policies and guidelines can be easily found and accessed from the Care Web.
- The following are links that may be useful to have quickly at hand for efficiency in decision making during care of trauma patients:
 - UKHealthCare Policy Web: <https://ukhealthcare.mc.uky.edu/policies>
 - Autopsies: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-120%20Autopsy%20Procedures.docx?Web=1>
 - Donation After Cardiac Death: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-075%20Donation%20after%20Cardiac%20Death.docx?Web=1>
 - Organ Donation and Transplant Procurement: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-125%20Organ%20and%20Tissue%20Procurement%20for%20Transplantation.docx?Web=1>
 - Protection of Minors; Reporting Abuse in Children: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-145%20Protection%20of%20Children%20-%20Reporting%20Child%20Abuse%20Neglect%20and%20Abandonment.docx?Web=1>
 - Post Death Procedures: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-155%20Post-Death%20Procedures.docx?Web=1>
 - Diagnosis of Death: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-010%20Diagnosis%20of%20Death.docx?Web=1>
 - Consent: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-000%20Consent%20to%20Treatment.docx?Web=1>
 - Family Presence in Resuscitation: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-165%20Reports%20to%20Law%20Enforcement%20Officials.docx?Web=1>
 Patient's Family Presence During Resuscitation: <https://ukhealthcare.mc.uky.edu/policies/enterprise/Enterprise/A06-180%20Presence%20of%20a%20Patient's%20Family%20during%20Resuscitation.docx?Web=1>
 - KCH Policies: <https://ukhealthcare.mc.uky.edu/policies/departamental/KY%20Childrens%20Hospital>

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: ETOH SCREENING

Screening and Brief Intervention Process for Alcohol Use in Pediatric Patients

- All patients **admitted*** to the UK HealthCare system are screened for the use of ETOH as part of the admission patient profile and in accordance with the standards of The Joint Commission regulations.
- The Pediatric Trauma Program requires that all pediatric patients age 12 years and older be screened (Resources for Optimal Care of the Injured Patient CD 18-3, 18-4)
 - All* patients who are admitted with an inpatient stay <24hours are included.
 - ED, Observation, and outpatients, while they may benefit from screening and intervention, are exempt from the ACS requirements.
- Pediatric Trauma Alerts and Trauma Alert Reds, age 12 years and above, are required to have an ETOH and UDS diagnostic tests ordered in the initial trauma lab order set.
- All admitted pediatric patients age 12 years and older shall receive screening via the nursing patient profile utilizing CRAFFT regardless of screening tool within the EHR (as use of the adult CAGE tool may result in false negatives). (www.med.dartmouth-hitchcock.org/documents/CRAFFT-adolescent-substance-use-screen.pdf)
 - Possible positive screens:
 - positive ETOH lab value
 - positive UDS lab value
 - positive CRAFFT screen
 - When a patient tests or screens positive:
 - Core Measure Nurses or Social Workers or other “appropriately trained” staff¹ conduct a brief intervention *within the inpatient area*, providing a written packet of information to the patient and parent or legal guardian.
 - The parent or legal guardian provides consent for any intervention; and ONLY the parent or legal guardian can decline intervention and should be clearly documented as such for any declination.
 - Child psychiatry¹ can be consulted at any time for risk assessment or intervention.
- Brief Interventions are to be documented in the patient’s medical record utilizing the electronic system.
- Through the Brief Intervention or SBIRT process, or at the discretion of the medical provider, a consult may be placed as needed for the patient’s further treatment.
- For individual’s testing positive and requiring further intervention after the SBIRT process, the medical provider can place an order for pediatric psychiatry consult or pediatric child psychologists consult.
- Social Workers play an active role in the process by assuring successful intervention for the patient when requested by the patient and/or the legal guardian.

Note: Any pediatric patient may have or require an ETOH or UDS performed based on injury or presenting illness. An example is: Non-Accidental Trauma Patients.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: PALLIATIVE CARE

Palliative Care for Pediatric Trauma Patients and Family

Overview

The American College of Surgeons on Trauma published the Palliative Care Best Practices Guidelines through their Trauma Quality Improvement Program (TQIP) in fall of 2017. Though not currently a mandate for Level I trauma centers, these guidelines are highly recommended for implementation in trauma programs to provide a continuum of parallel care for improving the quality of life for patients and families sustaining serious life threatening and or end of life circumstances.

The Kentucky Children's Hospital (KCH) currently has in place a Pediatric Advanced Care Team and Pediatric Palliative Care that incorporates all aspects of the ACS Palliative Care Best Practices recommendations. The KCH incorporates this practice and service for all KCH patients and families. This service is multidisciplinary and is delivered by all members of the pediatric health care team throughout the continuum of care. The KCH Pediatric Palliative Care also incorporates the needs of the healthcare providers into the program on an as needed basis.

KCH Pediatric Advanced Care Team/Pediatric Palliative Care

The mission of the Pediatric Advanced Care Team is to improve the quality of life of children with serious illness. We aim to provide palliative care by focusing on the physical, emotional, social and spiritual needs of our patients and their families. Patients can receive care that seeks to cure and extend life while at the same time use our services to improve quality of life. We work to accomplish our mission by:

1. Providing pain and symptom management
2. Assisting families with decision making and goals of care
3. Promoting communication with families
4. Facilitating continuity during multiple admissions
5. Providing emotional, social, spiritual and bereavement services
6. Referring to home nursing or hospice care agencies
7. Delivering education to health care staff
8. Performing outreach to community resources

Our team is an interdisciplinary consult team consisting of a physician, a nurse practitioner, a social worker and a pediatric psychologist. We share resources with the broader health care system including chaplaincy, child life, nutrition and ethics.

Available Multidisciplinary Resources

- Nurses, physicians, and all primary/clinical care providers and services
- UK Chaplaincy Service
- Social Workers, Case Managers
- Child Life
- Dietary/Nutritional Services
- Pain Management Team
- Psychiatry
- Pediatric Palliative Care/Pediatric Advanced Care Team Services
- UK HealthCare Policies and Procedures including end of life care

These services are not all inclusive but represent the primary multidisciplinary entities.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: PALLIATIVE CARE **continued**

Implementation

Palliative care is integrated in the overall culture of care for all KCH patients and is further integrated based on growth/age development. Therefore, there is no requirement of formed specific assessment completions to ascertain the needs of patients and or families. Needs of patient and families are identified through routine daily care and acted upon as identified. This practice encompasses the mission of the Pediatric Advanced Care Team/Pediatric Palliative Care.

Trauma Performance Improvement

- The appropriate utilization of palliative care services for the pediatric trauma patients/families is evaluated through the Trauma Performance Improvement Program (PIPS).
- PIPS are conducted by the Pediatric Trauma Coordinator reviews of all trauma identified patients.
- All trauma related deaths are reviewed for elements of incorporated Palliative Care and reported through the Pediatric Trauma Interdepartmental Quality Assurance Committee (Peds IDTQA).

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: PROCESS FOR DISCHARGE DISPOSITION FOR SPICA CASTED PATIENTS

(May 2019)

The process and equipment utilized for discharge transport of children in a spica cast is as follows:

1. Equipment Used EZ-vest (all sizes) and Radian r120 car seats. Pictures of the equipment are located at end of this guideline.
2. This equipment is routinely stocked equipment by KCH as this is part of the continuum of care process for these children. This decision was established by Enterprise Administration in 2012.
3. The ez vest are stocked in a locked cabinet just across the hall from the KCH DCN's office.
4. PT retrieves the key from the DCN's office to obtain the correct sized vest when consulted by provider to proceed with discharge process and fitting of appropriate device.
5. The radian r120 car seats are also stocked by KCH and are located with the clerk who manages entry to PICU/NICU. This is currently where storage space allows for these items.
6. A consult order is placed to PT by attending and or resident (typically Ortho or PDS) for proper equipment utilization, car seat or ez-vest, because are based on age, size, weight of the patients in need.
7. The stock maintenance of the equipment is managed by nursing in charge of the ez-vest and PT for the car seats.
8. This equipment has been vetted and approved through Pediatric Trauma Orthopedic Liaison since 2016.
9. For car seat fitting, that is evaluated by PT through the consult and they coordinate and advise attending for sizing and best fit purposes.
10. The Attending gives approval for utilization of equipment and the patient is discharged.
11. In the event that the attending cannot give approval for utilization of the ez-vest or the radian r120 car seat, the only other alternative for discharge is via EMS.



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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: NEONATAL TRAUMA CONSULTS

July 15, 2019

Process for Obtaining Neonatal Trauma Consults

Overview

As the only Level I Pediatric Trauma Center for most rural hospitals in central and eastern Kentucky it's imperative to have a guideline in place to outline the process, for at minimum, a Pediatric Trauma Consult for neonatal cases that may be born as a result of maternal trauma. As indicated by the American College of Surgeons Advanced Trauma Life Support, "the best initial trauma care of the fetus is to provide optimal resuscitation of the mother." (ATLS 10th Edition, pg.228). Though these cases are low volume, a clear guideline to trauma care for the neonate must be in place to assure comprehensiveness of care for this high-risk population.

The University of Kentucky, Kentucky Children's Hospital is a Level IV Neonatal Intensive Care Unit (NICU). They are equipped to care for the tiniest and most critically ill babies by their Neonatology team.

Through a networking communication with Pediatric Trauma Program Managers throughout the United States via the Pediatric Trauma Society Nursing Leadership Group, it has been determined that most Pediatric Trauma Centers have a guideline that incorporates at minimum a Pediatric Trauma Surgery consult on said patients.

Performance Improvement, Quality & Safety

Due to the low volume and high risk of these cases **ALL** identified cases will undergo review through the Pediatric Trauma Quality review program, beginning with the Trauma Program Manager and Trauma Medical Director.

Process for Neonatal Trauma Consult

- Patients will be identified through the daily trauma patient identification process conducted by the Trauma Program Office Nurses and Registrars.
- Patients may also be identified and reported to the Pediatric Trauma Medical Director and Pediatric Trauma Coordinator by medical or nursing personnel within the KCH system, including the Adult Trauma Service.
- The Pediatric Trauma Coordinator will notify the Pediatric Division of Surgery (PDS) of the patient and need for a Trauma Consult.
- A Trauma Consult may be ordered by the NICU provider.
- Upon the PDS service learning of the neonatal trauma consult, they shall notify the NICU provider and staff to inform they will be conducting a trauma consult. This is the process should PDS not receive the consult order from Neonatology.
- Any neonatal trauma patient being received through the Emergency Department point of entry, shall be treated as any other potential Pediatric Trauma Alert or Pediatric Trauma Alert Red activation should they meet that activation criteria.

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SECTION V: PEDIATRIC TRAUMA PATIENT CARE GUIDELINES: RADIOLOGY OVER-READ PROCESS

July 3, 2019, updated 7/17/2019

Process for Obtaining an Over-read by ER Radiologist for Outside Films on Non-Accidental Trauma Cases and Other Pediatric Trauma Cases

Overview

As a Level I Pediatric Trauma Center, we serve as the primary tertiary pediatric trauma center for most rural hospitals in central and eastern Kentucky. The majority of these transfers involve radiological imaging for which the outside facilities are required to send copies. The goal is to provide the best possible care with the least amount of radiation exposure to the pediatric patient population. However, in **rare** circumstances these images are not diagnostically conformational to the primary medical service, thus requiring an over read by the Emergency Department Radiologist. These rare occurrences may occur for child abuse cases in which there are head CT's from the OSH that may need over-read; and the other rare case is OSH imaging in a pediatric trauma case. These instances in either circumstance are deemed **RARE** and **ARE NOT** the norm.

In accordance with the American College of Surgeons (ACS) Trauma Quality Improvement Program (TQIP) Best Practice Guidelines on Imaging

(<https://www.facs.org/quality-programs/trauma/tqp/center-programs/tqip/best-practice>) the transferring facility should include all pertinent imaging and the initial radiologist interpretations of the images or the transferring facility must send radiology reports as soon as available. This ACS TQIP best practice concludes that in order to eliminate repeat filming, or to clarify any clinically significant discrepancies that may impact patient care, on occasion "best practice is to ensure that second-opinion radiology reads by the receiving radiologist are performed".

Over-Read Process for Non-Accidental Trauma Cases with Head CT's

- [Refer to the Non-Accidental Trauma Clinical Practice Guideline](#)
- If CT head obtained at outlying hospital, speak with radiologist for an official UK over- read.
- Must speak with attending radiologist (ER or Neuro radiology) and need to have images and report (if available).
- If radiologist agrees an over-read is needed, then either radiologist or requesting physician can call imaging records desk at 3-2131 to get the study assigned an accession number.

All Other Pediatric Trauma Cases

- Handled on a case by case process
- Radiology attending has to agree to the over-read first and there are situations when the radiologist would decline to over-read.
- A second opinion read on an outside study should be documented if the images are of good quality and there is medical necessity, defined by:
 1. Have outside images and no outside report is available (attempt has to be made to obtain outside report. Image records can assist in obtaining 3-2131).
 2. Have outside images and outside report but disagree with outside report.
- If the outside report is correct then that should be uploaded into our system and stand as the final read.

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